

May 1, 2026

Design Percent Completion: 100%

Statement of Work

CONSTRUCT METAL BUILDING, F5067

VANCE AIR FORCE BASE, OKLAHOMA

FACILITY PROJECT ID 1024054

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INDEX

PART 1 GENERAL 2

1.01. PROJECT DESCRIPTION 2

1.02. INSTRUCTIONS TO BIDDERS 2

1.03. LOCATION OF WORK 2

1.04. SCOPE OF WORK 2

1.05. CHANGES 11

1.06. WARRANTY 11

1.07. SCHEDULE OF WORK 12

1.08. WORK RESTRICTIONS 12

1.09. CONTRACTOR’S REPRESENTATIVE 14

1.10. PROVIDED SERVICES AND UTILITIES 15

1.11. INSPECTION 15

1.12. EXCESS MATERIALS (EXCLUDING HAZARDOUS MATERIALS OR WASTE) 15

1.13. SPECIAL MATERIALS 15

1.14. MATERIALS 16

PART 2 EXECUTION 17

2.01. SITE SAFETY REQUIREMENTS 17

2.02. INSTALLATION 17

PART 3 SITE INFORMATION 18

PART 4 APPENDICES 19

4.01. APPENDIX A – BID FORM 21

4.02. APPENDIX B – SPECIFICATIONS 23

4.03. APPENDIX C – DRAWINGS 79

4.04. APPENDIX D – SCHEDULE E 90

4.05. APPENDIX E – BASIS OF DESIGN DETAILS 94

PART 1 GENERAL

1.01. PROJECT DESCRIPTION

- A. The work covered under this Statement of Work shall consist of constructing a 40-foot by 60-foot, prefabricated metal building, complete-in-place (turnkey) and any incidental work necessary to provide a complete and usable finished product.

1.02. INSTRUCTIONS TO BIDDERS

- A. As part of the bid package the Contractor shall submit a completed copy of *APPENDIX A – BID FORM*, along with all required documentation as stated in Sam.gov.

1.03. LOCATION OF WORK

- A. Location of work to be accomplished under this statement of work is at Vance Air Force Base (AFB). Vance AFB is located three miles south of the City of Enid, Oklahoma. Facility 5067 will be constructed Northeast of the intersection of Conway Street and Macchi Avenue.

1.04. SCOPE OF WORK

- A. The Contractor shall furnish all supervision, labor, equipment, consumable materials, supplies, tools, and other items to complete the construction as listed in the Scope of Work. All quotes shall include all shipping charges for completion of the project. The contractor will coordinate all work with *ASRC Federal (ASRC)* Construction Management prior to execution. The list of items below are to be considered the general Scope of Work for the project.
- B. The Contractor shall field verify location of work, dimensions, and existing site conditions prior to commencing work. The Contractor shall contact ASRC Construction Management for any discrepancies with the Scope of Work.
- C. The Contractor shall furnish and construct a complete and usable, 40-foot wide by 60-foot-long pre-fabricated metal building with Portland Cement (P.C.) concrete slab, footings, doors, spray foam insulation, electrical lighting and power, electrical heating, PC concrete building approaches, sidewalks and entrance drive. Additional details are described below:

1. DEMOLITION:

- a. Refer to APPENDIX C – DRAWINGS.
- b. Tree Removal:
 - 1. Contractor to remove one existing tree on the northeast corner of the intersection of Conaway Street and Machi Avenue. Work includes grinding the stumps.
- c. Portland Cement (P.C.) concrete curb and gutter removal:
 - 1. Contractor to saw cut and remove approximately 70-foot of P.C. concrete curb and gutter to provide for the construction of the new driveway approach.

2. NEW CONSTRUCTION:

- a. Stamped Engineering Drawings:
 - 1. Contractor to submit stamped engineering drawings for the building slab, foundation and the pre-fabricated metal building to ASRC Construction Management for review and approval prior to construction. Submitted drawings must comply with all design requirements listed within this scope of work. Foundation drawings must meet building manufacturers' requirements.
- b. Site Preparation:
 - 1. The Owner will remove the existing topsoil and grass prior to construction.
 - 2. The Contractor shall place approximately 12-inches of selective fill material, or as necessary to provide positive drainage away from the building pad, and foundation.
 - a. *Includes compaction, grading and leveling the work site in preparation to construct the building pad and approaches.*

3. Compaction:

- a. Includes soil compaction testing to verify that requirements are met, per stamped engineering drawings.*
- b. Under paved areas – Compact to 95% Standard Proctor*
- c. Utility trenching - On grass areas, backfill/compact with existing soil in 12-inch lifts to 90% ASTM D698 minimum density.*

c. New P.C. Concrete:

- 1. Refer to *APPENDIX B – SPECIFICATIONS* and *APPENDIX C – DRAWINGS*.
- 2. Work in this section includes base material, geotextile fabric for soil stabilization, compaction of base and sub-base material, and reinforcement as applicable.
- 3. The Contractor shall provide positive drainage (2% grade) away from the building , concrete approaches and drive entrance.
- 4. Building Slab and Foundation/Piers/Footing
 - a. The Contractor shall construct per the building manufacturer requirements.*
 - b. Aggregate Base Thickness: 6-inches minimum*
 - c. Slab Thickness: 6-inches minimum*
 - d. Concrete Strength: 4,000 psi*
 - e. Surface finish: smooth*
 - f. Finish slab elevation: 1287.5 ft or as necessary to provide positive drainage away from the building and foundation.*
- 5. Building Approaches
 - a. The Contractor shall construct a reinforced concrete apron at each overhead door entrance on both sides of the building, as indicated on the plan sheets.*

6. Sidewalk/Entrance Landing

- a. *The Contractor shall construct an entrance landing and sidewalk on each side of the building, as indicated on the plan sheets, to provide pedestrian access from the building approach to the personnel entrances.*
- b. *Thickness: 5-inches minimum*

7. Driveway Approach

- a. *The Contractor shall construct a concrete driveway approach connecting Macchi Avenue to the new building.*
- b. *The Contractor shall strictly adhere to the minimum radii specified in the plan sheets to ensure adequate turning clearance and safe ingress/egress for all entering vehicles.*

d. Steel Pipe Bollards, P.C. Concrete Filled, Painted, 8-inch Diameter:

- 1. The Contractor shall install protective bollards on both sides of each overhead door entrance. Bollards must be positioned between the building approach and the overhead door entrance as specified in the plan sheets.
- 2. Length: 8 feet long x 4-foot depth
- 3. Quantity: 4
- 4. Color: Powder Coat Safety Yellow

e. Prefabricated Metal Building:

1. Assemble a 40-feet x 60-feet metal storage building as seen in *APPENDIX C – DRAWINGS*.

a. *Basis of Design: Mueller Metal Buildings (Refer to APPENDIX E – BASIS OF DESIGN DETAILS).*

b. *Proposed building must meet or exceed the following design requirements:*

1. *Building Details:*

i. Footprint (Length x Width): 60.0' x 40.0'

ii. Sidewall Bay Spacing: 3 @ 20.0'

iii. Eave Height: 12.0'

iv. Roof Slope: 2.0 in 12

v. Frame Type: Rigid Frame

2. *Building Code:*

i. Design Code: IBC'18

ii. Closed/Open: C

iii. Exposure: C

iv. Site Class: D

v. Seismic Coefficient: 0.32

vi. Importance – Wind: 1.00

vii. Importance – Seismic: 1.00

viii. Importance Snow: 1.00

3. *Building Loads:*

- i. Dead Load: 2.50 psf
- ii. Live Load: 20.00 psf
- iii. Load Reduction: Yes
- iv. Ground Snow: 15.00 psf
- v. Collateral: 3.00 psf
- vi. Wind Load: 111.00 mph

4. *Additional features and accessories:*

- i. Gutters and downspouts.
- ii. Awnings to cover over 3'x7' personnel doors.

5. *Sheets & Trim:*

- i. Roof and wall panels to be 26-gauge PBR.
- ii. Roof and trim colors to be "Colonial Red". Provide color samples to ASRC Construction Management for review and approval prior to construction.
- iii. Wall colors to be "Sandstone". Provide color samples to ASRC Construction Management for review and approval prior to construction.

c. Doors and Openings:

- 1. Contractor to install two 3' x 7' insulated metal doors. Door hardware to include knob or level style handles compatible with interchangeable core.*
 - i. Provide cylinders that are "BEST" compatible. Cores shall be "BEST" Cormax, stainless steel, 7 pin, keyway J. The cores shall be contractor provided and government installed in contractor provided and installed cylinders. Contractor shall contact the government to coordinate with hardware suppliers and real property specialist.
 - ii. Contractor shall coordinate keying requirements with Vance real properties. Cores shall be keyed and shipped directly to Vance real property.
 - iii. All door hardware to be 626 finish.
 - iv. Contractor to submit door and door hardware product data to ASRC Construction Management for review and approval prior to construction.
- 2. Contractor to install two motorized 10' tall x 20' wide insulated metal overhead doors.*
 - i. Electric motors for overhead doors to be powered by "PANEL A" (Refer to APPENDIX C – DRAWINGS for motor location and electrical details). Motors to be sized by overhead door vendor.
 - ii. Overhead doors to have chain pulley for manual roll up in the event of a power outage.
 - iii. Bottom of overhead door to be 0.5" below Top of Slab (T.O.S).
 - iv. Contractor to submit overhead door and door opener product data to ASRC Construction Management for review and approval prior to construction.

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- f. Spray-foam Insulation:
1. Insulate the interior of the building (walls and roof) with closed cell spray-foam insulation. Insulate all walls to 2" thick or to an R-value of 14, whichever is greater. Insulate roof to 3" thick or to an R-value of 21, whichever is greater.
 2. Contractor to submit product to ASRC Construction Management for review and approval prior to construction.
- g. Electrical (Refer to *APPENDIX C – DRAWINGS* for locations and details):
1. Place 200 Amp Main Distribution Panel (MDP) (Refer to *APPENDIX E – BASIS OF DESIGN DETAILS* for electrical panel).
 - a. *New MDP ("PANEL A") to be surface mounted square D (QO) Panel – 200A, 240V Single Phase – 30 Space, Main Breaker, Load Center. Refer to APPENDIX C – DRAWINGS, SHEET E701 for panel schedule.*
 - b. *Contractor to submit product to ASRC Construction Management for review and approval prior to construction.*
 2. Trench and wire from transformer to MDP.
 - a. *Transformer to be installed by owner prior to construction.*
 - b. *Contractor to install 2-INCH Schedule 40 PVC conduit and pull 3/0 copper wire and #6 grounding wire to connect and power "PANEL A" via 37.5 kVA transformer.*
 - c. *Contractor to submit product to ASRC Construction Management for review and approval prior to construction.*
 3. Mount and wire all lights, outlets, and three-way light switches (Refer to *APPENDIX E – BASIS OF DESIGN DETAILS* for lighting).
 - a. *Exterior lighting to be switched on/off via photocell mounted on northwest exterior of building.*
 - b. *Contractor to product data to ASRC Construction Management for review and approval prior to construction.*
 4. Run power for electric garage doors and electric unit heaters.
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- h. Forced-air Electric Unit Heaters (Refer to *APPENDIX C – DRAWINGS SHEET M101* for locations and details):
 - 1. Install and power four 4 kW (13,648 BTU/hr.) 240V, single phase, forced-air electric unit heaters in corners of building.
 - a. *Unit heaters are to be mounted from the ceiling and comply with manufacturers' installation recommendations.*
 - 1. *Ensure unit heaters are not located within 3' of any insulation, walls, or ceiling.*
 - 2. *Ensure unit heaters will not create obstructions with planned vehicle parking, other equipment clearances, and obvious pedestrian paths. Refer to APPENDIX C – DRAWINGS SHEET A101 for planned vehicle parking.*
 - 3. *Orientate unit heaters to maximize airflow through the building.*
 - b. *Unit heaters are to have a built-in thermostat set to maintain a minimal indoor air temperature of 45°F throughout the facility.*
 - c. *Basis of Design: King Electric KB2404-1-T-B1 (Refer to APPENDIX E – BASIS OF DESIGN DETAILS).*
 - d. *Contractor to submit proposed product to ASRC Construction Management for review and approval prior to construction.*
 - i. Fire extinguishers are required for this building, please comply with NFPA 10 for quantity, locations, and cabinet requirements.
 - D. The Contractor shall always maintain clean areas free of clutter and debris during construction. After the work is completed, clean site to preconstruction conditions, or better.

E. Protection of Government Property

Take special care to protect Government property. Return areas damaged as a result of construction under this contract to their original condition. Protection of existing vegetation, structures, equipment, utilities, and improvements, perform the following:

1. Remove or alter existing work or facilities in such a manner as to prevent injury or damage to any portion of the existing work or facilities to remain.
2. Repair or replace portions of existing work altered during construction operations to match existing or adjoining work, as approved by ASRC Construction Management. At the completion of operations, existing work must be in a condition equal or better than that which existed before any new work started.

F. Construction Fence (Temporary Fence)

The Contractor shall provide construction fencing around the job site and open trenches.

1. Provide 4-foot-high heavy duty plastic safety fence (orange fence).
2. Provide 4-foot-high chain-link fence.

G. Erosion Control

1. The Contractor shall provide 3-foot-high rolled erosion control filter fabric (snow fence) on steel posts.

1.05. CHANGES

- A. No oral statement by any person other than the Contracting Officer will in any manner or degree modify or otherwise affect the terms of this contract.

1.06. WARRANTY

- A. The Contractor shall provide a standard 1-year Warranty for all labor and material from date of the base's acceptance of completion. For warranted items furnish the Manufacturer's original warranty accompanied by a copy of the supplier's receipt showing place of purchase, telephone number of supplier, address, delivery order if applicable and ticket number.

1.07. SCHEDULE OF WORK

- A. The Contractor shall be responsible for establishing a schedule to meet construction time.
- B. The Contractor shall submit a schedule of all project work in advance to ASRC Construction Management indicating all work activities including required interruptions of utilities, facilities activities, or traffic flow.
- C. The Contractor shall complete work on this contract such that interferences to necessary activities within and adjacent to the construction site are minimized and all Government property and personnel are always protected from incidental damage.
- D. The Contractor shall coordinate work of all trades to prevent any conflicts. The Contractor shall resolve any conflicts of components before proceeding with construction.

1.08. WORK RESTRICTIONS

A. EXISTING UNDERGROUND UTILITIES

Any underground utilities within the work areas will initially be located and identified with paint markings on the surface by Vance AFB personnel. The Contractor shall maintain all utility markings, after base personnel have located them, for the duration of the contract. The Contractor shall obtain an approved Air Force Form 103, BASE CIVIL ENGINEERING WORK CLEARANCE REQUEST before beginning any underground work. ***The Contractor shall notify ASRC Construction Management at least two weeks in advance of desired commencement of work.*** The Contractor is responsible for repairing any utilities damaged during construction.

B. WORK HOURS, ACCESS, AND PASSES

All Contractor employees, including subcontractors, subcontractors' employees, suppliers, and suppliers' employees are required to comply with the Installation Security Requirements regarding personnel, vehicle, and equipment security passes and access to the jobsite. Nothing in the contract is to be construed in any way to limit the authority of the Commanding Officer to prescribe new, or to enforce existing security regulations governing the admission or exclusion of persons and the conduct of persons while on VAFB, including but not limited to, the rights of search of all persons or vehicles on the Installation.

1. Access to Buildings / Occupied Buildings: The Contractor may work in or around existing occupied buildings. The Contractor is responsible, via the ASRC Construction Management, to obtain access to building and facilities and arrange for them to be opened and closed. Do not enter the building(s) without prior approval of the ASRC Construction Management. Keep the existing buildings and their contents secure at all times. Provide temporary closures as required to maintain security.
2. Passes: Contractor employees and representatives performing work under this contract are required to be either United States citizens or documented legal residents (status verified by prime contractor) and have a Real ID driver's license. All Contractor employees must obtain the required passes. Failure to obtain security and base access passes must not be a cause for contract performance time extension. The Contractor is required to immediately turn in all terminated employee's passes to the issuing office.
3. Contractor Vehicles: All vehicles must display a valid state license plate and safety inspection sticker, if applicable, and must be maintained in good condition. The company name must be displayed in a clearly visible manner and size on each Contractor vehicle used in the course of work. Registration, proof of insurance and driver's licenses are required to obtain a station vehicle pass.
4. Work Hours: Normal working hours are from 7:30 a.m. to 4:30 p.m., Monday through Friday. Night, Weekend (Saturday and Sunday) and Federal Holiday work will be subject to approval by the ASRC Construction Management. ***The Contractor must provide a written request of their intention to work outside of normal working hours to ASRC Construction Management at least 48 hours in advance of the specified time.***
 - a. The Contractor shall perform all work during normal working hours except as otherwise approved to accommodate agreed upon "shutdowns" and other required "after hours" work. At no time will any facility have their services or utilities disrupted during normal working hours without prior approval from the ASRC Construction Management.
5. Contractor employees must conduct themselves in a proper, efficient, courteous, and businesslike manner. Remove from the site any individual whose continued employment is deemed by the ASRC Construction Management to be contrary to the public interest or inconsistent with the best interests of National Security.

1.09. CONTRACTOR'S REPRESENTATIVE

- A. Superintendent: A representative of the Contractor shall directly supervise accomplishment of this work. The representative will be responsible for coordinating all phases of work with the ASRC Construction Management. The representative shall always be on the job site when work is being performed and shall have authority to execute no cost field changes on behalf of the Contractor if agreed to by all parties involved.
 - 1. Qualifications: Must have a minimum of 3 years of experience as a Superintendent on previous projects of similar size and complexity. Provide a Superintendent Resume for the proposed on-site Project Superintendent describing experience with references and qualifications to ASRC Construction Management for approval. ASRC Construction Management reserves the right to interview the proposed on-site Project Superintendent at any time in order to verify the submitted qualifications.
- B. Quality Control (QC): A representative of the Contractor responsible for all quality control of the job site and will be held responsible by the ASRC Construction Management for the removal and installation of all products in accordance with the statement of work. The representative shall be competent and familiar with all phases of work under their authority.
 - 1. Qualifications: Must have a minimum of 2 years' combined experience as a Superintendent, Inspector, QC manager, Project Manager, or Construction Manager on similar size and type construction contracts which included major trades included in this statement of work. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance.
- C. A single representative of the Contractor for supervising the project, ensuring quality control (QC) and site safety is permitted provided that the representative is competent and familiar with all phases of work under their authority, it is qualified to perform these duties and meets all requirements as stated in sections 1.09.A.1 and 1.09.B.1.
- D. Contractor Personnel: Provide ASRC Construction Management the name(s) of the supervisory person(s) authorized to act for the Contractor. Provide, and update as required, a list of the key personnel for the Contractor and Subcontractors including addresses and telephone numbers (main and cell) for use in the event of an emergency.

1.10. PROVIDED SERVICES AND UTILITIES

- A. All reasonable required amounts of electric power will be made available to the Contractor at the discretion of ASRC Construction Management from the nearest suitable and available connection.
- B. All reasonable required amounts of water will be made available to the Contractor at the discretion of ASRC Construction Management.
- C. The Contractor shall be required to provide their own latrines that are staked down to prevent being turned over.

1.11. INSPECTION

- A. The work shall be conducted in accordance with the Statement of Work and is subject to the inspection of the ASRC Construction Management and Government Representative to ensure strict compliance with the terms of the statement of work.

1.12. EXCESS MATERIALS (EXCLUDING HAZARDOUS MATERIALS OR WASTE)

- A. All waste materials resulting from execution of work shall become the property of the Contractor. The Contractor shall haul all non-hazardous waste off Vance AFB and dispose of it in a legal manner.
- B. The Contractor shall be required to control all foreign objects debris (FOD) from exiting the job site at all times.
- C. At no time shall the Contractor dispose of any recyclable materials in the landfill. All recyclable material shall be taken to an approved recycling center at no cost to the government.

1.13. SPECIAL MATERIALS

- A. Prohibited Material: The manufacturer of materials to be used shall certify the products do not contain mercury, toluene, chlorinated solvents, hydrolysable chlorine derivatives, ethylene base glycol ethers and their acetates, nor any carcinogen, as defined in 29 CFR 1910.1200. Materials used shall not contain lead and hexavalent chromium compounds.
- B. At no time shall class 1 Ozone Depleting Chemical (ODC's) be specified and shall not be used unless otherwise allowed by ASRC Construction Management.

1.14. MATERIALS

- A. All materials must meet the Buy American Act.
- B. Material Submittals:
 - 1. Copies of all materials and shop drawings are to be submitted to ASRC Construction Management and must be approved prior to purchasing any items.
 - 2. All hazardous chemicals used on the project are required to have Safety Data Sheets (SDS) approved by the Vance Environmental Office prior to transport to the installation.
 - 3. The Contractor shall use the Short-Term Hazardous Material Inventory and Usage Report Form to get Hazardous Materials authorized for use inside the fence.
- C. List of Materials
 - 1. Refer to Section 1.04 Scope of Work, within this Statement of Work.
 - 2. Refer to Appendix B – Specifications
 - 3. Refer to Appendix C – Drawings
- D. Material Delivery: The Contractor shall have all materials delivered during normal working hours and will have a representative present to receive shipments. The Contractor shall ensure all delivery vehicles have sufficient project information and are prepared to undergo inspection upon entering the Installation. This includes name and address of the project site, name of prime contractor and a point of contact for delivery. Material will be delivered to a location for staging as approved by ASRC Construction Management.

PART 2 EXECUTION

2.01. SITE SAFETY REQUIREMENTS

- A. All work shall be done in compliance with the safety standards as defined in DAFMAN 91-203, Air Force Occupational Safety, Fire, and Health Standards.
- B. The Contractor shall comply with all applicable standards and provisions of the Occupational Safety and Health Act (OSHA) (29 CFR OSHA 1926).
- C. The Contractor shall take all necessary and prudent safety precautions to ensure safety of the workforce and other exposed personnel.
- D. The Contractor shall coordinate all requests for entry to the airfield or Air Force control/restricted areas through ASRC Construction Management.

2.02. INSTALLATION

- A. Project Duration:
 - 1. The total period of performance is 165 calendar days as described below.
 - a. Non-Performance Period: The Contractor shall have up to 45 calendar days of nonperformance for mobilization and material lead time. If material lead time exceeds this number of days, the nonperformance period may be extended with coordination of the ASRC Construction Management and approval by Contracting Officer.
 - b. Performance Period: All work shall be completed within 120 calendar days.
- B. Installation shall include all items listed in this statement of work, and any incidental work necessary to provide a complete and usable finished product.
- C. Contractor shall remove equipment and material from the site upon completion of work as directed by ASRC Construction Management.

PART 3 SITE INFORMATION

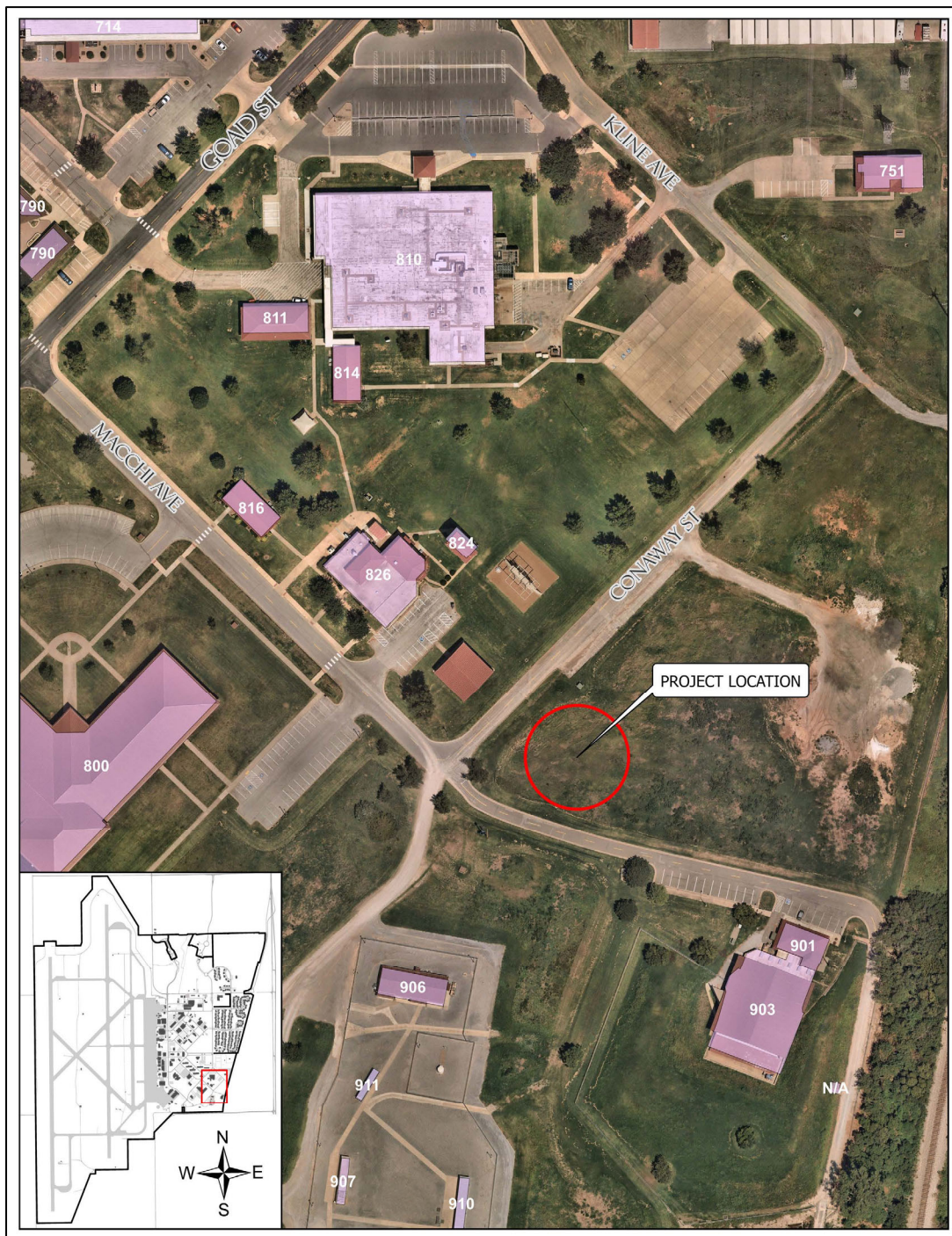


Figure 1: Project Location

PART 4 APPENDICES

4.01. APPENDIX A – BID FORM

4.02. APPENDIX B – SPECIFICATIONS

32 01 19.61 SEALING JOINTS IN RIGID PAVEMENTS

32 11 23 AGGREGATE BASE COURSES

32 13 13.03 PCC PAVEMENTS FOR ROADS AND SITE FACILITIES

32 16 19 CONCRETE CURBS, GUTTERS AND SIDEWALKS

4.03. APPENDIX C – DRAWINGS

A101 CONCEPTUAL FLOOR PLAN

A201 BUILDING ELEVATIONS

CS101 CONCEPTUAL SITE PLAN

CS102 NEW P.C. CONCRETE LOCATION

C501 ROADWAY PAVEMENT DETAILS

M101 MECHANICAL PLAN

E100 ELECTRICAL SITE PLAN

E101 ELECTRICAL POWER PLAN

E102 ELECTRICAL LIGHTING PLAN

E701 ELECTRICAL SCHEDULES

4.04. APPENDIX D – SCHEDULE E

4.05. APPENDIX E – BASIS OF DESIGN DETAILS

END OF STATEMENT OF WORK

APPENDIX A – BID FORM

APPENDIX A - BID FORM



Fac. Proj. ID FP1024054

Proj. Name: Construct Metal Building, F5067

ITEM	DESCRIPTION	UNIT	QTY	UNIT PRICE	TOTAL
Base Bid					
1	Tree Removal	LS	1		\$ -
2	Saw-cut and Remove Existing Portland Cement Concrete (P.C.C.) Curb and Gutter	LS	1		\$ -
3	P.C. Concrete - Complete-In-Place - Includes site preparation, base and sub-base material, compaction, geotextile fabric, rebar, tie-bars, joint sealant and primers	LS	1		\$ -
4	Steel Pipe Bollards, P.C. Concrete Filled, Painted, 8-inch Diameter - Complete-In-Place	LS	1		\$ -
5	Prefabricated Metal Building - Complete-In-Place - Includes Engineering Stamped Drawings, Building Materials and Personnel Doors.	LS	1		\$ -
6	Overhead Door and Door Openers - Complete-In-Place	LS	1		\$ -
7	Spray Foam Insulation	LS	1		\$ -
8	Electrical	LS	1		\$ -
9	Electric Unit Heaters	LS	1		\$ -
10	Fire Extinguishers	LS	1		\$ -
11	Construction Fence	LS	1		\$ -
12	Erosion Control - Silt Fence	LS	1		\$ -
13	Final Clean-Up	LS	1		\$ -
	Total Cost Base Bid (Sum of lines 1 through 13)				\$ -

Contractor's Name

Date

Contractor's Signature

Address:

Phone Number: _____

E-mail address: _____

APPENDIX B – SPECIFICATIONS

SECTION 32 01 19.61

SEALING OF JOINTS IN RIGID PAVEMENT

11/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C603	(2014; R 2019) Standard Test Method for Extrusion Rate and Application Life of Elastomeric Sealants
ASTM C639	(2015; R 2020) Standard Test Method for Rheological (Flow) Properties of Elastomeric Sealants
ASTM C661	(2015) Indentation Hardness of Elastomeric-Type Sealants by Means of a Durometer
ASTM C679	(2015) Tack-Free Time of Elastomeric Sealants
ASTM C719	(2014; R 2019) Standard Test Method for Adhesion and Cohesion of Elastomeric Joint Sealants Under Cyclic Movement (Hockman Cycle)
ASTM C792	(2015; R 2020) Effects of Heat Aging on Weight Loss, Cracking, and Chalking of Elastomeric Sealants
ASTM C793	(2005; R 2017) Standard Test Method for Effects of Laboratory Accelerated Weathering on Elastomeric Joint Sealants
ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM C1016	(2014) Standard Test Method for Determination of Water Absorption of Sealant Backing (Joint Filler) Material
ASTM C1193	(2013) Standard Guide for Use of Joint Sealants
ASTM D412	(2016) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension

APPENDIX B

ASTM D789	(2015) Determination of Relative Viscosity and Moisture Content of Polyamide (PA)
ASTM D903	(1998; R 2017) Standard Test Method for Peel or Stripping Strength of Adhesive Bonds
ASTM D5249	(2010; R 2016) Standard Specification for Backer Material for Use with Cold-and Hot-Applied Joint Sealants in Portland-Cement Concrete and Asphalt Joints
ASTM D5329	(2016) Standard Test Methods for Sealants and Fillers, Hot-Applied, for Joints and Cracks in Asphalt Pavements and Portland Cement Concrete Pavements
ASTM D5893/D5893M	(2016) Standard Specification for Cold Applied, Single Component, Chemically Curing Silicone Joint Sealant for Portland Cement Concrete Pavements
ASTM D6690	(2015) Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements

U.S. ARMY CORPS OF ENGINEERS (USACE)

COE CRD-C 525	(1989) Corps of Engineers Test Method for Evaluation of Hot-Applied Joint Sealants for Bubbling Due to Heating
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.

SD-03 Product Data

Sealants, Blocking Media/Back-up material, Backer Rod, Bond Breaking Tape; G

Submit catalog cuts, specifications, Safety Data Sheets and other information documenting conformance to Contract requirements.

Manufacturer's Recommendations

SD-07 Certificates

Equipment List

SD-08 Manufacturer's Instructions

Sealants

APPENDIX B

Provide instructions that include, but not limited to: storage requirements, ambient temperature and humidity ranges, and moisture condition of joints for successful installation; requirements for preparation of joints; safe heating temperature; mixing instructions; installation equipment and procedures; application and disposal requirements; compatibility of sealant with filler material; curing requirements; and restrictions to be adhered to in order to reduce hazards to personnel or to the environment. Submit instructions at least 30 days prior to use.

1.3 DELIVERY, STORAGE, AND HANDLING

Inspect materials delivered to the site for visible damage, and unload and store with a minimum of handling. Deliver joint materials in original sealed containers and protect from freezing or overheating. Provide jobsite storage facilities capable of maintaining temperature ranges within manufacturers recommendations.

1.4 ENVIRONMENTAL REQUIREMENTS

Do not proceed with work when weather conditions detrimentally affect the quality of cleaning joints or applying sealants. Proceed with joint preparation and sealing only when weather conditions are in accordance with manufacturer's instructions. Install joint sealant to dry surfaces and protect sealant and bond breakers from moisture.

1.5 TRAFFIC CONTROL

Do not permit vehicular or heavy equipment traffic on the pavement in the area of the joints being sealed during the protection and curing period of the sealant. Permit traffic on the pavement at the end of the curing period.

PART 2 PRODUCTS

2.1 SEALANTS

Use self leveling, non-acid curing silicone sealant meeting the following requirements in accordance with [ASTM C920](#) or [ASTM C1193](#):

TEST	TEST METHOD	REQUIREMENTS
Weight Loss	ASTM C792 Modified (see Note 1 below)	10 percent max.
Flow	ASTM C639 (Type I)	Smooth and level
Extrusion Rate	ASTM C603	30 sec. max.

APPENDIX B

TEST	TEST METHOD	REQUIREMENTS
Tack Free Time	ASTM C679	5 hours max.
Hardness (Shore 00) (see Note 2 below)	ASTM C661	30 - 80
Tensile Stress at 150 Percent Elongation (see Note 2 below)	ASTM D412 (Die C)	30 psi max.
Percent Elongation (see Note 2 below)	ASTM D412 (Die C)	700 min.
Accelerated Weathering	ASTM C793	Pass 5000 hours
Bond and Movement Capability	ASTM C719	Pass 10 cycles at plus 50 percent movement (no adhesion or cohesion failure)
Peel	ASTM D903	Minimum 20 psi of width with at least 75 percent cohesive failure
NOTES: 1. Percent weight loss of wet (uncured) sample after placing in forced-draft oven maintained at 158 degrees plus 1 degree F for two hours. 2. Specimen cured 21 days at 73 degrees plus 1 degree F and 50 percent plus 5 percent humidity.		

ACCELERATED WEATHERING FACTORY TEST REPORT. For the Accelerated Weathering test, in lieu of testing of actual sealant to be used on the project, it is permitted to submit a report of a factory test, performed within two years of Contract award.

2.2 PRIMERS

Use primers in accordance with the recommendation of the manufacturer.

2.3 BOND BREAKERS

2.3.1 Blocking Media/Backup Materials

Provide backup (joint filler) material that is a compressible, nonshrinking, nonstaining, nonabsorbing, nonreactive material with the sealant. Use backup material compliant with ASTM D5249. Use material with a melting point at least 5 degrees F greater than the pouring temperature of the sealant being used when tested in accordance with ASTM D789. Use material with a water absorption of not more than 5 percent of the sample weight when tested in accordance with ASTM C1016. Use backup (joint filler) material that is 25 plus or minus 5 percent larger in diameter than the nominal width of the crack. Use blocking media consistent with the sealant manufacturer's installation instructions.

2.3.2 Bond Breaking Tapes

Provide a bond breaking tape or separating material that is a flexible, nonshrinkable, nonabsorbing, nonstaining, and nonreacting adhesive-backed tape. Use material with a melting point at least 5 degrees F greater than the pouring temperature of the sealant being used when tested in accordance with ASTM D789. Use bond breaker tape approximately 1/8 in wider than the nominal width of the joint and that does not bond to the sealant. Use bond

breaking tape shall be consistent with the sealant manufacturer's installation instructions.

PART 3 EXECUTION

3.1 EXECUTING EQUIPMENT

Submit [equipment list](#) and description of the equipment to be used and a statement from the supplier of the sealant that the proposed equipment is acceptable for installing the specified sealant. Use equipment for heating, mixing, and installing seals in accordance with the instructions provided by the sealant manufacturer. Provide equipment, tools, and accessories necessary to clean existing joints and install liquid joint sealants. Maintain machines, tools, and other equipment in proper working condition. Submit a list of proposed equipment to be used in performance of construction work including descriptive data, 30 days prior to use on the project.

3.1.1 Joint Cleaning Equipment

3.1.1.1 Tractor-Mounted Routing and Plowing Tool

Use routing tools for removing old sealant from the joints, of such shape and dimensions and so mounted on the tractor that do not damage the sides of the joints. Use tools designed to be adjusted to remove the old material to varying depths and widths as required. Use equipment capable of maintaining accurate cutting depth and width control. Use a joint plow equipped with a spring or hydraulic mechanism to release pressure on the tool prior to spalling the concrete. Do not permit the use of V-shaped tools or rotary impact routing devices. Permit the use of hand-operated spindle routing devices to clean and enlarge random cracks.

3.1.1.2 Concrete Saw

Provide a self-propelled power saw, with water-cooled diamond or abrasive saw blades, for cutting joints to the depths and widths specified, for refacing joints, cleaning sawed joints where sandblasting does not provide a clean joint, widening, or deepening existing joints as specified without damaging the sides, bottom, or top edge of joints. Permit single or gang type blades with one or more blades mounted in tandem for fast cutting. Select saw adequately powered and sized to cut specified opening with not more than two passes of the saw through the joint.

3.1.1.3 Sandblasting Equipment

Include with the sandblasting equipment an air compressor, hose, and long-wearing venturi-type nozzle of proper size, shape and opening. Do not permit the maximum nozzle opening to exceed 1/4 in. Use a portable air compressor capable of providing not less than 150 cfm and maintaining a line pressure of not less than 90 psi at the nozzle while in use. Demonstrate compressor capability, under job conditions, before approval. Use a compressor equipped with traps that maintain the compressed air free of oil and water. Use a nozzle with an adjustable guide that holds the nozzle aligned with the joint approximately 1 in above the pavement surface. Adjust the height, angle of inclination and the size of the nozzle to secure satisfactory results.

3.1.1.4 Waterblasting Equipment

Include with the waterblasting equipment a trailer-mounted water tank, pumps, high-pressure hose, wand with safety release cutoff control, nozzle, and auxiliary water resupply equipment. Provide water tank and auxiliary resupply equipment of sufficient capacity to permit continuous operations. Use a nozzle with an adjustable guide that holds the nozzle aligned with the joint approximately 1 in above the pavement surface. Adjust the height, angle of inclination and the size of the nozzle to obtain satisfactory results. Use a pressure gauge mounted at the pump that shows the pressure in psi at which the equipment is operating.

3.1.1.5 Air Compressor

Use a portable air compressor capable of operating the sandblasting equipment and capable of blowing out sand, water, dust adhering to sidewalls of concrete, and other objectionable materials from the joints. Use a compressor that provides air at a pressure not less than 90 psi and a minimum rate of 150 cubic ft of air per minute at the nozzles and free of oil.

3.1.1.6 Vacuum Sweeper

Use a self-propelled, vacuum pickup sweeper capable of completely removing loose sand, water, joint material, and debris from pavement surface.

3.1.1.7 Hand Tools

Permit the use of hand tools, such as brooms and chisels, when approved, for removing defective sealant from a crack and repairing or cleaning the crack faces.

3.1.2 Sealing Equipment

Use joint sealing equipment of a type required by the sealant manufacturer's installation instructions. Use equipment capable of installing sealant to the depths, widths and tolerances indicated. Do not proceed with joint sealing when malfunctions are noted until the malfunctions are corrected.

3.1.2.1 Hot-Poured Sealing Equipment

Use mobile unit applicators equipped with a double-boiler, agitator-type kettle with an oil medium in the outer space for heat transfer for heating and installing ASTM D6690 joint sealant materials; a direct-connected pressure-type extruding device with a nozzle shaped for inserting in the joint to be filled; positive temperature devices for controlling the temperature of the transfer oil and sealant; and a recording thermometer for indicating the temperature of the sealant. Use an applicator unit designed so that the sealant circulates through the delivery hose and returns to the inner kettle when not in use.

3.1.2.2 Cold-Applied, Single-Component Sealing Equipment

Use equipment for installing ASTM D5893/D5893M single component joint sealants that consists of an extrusion pump, air compressor, following

plate, hoses, and nozzle for transferring the sealant from the storage container into the joint opening. Use a nozzle with dimensions that allows the tip of the nozzle to extend into the joint to allow sealing from the bottom of the joint to the top. Maintain the initially approved equipment in good working condition, serviced in accordance with the supplier's instructions, and unaltered in any way without obtaining prior approval. Use lined hoses and seals to prevent moisture penetration and withstand pumping pressures. Use equipment free of contamination from previously used or other type sealant. Permit use of small hand-held air-powered equipment (i.e., caulking guns) for small applications.

3.2 SAFETY

In accordance with the provisions of the Contract respecting "Accident Prevention," take appropriate measures to control worker exposure to toxic substances during the work. Provide personnel protective equipment as required. Make Material Safety Data Sheets (Department of Labor Form OSHA-20 or comparable form) available on the site.

3.3 PREPARATION OF JOINTS

Unless otherwise indicated, remove existing material, saw, clean and reseal joints. Do not proceed with final cleaning operations by more than one working day in advance of sealant. Clean joints by removing existing joint sealing compound, bond-breakers, dirt, laitance, curing compound, filler, and protrusions of hardened concrete from the sides and upper edges of the joint space to be sealed and other foreign material with the equipment. Do not permit cleaning procedures that damage joints or previously repaired patches by chipping or spalling. Remove existing sealant to the required depth. Precise shape and size of existing joints vary, and conditions of joint walls and edges vary and include but are not limited to rounding, square edges, sloping, chips, voids, depressions, and projections.

3.3.1 Existing Sealant Removal

Cut loose the in-place sealant from both joint faces and to the required depth, using the concrete saw as specified in paragraph EQUIPMENT. Provide a depth sufficient to accommodate blocking media and bond breakers that are required to maintain the depth of new sealant to be installed. For expansion joints, remove existing sealant to a depth of not less than the indicated depth. When existing preformed expansion-joint material is more than 1 in below the surface of the pavement, remove existing sealant to the top of the preformed joint filler. Prior to further cleaning operations, remove loose old sealant remaining in the joint opening by blowing with compressed air. Permit use of hand tools to remove sealant from random cracks. Do not permit chipping, spalling, or other damage to the concrete. Clean pavement surface with vacuum sweeper. Protect previously cleaned joints from being contaminated by subsequent cleaning operations.

3.3.2 Sawing

3.3.2.1 Refacing of Joints

Accomplish refacing of joints using a concrete saw as specified in paragraph EQUIPMENT to remove residual old sealant and a minimum of

concrete from the joint face to provide exposure of newly cleaned concrete, and, if required, to enlarge the joint opening to the width and depth shown on the drawings. Provide exposure of newly clean concrete through removal. Remove burrs and irregularities from sides of joint faces. Stiffen the blade with a sufficient number of dummy (used) blades or washers. Clean, immediately following the sawing operation, the joint opening using a water jet to remove saw cuttings and debris and adjacent concrete surface. Protect adjacent previously cleaned joint spaces from receiving water and debris during the cleaning operation.

- a. Joint Widening (Except Expansion Joints): Saw joints having grooves less than $\frac{3}{8}$ in wide and less than 1 in deep to a minimum width of $\frac{1}{2}$ in and to the minimum depth as indicated.

3.3.2.2 Refacing of Random Cracks

Accomplish sawing of the cracks using a power-driven concrete saw as specified in paragraph EQUIPMENT. Use a saw blade 6 in or less in diameter to enable the saw to follow the trace of the crack. Stiffen the blade with dummy (or used) blades or washers. Immediately following the sawing operation, clean the crack opening using a water jet to remove saw cuttings and debris.

3.3.3 Final Cleaning of Joints

3.3.3.1 Sandblasting

Following removal of existing sealant, and sawing, and immediately before resealing, clean newly exposed concrete joint faces and pavement surface extending to a minimum of $\frac{1}{2}$ in up to 2 in from each joint edge by sandblasting until concrete surfaces in the joint space are free of sealants, dust, dirt, water and other foreign materials that prevent bonding of new sealants to the concrete. Use sand particles of the proper size and quality for the work. Perform sandblasting with specified nozzles, air compressor, and other appurtenant equipment. Position nozzles to clean the joint faces. Make at least two passes; one for each joint face. Make as many passes as required for proper cleaning. Immediately prior to sealing the joint, blow out the joint spaces with compressed air until completely free of sand, water, and dust. Install joint sealants to dry joints. Replace expansion joint filler material damaged in performing the work with new materials of the same type and dimensions as the existing material, or with appropriate blocking media.

3.3.4 Bond Breaker

At the time the joints receive the final cleaning and are dry, install bond breaker material as indicated with a steel wheel or other approved device.

3.3.4.1 Blocking Media (Backer Rod) (Except for Expansion Joints)

When the joint opening is of a greater depth than indicated for the sealant depth, plug or seal off the lower portion of the joint opening using a blocking media/back-up material to prevent the entrance of the sealant below the specified depth. Take care to ensure that the blocking media/backup material is placed at the specified depth and is not stretched or twisted during installation.

3.3.4.2 Bond Breaking Tape

Where inserts or filler materials contain bitumen, or the depth of the joint opening does not allow for the use of a backup material, insert a bond breaker separating tape to prevent incompatibility with the filler materials and three-sided adhesion of the sealant. Bond the tape to the bottom of the joint opening to prevent it from floating up into the new sealant.

3.3.5 Rate of Progress of Joint Preparation

Limit the stages of joint preparation, including sandblasting, air pressure cleaning and placing of the back-up material to only that lineal footage that can be sealed during the same day.

3.3.6 Disposal of Debris

Sweep pavement surface to remove excess joint material, dirt, water, sand, and other debris by vacuum sweepers or hand brooms. Remove the debris immediately to a point off station.

3.4 PREPARATION OF SEALANT

3.4.1 Hot-Poured Sealants - (Not Use for this Project)

Heat hot-poured sealing materials in accordance with [ASTM D6690](#) and with safe heating temperature ranges recommended by the manufacturer. Withdraw and waste sealant that has been overheated or subjected to heating for over 3 hours or that remain in the applicator at the end of the day's operation. Heat sealant in specified equipment.

3.4.2 Single-Component, Cold-Applied Sealants

Inspect the [ASTM D5893/D5893M](#) sealant and containers prior to use. Reject materials that contain water, hard caking of any separated constituents, nonreversible jell, or materials that are otherwise unsatisfactory. Do not reject sealants that exhibit settlement of constituents in a soft mass that can be readily and uniformly remixed in the field with simple tools.

3.5 INSTALLATION OF SEALANT

3.5.1 Time of Application

After approval of the test section, seal joints immediately following final cleaning and placing of bond breakers. Commence sealing joints when walls are dust free and dry, and when weather conditions meet sealant manufacturer's instructions. If the above conditions cannot be met, or when rain interrupts sealing operations, reclean and permit the joints to dry prior to installing the sealant.

3.5.2 Sealing Joints

Do not install joint sealant until joints to be sealed have been inspected and approved. Install bond breaker just prior to pouring sealant. Fill the joints with sealant from bottom up until joints are uniformly filled solid from bottom to top using the specified equipment for the type of sealant required. Fill joints to $1/8$ in plus or minus $1/16$ in below top of

pavement, and without formation of voids or entrapped air. Do not permit gravity methods or pouring pots to be used to install the sealant material. Except as otherwise permitted, tool the sealant immediately after application to provide firm contact with the joint walls and to form the indicated sealant profile below the pavement surface. Remove excess sealant that has been inadvertently spilled on the pavement surface. Do not permit traffic over newly sealed pavement until authorized. When a primer is recommended by the manufacturer, apply it evenly to the joint faces in accordance with the [manufacturer's recommendations](#). Check sealed joints frequently to ensure that newly installed sealant is cured to a tack-free condition within the specified time. Protect new sealant from rain during curing period.

3.6 INSPECTION/FIELD QUALITY CONTROL

3.6.1 Joint Cleaning

Inspect joints during the cleaning process to correct improper equipment and cleaning techniques that damage the concrete pavement in any manner. Approve cleaned joints prior to installation of the separating or back-up material and joint sealant.

3.6.2 Sealant Application Equipment

Inspect the application equipment to ensure conformance to temperature requirements, proper proportioning and mixing (if two-component sealant) and proper installation. Suspend operations if there is evidence of bubbling, improper installation, or failure to cure or set until causes of the deficiencies are determined and corrected.

3.6.3 Joint Sealant

Inspect the joint sealant for proper rate of cure and set, bonding to the joint walls, cohesive separation within the sealant, reversion to liquid, entrapped air and voids. Remove sealants exhibiting these deficiencies prior to the final acceptance of the project from the joint, wasted, and replace at no additional cost to the Government. Obtain approval for each joint seal installation.

3.7 ACCEPTANCE

Reject sealer that fails to cure properly, or fails to bond to joint walls, or reverts to the uncured state, or fails in cohesion, or shows excessive air voids, blisters, surface defects, swelling, or other deficiencies, or

is not properly recessed within indicated tolerances. Remove rejected sealer and reclean and reseal joints. Perform removal and reseal work promptly by and at the expense of the Contractor.

3.8 CLEAN-UP

Upon completion of the project, remove unused materials from the site and leave the pavement in a clean condition.

-- End of Section --

SECTION 32 11 23

AGGREGATE BASE COURSES

08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

- | | |
|--------------|--|
| AASHTO T 88 | (2013) Standard Method of Test for Particle Size Analysis of Soils |
| AASHTO T 180 | (2017) Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop |
| AASHTO T 224 | (2010) Standard Method of Test for Correction for Coarse Particles in the Soil Compaction Test |

ASTM INTERNATIONAL (ASTM)

- | | |
|-----------------|--|
| ASTM C29/C29M | (2017a) Standard Test Method for Bulk Density ("Unit Weight") and Voids in Aggregate |
| ASTM C88 | (2018) Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate |
| ASTM C117 | (2017) Standard Test Method for Materials Finer than 75-um (No. 200) Sieve in Mineral Aggregates by Washing |
| ASTM C127 | (2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Coarse Aggregate |
| ASTM C128 | (2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate |
| ASTM C131/C131M | (2020) Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine |

APPENDIX B

ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM D75/D75M	(2019) Standard Practice for Sampling Aggregates
ASTM D1556/D1556M	(2015; E 2016) Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method
ASTM D1557	(2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft ³) (2700 kN-m/m ³)
ASTM D2167	(2015) Density and Unit Weight of Soil in Place by the Rubber Balloon Method
ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D4318	(2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D5821	(2013; R 2017) Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)
ASTM E11	(2020) Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

1.2 DEFINITIONS

For the purposes of this specification, the following definitions apply.

1.2.1 Aggregate Base Course

Aggregate base course (ABC) is well graded, durable aggregate uniformly moistened and mechanically stabilized by compaction.

1.2.2 Degree of Compaction

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum laboratory dry density obtained by the test procedure presented in [ASTM D1557](#) abbreviated as a percent of laboratory maximum dry density. Since [ASTM D1557](#) applies only to soils that have 30 percent or less by weight of their particles retained on the [3/4 inch](#) sieve, the degree of compaction for material having more than 30 percent by weight of their particles retained on the [3/4 inch](#) sieve will be

APPENDIX B

expressed as a percentage of the laboratory maximum dry density in accordance with AASHTO T 180 Method D and corrected with AASHTO T 224.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Aggregates; G

Waybills and Delivery Tickets; G

SD-06 Test Reports

Initial Tests; G

1.4 EQUIPMENT, TOOLS, AND MACHINES

All plant, equipment, and tools used in the performance of the work will be subject to approval by the Contracting Officer before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Use equipment capable of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

1.5 QUALITY ASSURANCE

Sampling and testing are the responsibility of the Contractor. Perform sampling and testing using a laboratory approved in accordance with Section 01 45 00.00 1001 45 00.00 2001 45 00.00 40 QUALITY CONTROL. Work requiring testing will not be permitted until the testing laboratory has been inspected and approved. Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. The Contracting Officer may specify the time and location of the tests. Furnish copies of test results to the Contracting Officer within 24 hours of completion of the tests.

1.5.1 Sampling

Take samples for laboratory testing in conformance with ASTM D75/D75M. When deemed necessary, the sampling will be observed by the Contracting Officer.

1.5.2 Tests

1.5.2.1 Sieve Analysis

Perform sieve analysis in conformance with ASTM C117 and ASTM C136/C136M using sieves conforming to ASTM E11. Perform particle-size analysis of the soils in conformance with AASHTO T 88.

APPENDIX B

1.5.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with ASTM D4318.

1.5.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture content in accordance with paragraph DEGREE OF COMPACTION.

1.5.2.4 Field Density Tests

Measure field density in accordance with ASTM D1556/D1556M, ASTM D2167 or ASTM D6938. For the method presented in ASTM D1556/D1556M use the base plate as shown in the drawing. For the method presented in ASTM D6938 check the calibration curves and adjust them, if necessary, using only the sand cone method as described in paragraph Calibration, of the ASTM publication. Tests performed in accordance with ASTM D6938 result in a wet unit weight of soil and ASTM D6938 will be used to determine the moisture content of the soil. Also check the calibration curves furnished with the moisture gauges along with density calibration checks as described in ASTM D6938. Make the calibration checks of both the density and moisture gauges using the prepared containers of material method, as described in paragraph Calibration of ASTM D6938, on each different type of material being tested at the beginning of a job and at intervals as directed. Submit calibration curves and related test results prior to using the device or equipment being calibrated.

1.5.2.5 Wear Test

Perform wear tests on ABC course material in conformance with ASTM C131/C131M.

1.6 ENVIRONMENTAL REQUIREMENTS

Perform construction when the atmospheric temperature is above 35 degrees F. When the temperature falls below 35 degrees F, protect all completed areas by approved methods against detrimental effects of freezing. Correct completed areas damaged by freezing, rainfall, or other weather conditions to meet specified requirements.

PART 2 PRODUCTS

2.1 AGGREGATES

Provide ABC consisting of clean, sound, durable particles of crushed stone, crushed gravel, angular sand, or other approved material. Provide ABC that is free of lumps of clay, organic matter, and other objectionable materials or coatings. The portion retained on the No. 4 sieve is known as coarse aggregate; that portion passing the No. 4 sieve is known as fine aggregate. When the coarse and fine aggregate is supplied from more than one source, provide aggregate from each source that meets the specified requirements.

2.1.1 Coarse Aggregate

Provide coarse aggregates with angular particles of uniform density. Separately stockpile coarse aggregate supplied from more than one source.

APPENDIX B

- a. Crushed Gravel: Provide crushed gravel that has been manufactured by crushing gravels and that meets all the requirements specified below.
- b. Crushed Stone: Provide crushed stone consisting of freshly mined quarry rock, meeting all the requirements specified below.

2.1.1.1 Aggregate Base Course

The percentage of loss of ABC coarse aggregate must not exceed 50 percent when tested in accordance with [ASTM C131/C131M](#). Provide aggregate that contains no more than 30 percent flat and elongated particles. A flat particle is one having a ratio of width to thickness greater than 3; an elongated particle is one having a ratio of length to width greater than 3. In the portion retained on each sieve specified, the crushed aggregates must contain at least 50 percent by weight of crushed pieces having two or more freshly fractured faces determined in accordance with [ASTM D5821](#). When two fractures are contiguous, the angle between planes of the fractures must be at least 30 degrees in order to count as two fractured faces. Manufacture crushed gravel from gravel particles 50 percent of which, by weight, are retained on the maximum size sieve listed in TABLE 1.

2.1.2 Fine Aggregate

Provide fine aggregates consisting of angular particles of uniform density.

2.1.2.1 Aggregate Base Course

Provide ABC fine aggregate that consists of screenings, angular sand, crushed recycled concrete fines, or other finely divided mineral matter processed or naturally combined with the coarse aggregate.

2.1.3 Gradation Requirements

Apply the specified gradation requirements to the completed base course. Provide aggregates that are continuously well graded within the limits specified in TABLE 1. Use sieves that conform to [ASTM E11](#).

TABLE 1. GRADATION OF AGGREGATES

Percentage by Weight Passing Square-Mesh Sieve

Sieve Designation	No. 1	No. 2	No. 3
2 inch	100	----	----
1-1/2 inch	70-100	100	----
1 inch	45-80	60-100	100
1/2 inch	30-60	30-65	40-70
No. 4	20-50	20-50	20-50
No. 10	15-40	15-40	15-40
No. 40	5-25	5-25	5-25
No. 200	0-8	0-8	0-8

NOTE 1: Particles having diameters less than 0.02 mm must not be in excess

of 3 percent by weight of the total sample tested as determined in accordance with AASHTO T 88.

NOTE 2: The values are based on aggregates of uniform specific gravity. If materials from different sources are used for the coarse and fine aggregates, test the materials in accordance with ASTM C127 and ASTM C128 to determine their specific gravities. Correct the percentages passing the various sieves as directed by the Contracting Officer if the specific gravities vary by more than 10 percent.

2.2 LIQUID LIMIT AND PLASTICITY INDEX

Apply liquid limit and plasticity index requirements to the completed course and to any component that is blended to meet the required gradation. The portion of any component or of the completed course passing the No. 40 sieve must be either nonplastic or have a liquid limit not greater than 25 and a plasticity index not greater than 5.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

2.3.1 Initial Tests

Perform one of each of the following tests, on the proposed material prior to commencing construction, to demonstrate that the proposed material meets all specified requirements when furnished. Complete this testing for each source if materials from more than one source are proposed.

- a. Sieve Analysis .
- b. Liquid limit and plasticity index.
- c. Moisture-density relationship.
- d. Wear.

Submit certified copies of test results for approval not less than 30 days before material is required for the work.

2.3.2 Approval of Material

Tentative approval of material will be based on initial test results.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

When the ABC is constructed in more than one layer, clean the previously constructed layer of loose and foreign matter by sweeping with power sweepers or power brooms, except that hand brooms may be used in areas where power cleaning is not practicable. Provide adequate drainage during the entire period of construction to prevent water from collecting or standing on the working area.

3.2 OPERATION OF AGGREGATE SOURCES

Condition aggregate sources on private lands in accordance with local laws or authorities.

3.3 STOCKPILING MATERIAL

Clear and level storage sites prior to stockpiling of material. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates on the cleared and leveled areas designated by the Contracting Officer to prevent segregation. Stockpile materials obtained from different sources separately.

3.4 PREPARATION OF UNDERLYING COURSE OR SUBGRADE

Clean the underlying course or subgrade of all foreign substances prior to constructing the base course(s). Do not construct base course(s) on underlying course or subgrade that is frozen. Construct the surface of the underlying course or subgrade to meet specified compaction and surface tolerances. Correct ruts or soft yielding spots in the underlying courses, areas having inadequate compaction, and deviations of the surface from the specified requirements set forth herein by loosening and removing soft or unsatisfactory material and adding approved material, reshaping to line and grade, and recompacting to specified density requirements. For cohesionless underlying courses or subgrades containing sands or gravels, as defined in [ASTM D2487](#), stabilize the surface prior to placement of the base course(s). Stabilize by mixing ABC into the underlying course and compacting by approved methods. Consider the stabilized material as part of the underlying course and meet all requirements of the underlying course. Do not allow traffic or other operations to disturb the finished underlying course and maintain in a satisfactory condition until the base course is placed.

3.5 GRADE CONTROL

Provide a finished and completed base course conforming to the lines, grades, and cross sections shown. Place line and grade stakes as necessary for control.

3.6 MIXING AND PLACING MATERIALS

Mix the coarse and fine aggregates in a stationary plant. Make adjustments in mixing procedures or in equipment, as directed, to obtain true grades, to minimize segregation or degradation, to obtain the required water content, and to insure a satisfactory base course meeting all requirements of this specification. Place the mixed material on the prepared subgrade or subbase in layers of uniform thickness with an approved spreader. Place the layers so that when compacted they will be true to the grades or levels required with the least possible surface disturbance. Where the base course is placed in more than one layer, clean the previously constructed layers of loose and foreign matter by sweeping with power sweepers, power brooms, or hand brooms, as directed. Make adjustments in placing procedures or equipment as may be directed by the Contracting Officer to obtain true grades, to minimize segregation and degradation, to adjust the water content, and to insure an acceptable base course.

3.7 LAYER THICKNESS

Compact the completed base course to the thickness indicated. No

individual layer may be thicker than 6 inches nor be thinner than 3 inches in compacted thickness. Compact the base course(s) to a total thickness that is within 1/2 inch of the thickness indicated. Where the measured thickness is more than 1/2 inch deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompacting as directed. Where the measured thickness is more than 1/2 inch thicker than indicated, the course will be considered as conforming to the specified thickness requirements. The average job thickness will be the average of all thickness measurements taken for the job and must be within 1/4 inch of the thickness indicated.

3.8 COMPACTION

Compact each layer of the base course, as specified, with approved compaction equipment. Maintain water content during the compaction procedure to within plus or minus 2 percent of the optimum water content determined from laboratory tests as specified in this Section. Begin rolling at the outside edge of the surface and proceed to the center, overlapping on successive trips at least one-half the width of the roller. Slightly vary the length of alternate trips of the roller. Adjust speed of the roller as needed so that displacement of the aggregate does not occur. Compact mixture with hand-operated power tampers in all places not accessible to the rollers. Continue compaction until each layer is compacted through the full depth to at least 95 percent of laboratory maximum density. Make such adjustments in compacting or finishing procedures as may be directed by the Contracting Officer to obtain true grades, to minimize segregation and degradation, to reduce or increase water content, and to ensure a satisfactory base course. Remove any materials found to be unsatisfactory and replace with satisfactory material or rework, as directed, to meet the requirements of this specification.

3.9 PROOF ROLLING

In addition to the compaction specified, proof roll areas designated on the drawings by application of 2 coverages of a heavy pneumatic-tired roller having four or more tires abreast, each tire loaded to a minimum of 30,000 pounds and inflated to a minimum of 125 psi. A coverage is defined as the application of one tire print over the designated area. In the areas designated, apply proof rolling to the top of the underlying material on which the base course is laid and to the top of the completed base course. Maintain water content of the underlying material and each layer of the base course as specified in Paragraph COMPACTION from start of compaction to completion of proof rolling of that layer. Remove any base course materials or any underlying materials that produce unsatisfactory results by proof rolling and replace with satisfactory materials. Then recompact and proof roll to meet these specifications.

3.10 EDGES OF BASE COURSE

Place the base course(s) so that the completed section will be a minimum of 1 foot wider, on all sides, than the next layer that will be placed above it. Place approved material along the outer edges of the base course in sufficient quantity to compact to the thickness of the course being constructed. When the course is being constructed in two or more layers, simultaneously roll and compact at least a 2 foot width of this shoulder material with the rolling and compacting of each layer of the base course, as directed.

3.11 FINISHING

Finish the surface of the top layer of base course after final compaction and proof rolling by cutting any overbuild to grade and rolling with a steel-wheeled roller. Do not add thin layers of material to the top layer of base course to meet grade. If the elevation of the top layer of base course is $1/2$ inch or more below grade, scarify the top layer to a depth of at least 3 inches and blend new material in and compact to bring to grade. Make adjustments to rolling and finishing procedures as directed by the Contracting Officer to minimize segregation and degradation, obtain grades, maintain moisture content, and insure an acceptable base course. Should the surface become rough, corrugated, uneven in texture, or traffic marked prior to completion, scarify the unsatisfactory portion and rework and recompact it or replace as directed.

3.12 SMOOTHNESS TEST

Construct the top layer so that the surface shows no deviations in excess of $3/8$ inch when tested with a 12 foot straightedge. Take measurements in successive positions parallel to the centerline of the area to be paved. Also take measurements perpendicular to the centerline at 50 foot intervals. Correct deviations exceeding this amount by removing material and replacing with new material, or by reworking existing material and compacting it to meet these specifications.

3.13 FIELD QUALITY CONTROL

3.13.1 In-Place Tests

Perform each of the following tests on samples taken from the placed and compacted ABC . Take samples and test at the rates indicated.

- a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square yards, or portion thereof, of completed area.
- b. Perform sieve analysis on every lift of material placed and at a frequency of one sieve analysis for every 500 square yards, or portion thereof, of material placed.
- c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.
- d. Measure the thickness of the base course at intervals providing at least one measurement for each 500 square yards of base course or part thereof. Measure the thickness using test holes, at least 3 inch in diameter through the base course.

3.13.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and fully compacted course(s).

3.14 TRAFFIC

Do not allow traffic on the completed base course.

3.15 MAINTENANCE

Maintain the base course in a satisfactory condition until the full pavement section is completed and accepted. Immediately repair any defects and repeat repairs as often as necessary to keep the area intact. Retest any base course that was not paved over prior to the onset of winter to verify that it still complies with the requirements of this specification. Rework or replace any area of base course that is damaged as necessary to comply with this specification.

3.16 DISPOSAL OF UNSATISFACTORY MATERIALS

Dispose of any unsuitable materials that have been removed as directed . No additional payments will be made for materials that have to be replaced.

-- End of Section --

SECTION 32 13 13.06

PORTLAND CEMENT CONCRETE PAVEMENT FOR ROADS AND SITE FACILITIES
05/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 211.1	(1991; R 2009) Standard Practice for Selecting Proportions for Normal, Heavyweight and Mass Concrete
ACI 305R	(2010) Guide to Hot Weather Concreting
ACI 306R	(2016) Guide to Cold Weather Concreting

ASTM INTERNATIONAL (ASTM)

ASTM A184/A184M	(2019) Standard Specification for Welded Deformed Steel Bar Mats for Concrete Reinforcement
ASTM A615/A615M	(2020) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A775/A775M	(2017) Standard Specification for Epoxy-Coated Steel Reinforcing Bars
ASTM A966/A966M	(2015; R 2020) Standard Test Method for Magnetic Particle Examination of Steel Forgings Using Alternating Current
ASTM C31/C31M	(2019a) Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C33/C33M	(2018) Standard Specification for Concrete Aggregates
ASTM C42/C42M	(2020) Standard Test Method for Obtaining and Testing Drilled Cores and Sawed Beams of Concrete
ASTM C78/C78M	(2018) Standard Test Method for Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading)
ASTM C88	(2018) Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or

APPENDIX B

	Magnesium Sulfate
ASTM C94/C94M	(2020) Standard Specification for Ready-Mixed Concrete
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C150/C150M	(2020) Standard Specification for Portland Cement
ASTM C171	(2020) Standard Specification for Sheet Materials for Curing Concrete
ASTM C172/C172M	(2017) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C231/C231M	(2017a) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C260/C260M	(2010a; R 2016) Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C309	(2011) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM C494/C494M	(2019) Standard Specification for Chemical Admixtures for Concrete
ASTM C595/C595M	(2020) Standard Specification for Blended Hydraulic Cements
ASTM C618	(2019) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C881/C881M	(2020a) Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete
ASTM C989/C989M	(2018a) Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM C1017/C1017M	(2013; E 2015) Standard Specification for Chemical Admixtures for Use in Producing Flowing Concrete
ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM C1240	(2020) Standard Specification for Silica Fume Used in Cementitious Mixtures

APPENDIX B

ASTM C1260	(2014) Standard Test Method for Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)
ASTM C1542/C1542M	(2019) Standard Test Method for Measuring Length of Concrete Cores
ASTM C1549	(2016) Standard Test Method for Determination of Solar Reflectance Near Ambient Temperature Using a Portable Solar Reflectometer
ASTM C1567	(2013) Standard Test Method for Potential Alkali-Silica Reactivity of Combinations of Cementitious Materials and Aggregate (Accelerated Mortar-Bar Method)
ASTM C1602/C1602M	(2018) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM D1751	(2004; E 2013; R 2013) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D1752	(2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction
ASTM D2995	(1999; R 2009) Determining Application Rate of Bituminous Distributors
ASTM D6155	(2019) Nontraditional Coarse Aggregate for Bituminous Paving Mixtures
ASTM E1274	(2018) Standard Test Method for Measuring Pavement Roughness Using a Profilograph

NATIONAL READY MIXED CONCRETE ASSOCIATION (NRMCA)

NRMCA QC 3	(2015) Quality Control Manual: Section 3, Plant Certifications Checklist: Certification of Ready Mixed Concrete Production Facilities
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are [for Contractor Quality Control approval. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.

SD-03 Product Data

Curing Materials

Reinforcement; G

Epoxy Resin

Cementitious Materials; G

Dowel Bars

Expansion Joint Filler

SD-05 Design Data

Mix Design Report; G

SD-06 Test Reports

Concrete Slump Tests

Concrete Uniformity

Flexural Strength; G

Air Content

SD-07 Certificates

Batch Tickets

NRMCA Certificate Of Conformance

SD-08 Manufacturer's Instructions

Diamond Grinding Plan

1.3 QUALITY CONTROL

1.3.1 Qualifications

1.3.1.1 Laboratory Accreditation

Perform sampling and testing using an approved commercial testing laboratory or on-site facilities that are accredited in accordance with [ASTM C1077](#). Do not start work requiring testing until the facilities have been inspected and approved. The Government will inspect all laboratories requiring validation for equipment and test procedures prior to the start of any concreting operations for conformance to [ASTM C1077](#). Schedule and provide payment for laboratory inspections. Additional payment or a time extension due to failure to acquire the required laboratory validation is not allowed. Maintain this certification for the duration of the project.

1.3.1.2 Field Technicians

APPENDIX B

Provide field technicians meeting one of the following criteria:

- a. Have at least one National Ready Mixed Concrete Association (NRMCA) certified concrete craftsman and at least one American Concrete Institute (ACI) Flatwork Finisher Certified craftsman on site, overseeing each placement crew during all concrete placement.
- b. Have no less than three NRMCA certified concrete installers and at least two American Concrete Institute (ACI) Flatwork Finisher Certified installers on site working as members of each placement crew during all concrete placement.

1.3.2 Batch Tickets

Submit [batch tickets](#) for each load of ready-mixed concrete in accordance with [ASTM C94/C94M](#).

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver concrete paving in accordance with [ASTM C94/C94M](#).

1.5 ACCEPTANCE

1.5.1 Tolerances

Acceptance of Portland cement concrete pavement is based on compliance with the tolerances presented in Table 1. Remove and replace concrete pavement represented by the failing tests or submit repair plan for approval.

Table 1	
Measurement	Tolerance
PLASTIC CONCRETE	
Slump	plus 0, minus 1.5 inches
Air Content	plus/minus 1.5 percent
Flexural Strength	No individual specimen less than 100 psi below specified strength.
HARDENED CONCRETE	
Grade	plus/minus 0.05 feet from plan
Smoothness	No abrupt change exceeding 1/8 inch
Straightedge	Not more than 1/8 in for roads. Not more than 1/4 in for open storage areas.
Profilograph	Not more than 9 inches/mile
Thickness	minus 1/2 inch for pavement less than 8 inches thick.

Table 1	
Edge Slump	85 percent less than 1/4 inch and 100 percent less than 3/8 inch.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Cementitious Materials

2.1.1.1 Portland Cement

Conforming to ASTM C150/C150M, Type I or II .

2.1.1.2 Fly Ash and Pozzolan

Conforming to ASTM C618, Type F, or N, with a loss on ignition not exceeding 3 percent. Include test results in accordance with ASTM C618.

2.1.2 Water

Water conforming to ASTM C1602/C1602M.

2.1.3 Aggregate

2.1.3.1 Durability

Evaluate and test all fine and coarse aggregates to be used in all concrete for durability in accordance with ASTM C88. Provide fine and coarse aggregates with a maximum of 18 percent loss when subjected to 5 cycles using Magnesium Sulfate or a maximum of 12 percent loss when subjected to 5 cycles of Sodium Sulfate.

2.1.3.2 Alkali Reactivity Test

Evaluate and test fine and coarse aggregates to be used in all concrete for alkali-aggregate reactivity. Test all size groups and sources proposed for use.

- a. Evaluate the fine and coarse aggregates separately, using ASTM C1260. Reject individual aggregates with test results that indicate an expansion of greater than 0.08 percent after 28 days of immersion in 1N NaOH solution, or perform additional testing as follows: utilize the proposed low alkali portland cement, blended cement, or SCM in combination with each individual aggregate. Test in accordance with ASTM C1567. Determine the quantity that meets all the requirements of these specifications and that lowers the expansion equal to or less than 0.08 percent after 28 days of immersion in a 1N NaOH solution. Base the mixture proportioning on the highest percentage of SCM required to mitigate ASR-reactivity.
- b. If any of the above options does not lower the expansion to less than

APPENDIX B

0.08 percent after 28 days of immersion in a 1N NaOH solution, reject the aggregate(s) and submit new aggregate sources for retesting. Submit the results of testing for evaluation and acceptance.

2.1.3.3 Fine Aggregates

Conforming to the quality and gradation of [ASTM C33/C33M](#).

2.1.3.4 Coarse Aggregates

Coarse aggregate consisting of crushed or uncrushed gravel, crushed stone, or a combination thereof. Provide aggregates, as delivered to the mixers, consisting of clean, hard, uncoated particles. Wash coarse aggregate sufficient to remove dust and other coatings. Provide fine aggregate consisting of natural sand, manufactured sand, or a combination of the two, and composed of clean, hard, durable particles. Provide both coarse and fine aggregates meeting the requirements of [ASTM C33/C33M](#).

- a. Gradation: Provide coarse aggregate with a nominal maximum size of 1.5 inches. Grade and provide the individual aggregates in two or more size groups meeting the individual grading requirements of [ASTM C33/C33M](#), Size No. 4 (1.5 to 0.75 inch) and Size No. 67 (0.75 inch to No. 4).
- b. Quality: Conforming to [ASTM C33/C33M](#), Class 4S.

2.1.4 Chemical Admixtures

2.1.4.1 Water Reducing Admixtures

Provide admixture conforming to [ASTM C494/C494M](#): Type A, water reducing; Type B, retarding; Type C, accelerating; Type D, water-reducing and retarding; and Type E, water-reducing and accelerating admixture. Do not use calcium chloride admixtures. [ASTM C494/C494M](#) Type S specific performance admixtures and [ASTM C1017/C1017M](#) flowable admixtures are not allowed.

2.1.4.2 Air Entraining Admixture

Conforming to [ASTM C260/C260M](#): Air-entraining.

2.1.4.3 High Range Water Reducing Admixture

Provide a high-range water-reducing admixture that meets the requirements of [ASTM C494/C494M](#), Type F or G, that is free from chlorides, alkalis, and is of the synthesized, sulfonated complex polymer type. Add the HRWRA to the concrete as a single component at the batch plant. Add the admixture to the concrete mixture only when its use is approved or directed, and only when it has been used in mixture proportioning studies to arrive at approved mixture proportions. Submit certified copies of the independent laboratory test results required for compliance with [ASTM C494/C494M](#).

2.1.5 Reinforcement

2.1.5.1 Dowel Bars

Dowel bars conforming to [ASTM A615/A615M](#), Grade 60 for plain billet-steel bars of the size and length indicated. Remove all burrs and projections

APPENDIX B

from the bars. Epoxy coat in accordance with ASTM A775/A775M.

2.1.5.2 Tie Bars

Billet or axle steel deformed bars conforming to ASTM A615/A615M or

2.1.5.3 Reinforcement 60. Epoxy coat in accordance with ASTM A775/A775M.

Deformed steel bar mats conforming to ASTM A184/A184M. Bar reinforcement conforming to ASTM A615/A615M, Grade 60. Epoxy coat in accordance with ASTM A775/A775M.

2.1.6 Curing Materials

Provide curing materials consisting of:

2.1.6.1 White-Burlap-Polyethylene Sheet

Conforming to ASTM C171, 0.004 inch thick white opaque polyethylene bonded to 10 oz/linear yard (40 inch) wide burlap.

2.1.6.2 Liquid Membrane-Forming Compound

Conforming to ASTM C309, white pigmented, Type 2, Class B, free of paraffin or petroleum.

2.1.6.3 Liquid Chemical Sealer-Hardener Compound

Compound consisting of magnesium fluosilicate which when mixed with water seals and hardens the surface of the concrete. Do not use on exterior slabs exposed to freezing conditions.

2.1.7 Joint Fillers and Sealants

Provide as specified in Section 32 01 19.61 SEALING OF JOINTS IN RIGID PAVEMENT Match new joints with existing alignment.

2.1.8 Biodegradable Form Release Agent

Provide form release agent that is colorless and biodegradable. Provide product that does not bond with, stain, or adversely affect concrete surfaces and does not impair subsequent treatments of concrete surfaces. Provide form release agent with a minimum of 87 percent biobased material and does not contain diesel fuel, petroleum-based lubricating oils, waxes, or kerosene.

2.1.9 Epoxy Resin

Provide epoxy-resin materials that consist of two-component materials conforming to the requirements of ASTM C881/C881M, Class as appropriate for each application temperature to be encountered, except that in addition, the materials meet the following requirements:

- a. Type IV, Grade 3, for use for embedding dowels and anchor bolts.
- b. Type III, Grade as approved, for use as patching materials for complete filling of spalls and other voids and for use in preparing epoxy resin mortar.

APPENDIX B

- c. Type IV, Grade 1, for use for injecting cracks.
- d. Type V, Grade as approved, for bonding freshly mixed portland cement concrete or mortar or freshly mixed epoxy resin concrete or mortar to hardened concrete.

2.1.10 Joint Materials

2.1.10.1 Expansion Joint Materials

Provide preformed [expansion joint filler](#) material conforming to [ASTM D1751](#). Provide expansion joint filler that is [3/4 inch](#) thick, unless otherwise indicated, and provided in a single full depth piece.

2.1.10.2 Slip Joint Material

Provide slip joint material that is [1/4 inch](#) thick expansion joint filler, unless otherwise indicated, conforming to paragraph EXPANSION JOINT MATERIAL.

2.2 MIX DESIGN

Proportion concrete mix in accordance with [ACI 211.1](#) except as modified herein.

2.2.1 Specified Concrete Properties

2.2.1.1 Flexural Strength

Provide concrete with a minimum flexural strength of [650 psi](#) at 28 days of age.

2.2.1.2 Air Entrainment

Provide an entrained air content of 5.5 percent.

2.2.1.3 Slump

For fixed form and hand placement, provide a maximum slump of [3 inches](#).

2.2.1.4 Water/Cementitious Materials Ratio

Maximum allowable water-cementitious material ratio is 0.45. The water-cementitious material ratio is based on absolute volume equivalency, where the ratio is determined using the weight of cement for a cement only mix, or using the total volume of cement plus pozzolan converted to an equivalent weight of cement by the absolute volume equivalency method described in [ACI 211.1](#).

2.2.2 Mix Design Report

Perform trial design batches, mixture proportioning studies, testing, and include test results demonstrating that the proposed mixture proportions produce concrete of the qualities indicated. An existing mix design may be submitted if developed within the previous 12 months. Submit test results in a [mix design report](#) to include:

- a. Coarse and fine aggregate gradations and plots.
- b. Coarse and fine aggregate quality test results, include deleterious materials and ASR testing.
- c. Mill certificates for cement and supplemental cementitious materials.
- d. Certified test results for all proposed admixtures.
- e. Specified flexural strength, slump, and air content.
- f. Recommended proportions and volumes for proposed mixture and each of three trial water-cementitious materials ratios.
- g. Individual beam breaks.
- h. Flexural strength summaries and plots.
- i. Historical record of test results, documenting production standard deviation (if available).
- j. Narrative discussing methodology on how the mix design was developed.

2.2.3 Mix Verification

Mix verification tests may be performed by the Government. Provide quantities of cementitious materials, aggregates and admixtures as requested.

2.3 EQUIPMENT

2.3.1 Transporting Equipment

Provide transporting equipment in conformance with ASTM C94/C94M and as specified herein. Transport concrete to the paving site in rear-dump trucks, in truck mixers designed with extra large blading and rear opening specifically for low slump concrete, or in agitators. Do not permit bottom-dump trucks for delivery of concrete.

2.3.2 Delivery Equipment

When concrete transport equipment cannot operate on the paving lane, provide side-delivery transport equipment consisting of self-propelled moving conveyors to deliver concrete from the transport equipment and discharge it in front of the paver. Do not permit front-end loaders, dozers, or similar equipment to distribute the concrete.

2.3.3 Paver-Finisher

Provide a heavy-duty, self-propelled paver-finisher machine designed specifically for paving and finishing high quality pavement and capable of spreading, consolidating, and shaping the plastic concrete to the desired cross section in one pass. Equip the paver-finisher with a full width "knock-down" auger, capable of operating in both directions, which will evenly spread the fresh concrete in front of the screed or extrusion plate. Gang-mount immersion vibrators at the front of the paver on a frame equipped with suitable controls so that all vibrators can be operated at

any desired depth within the slab or completely withdrawn from the concrete. Automatically control the vibrators so they will be immediately stopped as forward motion of the paver ceases. Space the immersion vibrators across the paving lane as necessary to properly consolidate the concrete, but limit the clear distance between vibrators not to exceed 30 inches, and the outside vibrators not to exceed 12 inches from the edge of the lane. Vibrators may be pneumatic, gas driven, or electric, and operated at frequencies within the concrete between 6,000 and 7,000 vibrations per minute, with an amplitude of vibration such that noticeable vibrations occur at 1.5 foot radius when the vibrator is inserted in the concrete to the depth specified. Equip the paver-finisher with a transversely oscillating screed or an extrusion plate to shape, compact, and smooth the surface.

2.3.3.1 Paver-Finisher with Fixed Forms

Equip the paver-finisher with wheels designed to ride the forms, keep it aligned with the forms, and to prevent deformation of the forms.

2.3.3.2 Slipform Paver-Finisher

Provide a track-mounted slipform paver-finisher with automatic controls and padded tracks. Electronically reference horizontal alignment to a taut wire guideline. Electronically reference vertical alignment on both sides of the paver to a taut wire guideline, to an approved laser control system, or to a ski operating on a completed lane. Do not control from a slope-adjustment control or from the underlying material.

2.3.3.3 Other Types of Finishing Equipment

Heavy duty vibratory truss screeds may be approved for use if successfully demonstrated on the test section to consolidate the slab full depth and without segregation. Clary screeds, rotating tube floats, or laser screeds will not be allowed on the project. Provide hand floats that are not less than 12 feet long and 6 inches wide and stiffened to prevent flexing and warping.

2.3.4 Texturing Equipment

Provide texturing equipment as specified below.

2.3.4.1 Deep Texturing Equipment

Provide texturing equipment consisting of a stiff bristled broom which will produce true, even grooves. Mount this drag in a wheeled frame spanning the paving lane and constructed to mechanically pull the drag in a straight line across the paving lane perpendicular to the centerline.

2.3.5 Curing Equipment

Provide equipment for applying membrane-forming curing compound mounted on a self-propelled frame that spans the paving lane. Constantly agitate the curing compound reservoir mechanically (not air) during operation and provide a means for completely draining the reservoir. Provide a spraying system that consists of a mechanically powered pump which maintains constant pressure during operation, an operable pressure gauge, and either a series of spray nozzles evenly spaced across the lane to provide

uniformly overlapping coverage or a single spray nozzle which is mounted on a carriage which automatically traverses the lane width at a speed correlated with the forward movement of the overall frame. Protect all spray nozzles with wind screens. Calibrate the spraying system in accordance with [ASTM D2995](#), Method A, for the rate of application required in subpart CURING AND PROTECTION. Provide hand-operated sprayers powered by compressed air supplied by a mechanical air compressor. Immediately replace curing equipment if it fails to apply an even coating of compound at the specified rate.

2.3.6 Sawing Equipment

Provide equipment for sawing joints and for other similar sawing of concrete consisting of standard diamond-type concrete saws mounted on a wheeled chassis which can be easily guided to follow the required alignment. Provide diamond tipped blades. If demonstrated to operate properly, abrasive blades may be used. Provide spares as required to maintain the required sawing rate. Early-entry saws may be used, subject to demonstration and approval. No change to the initial sawcut depth is permitted.

2.3.7 Straightedge

Furnish one [12 foot](#) straightedge constructed of aluminum or magnesium alloy, having blades of box or box-girder cross section with flat bottom, adequately reinforced to insure rigidity and accuracy. Provide handles for operation on the pavement.

PART 3 EXECUTION

3.1 PREPARATION FOR PAVING

3.1.1 Weather Limitations

When windy conditions during paving appear probable, have equipment and material at the paving site to provide windbreaks, shading, fogging, or other action to prevent plastic shrinkage cracking or other damaging drying of the concrete.

3.1.1.1 Inclement Weather

Do not commence placing operations when heavy rain or other damaging weather conditions appear imminent. At all times when placing concrete, maintain on-site sufficient waterproof cover and means to rapidly place it over all unhardened concrete or concrete that might be damaged by rain. Suspend placement of concrete whenever rain, high winds, or other damaging weather commences to damage the surface or texture of the placed unhardened concrete, washes cement out of the concrete, or changes the water content of the surface concrete. Immediately cover and protect all unhardened concrete from the rain or other damaging weather. Completely remove and replace any slab damaged by rain or other weather full depth, by full slab width, to the nearest original joint.

3.1.1.2 Hot Weather

Maintain required concrete temperature in accordance with [ACI 305R](#) to prevent evaporation rate from exceeding [0.2 pound of water per square foot](#)

of exposed concrete per hour. Cool ingredients before mixing, place concrete during cooler night time hours, or use other suitable means to control concrete temperature and prevent rapid drying of newly placed concrete. Water is not allowed to be added after the initial introduction of mixing water except, when on arrival at the job site, the slump is less than specified and the water-cement ratio is less than that given as a maximum in the approved mixture. Additional water may be added to bring the slump within the specified range provided the approved water-cement ratio is not exceeded. Inject water into the head of the mixer (end opposite the discharge opening) drum under pressure, and turn the drum or blades a minimum of 30 additional revolutions at mixing speed. The addition of water to the batch at any later time is not allowed. After placement, use fog spray, apply monomolecular film, or use other suitable means to reduce the evaporation rate. Start curing when surface of fresh concrete is sufficiently hard to permit curing without damage. Cool underlying material by sprinkling lightly with water before placing concrete. Follow practices found in [ACI 305R](#).

3.1.1.3 Prevention of Plastic Shrinkage Cracking

During weather with low humidity, and particularly with high temperature and appreciable wind, develop and institute measures to prevent plastic shrinkage cracks from developing. If plastic shrinkage cracking occurs, halt further placement of concrete until protective measures are in place to prevent further cracking. Periods of high potential for plastic shrinkage cracking can be anticipated by use of [ACI 305R](#). In addition to the protective measures specified in the previous paragraph, the concrete placement may be further protected by erecting shades and windbreaks and by applying fog sprays of water, the addition of monomolecular films, or wet covering. Apply monomolecular films after finishing is complete, do not use in the finishing process. Immediately commence curing procedures when such water treatment is stopped.

3.1.1.4 Cold Weather

Do not place concrete when ambient temperature is below 40 degrees F or when concrete is likely to be subjected to freezing temperatures within 24 hours. When authorized, when concrete is likely to be subjected to freezing within 24 hours after placing, heat concrete materials so that temperature of concrete when deposited is between 65 and 80 degrees F. Methods of heating materials are subject to approval. Do not heat mixing water above 165 degrees F. Remove lumps of frozen material and ice from aggregates before placing aggregates in mixer. Follow practices found in [ACI 306R](#).

3.1.2 Conditioning of Underlying Material

Verify the underlying material, upon which concrete is to be placed is clean, damp, and free from debris, waste concrete or cement, frost, ice, and standing or running water. Prior to setting forms or placement of concrete, verify the underlying material is well drained and has been satisfactorily graded by string-line controlled, automated, trimming machine and uniformly compacted in accordance with the applicable Section of these specifications. Test the surface of the underlying material to crown, elevation, and density in advance of setting forms or of concrete placement using slip-form techniques. Trim high areas to proper elevation. Fill and compact low areas to a condition similar to that of surrounding grade, or fill with concrete monolithically with the pavement.

Low areas filled with concrete are not to be cored for thickness to avoid biasing the average thickness used for evaluation and payment adjustment. Rework and compact any underlying material disturbed by construction operations to specified density immediately in front of the paver. If a slipform paver is used, continue the same underlying material under the paving lane beyond the edge of the lane a sufficient distance that is thoroughly compacted and true to grade to provide a suitable trackline for the slipform paver and firm support for the edge of the paving lane.

3.1.3 Forms

Use steel forms, except that wood forms may be used for curves having a radius of 150 feet or less, and for fillets. Forms may be built up with metal or wood, added only to the base, to provide an increase in depth of not more than 25 percent. Provide forms with the base width not less than eight-tenths of the vertical height of the form, except that for forms 8 inches or less in vertical height, provide forms with a base width not less than the vertical height of the form. Provide wood forms adequate in strength and rigidly braced for curves and fillets. Set forms on firm material cut true to grade so that each form section when placed will be firmly in contact with the underlying layer for its entire base. Do not set forms on blocks or on built-up spots of underlying material.[Set and secure forms in place with stakes or by other approved methods for overlay pavements and for other locations where forms are set on existing pavements. Carefully drill holes in existing pavements for form stakes without cracking or spalling the existing pavement. Prior to setting forms for paving operations, demonstrate the proposed form setting procedures at an approved location and do not proceed further until the proposed method is approved.] Before placing the concrete, coat the contact surfaces of forms[except existing pavement sections where bonding is required,] with a non-staining mineral oil, non-staining form coating compound, biodegradable form release agent, or two coats of nitro-cellulose lacquer.[When using existing pavement as a form, clean existing concrete and then coat with asphalt emulsion bondbreaker before concrete is placed.] Check and correct grade elevations and alignment of the forms immediately before placing concrete.

3.1.4 Reinforcement

3.1.4.1 Dowel Bars

Install dowels with horizontal and vertical alignment plus or minus 1 inch. Except as otherwise specified, maintain location of dowels within a skew alignment of 1/4 inch over 1 foot length. Omit Dowels and tie bars when the center of the dowel is located within a horizontal distance from an intersecting joint equal to or less than one-fourth of the slab thickness. Maintain dowels in position during concrete placement and curing. Before concrete placement, thoroughly grease the entire length of each dowel secured in a dowel basket or fixed form.

3.1.4.2 Tie Bars

Install bars, accurately aligned horizontally and vertically, and to the tolerances shown on the drawings, at indicated locations.

3.1.4.3 Setting Slab Reinforcement

Position reinforcement on suitable chairs prior to concrete placement. At

APPENDIX B

expansion, contraction and construction joints, place the reinforcement as indicated. Clean reinforcement free of mud, oil, scale or other foreign materials. Place reinforcement accurately and wire securely. Lap splices 12 inches minimum. Maintain the bar spacing from ends and sides of slabs and joints as indicated.

3.2 MEASURING, MIXING, CONVEYING, AND PLACING CONCRETE

3.2.1 Measuring

Conform to ASTM C94/C94M.

3.2.2 Mixing

Conform to ASTM C94/C94M, except as modified herein. Begin mixing within 30 minutes after cement has been added to aggregates. When the air temperature is greater than 85 degrees F, place concrete within 60 minutes. With approval, a hydration stabilizer admixture meeting the requirements of ASTM C494/C494M Type D, may be used to extend the placement time to 90 minutes. Additional water may be added to bring slump within required limits as specified in ASTM C94/C94M, provided that the specified water-cement ratio is not exceeded.

3.2.3 Conveying

Conform to ASTM C94/C94M.

3.2.4 Placing

Do not exceed a free vertical drop of 5 feet from the point of discharge. Deposit concrete either directly from the transporting equipment or by conveyor on to the pre-wetted subgrade or subbase, unless otherwise specified. Deposit the concrete between the forms to an approximately uniform height. Place concrete continuously at a uniform rate, without damage to the grade and without unscheduled stops except for equipment failure or other emergencies. If an unscheduled stop occurs within 10 feet of a previously placed expansion joint, remove concrete back to joint, repair any damage to grade, install a construction joint and continue placing concrete only after cause of the stop has been corrected.

3.3 PAVING

Construct pavement with paving and finishing equipment utilizing fixed forms.

3.3.1 Operation

When the paver is operated between or adjacent to previously constructed pavement (fill-in lanes), make provisions to prevent damage to the previously constructed pavement, including keeping the existing pavement surface free of debris, and placing rubber mats beneath the paver tracks. Operate transversely oscillating screeds and extrusion plates to overlap the existing pavement the minimum possible, but in no case more than 8 inches.

3.3.2 Consolidation

Immediately after spreading concrete, consolidate full depth with internal type vibrating equipment along the boundaries of all slabs regardless of slab thickness, and interior of all concrete slabs. For pavements less than 10 inches thick, operate vibrators at mid-depth parallel with or at a slight angle to the base course. For thicker pavements, angle vibrators toward the vertical, with vibrator tip preferably about 2 inches above the base course, and top of vibrator a few inches below pavement surface. Automatically control the vibrators or tamping units in front of the paver so that they stop immediately as forward motion ceases. Limit duration of vibration to that necessary to produce consolidation of concrete. Do not permit excessive vibration. Vibrate concrete in small, odd-shaped slabs or in locations inaccessible to the paver mounted vibration equipment with a hand-operated immersion vibrator operated from a bridge spanning the area. Do not operate vibrators at one location for more than 15 seconds. Do not use vibrators to transport or spread the concrete.

3.3.3 Fixed Form Paving

Spread and strike off concrete with with the paver. Shape the concrete to the specified and indicated cross section in one pass, and finish the surface and edges so that only a very minimum amount of hand finishing is required. Use single spud hand vibrators to consolidate the concrete adjacent to fixed forms as required to achieve a void-free formed edge. Do not allow vibrators to contact reinforcement, forms, or the grade during vibration.

3.4 JOINTS

3.4.1 Contraction Joints

Hold dowels and tie bars in longitudinal and transverse contraction joints within the paving lane securely in place by means of rigid metal basket assemblies. Weld the dowels and tie bars to the assembly or hold firmly by mechanical locking arrangements that will prevent them from becoming distorted during paving operations. Anchor the basket assemblies securely in the proper location.

3.4.2 Construction Joints - Fixed Form Paving

Install dowels and tie bars by the bonded-in-place method, supported by means of devices fastened to the forms. Do not permit installation by removing and replacing in preformed holes.

3.4.3 Dowels Installed In Hardened Concrete

Install by bonding the dowels into holes drilled into the hardened concrete. Drill holes into the hardened concrete approximately 1/8 inch greater in diameter than the dowels. Bond the dowels in the drilled holes using epoxy resin injected at the back of the hole before installing the dowel and extruded to the collar during insertion of the dowel so as to completely fill the void around the dowel. Application by buttering the dowel is not permitted. Hold the dowels in alignment at the collar of the hole, after insertion and before the epoxy resin hardens, by means of a suitable metal or plastic collar fitted around the dowel. Check the vertical alignment of the dowels by placing the straightedge on the surface of the pavement over the top of the dowel and measuring the vertical distance between the straightedge and the beginning and ending point of the exposed part of the dowel.

3.5 FINISHING CONCRETE

Start finishing operations immediately after placement of concrete. Use finishing machine, except hand finishing may be used in emergencies and for concrete slabs in inaccessible locations or of such shapes or sizes that machine finishing is impracticable. Immediately halt any operations which produce more than 1/8 inch of mortar-rich surface (defined as deficient in plus U.S. No. 4 sieve size aggregate) and modify the equipment, mixture, or procedures. Finish pavement surface on both sides of a joint to the same grade. Finish formed joints from a securely supported transverse bridge. Provide hand finishing equipment for use at all times.

3.5.1 Machine Finishing

Strike off and screed concrete to the required slope and cross-section by a power-driven transverse finishing machine. A transverse rotating tube or pipe is not permitted. Maintain elevation of concrete such that, when consolidated and finished, pavement surface will be adequately consolidated and at the required grade. Equip finishing machine with a screed which is readily and accurately adjustable for changes in pavement slope and compensation for wear and other causes. Do not permit excessive operation over an area, which will result in an excess of mortar and water being brought to the surface.

3.5.1.1 Equipment Operation

Maintain the travel of machine on the forms without lifting, wobbling, or other variation of the machine which tend to affect the precision of concrete finish. Keep the tops of the forms clean by a device attached to the machine. Maintain a uniform ridge of concrete ahead of the front screed for its entire length.

3.5.1.2 Joint Finish

Before concrete is hardened, correct edge slump of pavement, exclusive of edge rounding, in excess of 0.25 inches. Finish concrete surface on each side of construction joints to the same plane, and correct deviations before newly placed concrete has hardened.

3.5.1.3 Hand Finishing

Strike-off and screed surface of concrete to elevations slightly above finish grade so that when concrete is consolidated and finished, the pavement surface is at the indicated elevation. Vibrate entire surface until required compaction and reduction of surface voids is secured with a strike-off template. After initial finishing, further smooth and consolidate concrete by means of hand-operated longitudinal floats.

3.5.2 Texturing

Before the surface sheen has disappeared and before the concrete hardens, provide a texture to the surface of the pavement as described herein. After curing is complete, thoroughly broom all textured surfaces to remove all debris. Finish the concrete in areas of recesses for tie-down anchors, lighting fixtures, and other outlets in the pavement to provide a surface of the same texture as the surrounding area.

3.5.2.1 Brooming

Finish the surface of the slab by brooming the surface with a new wire broom at least 18 inches wide. Gently pull the broom over the surface of the pavement from edge to edge just before the concrete becomes non-plastic. Slightly overlap adjacent strokes of the broom. Broom perpendicular to centerline of pavement so that corrugations produced will be uniform in character and width, and not more than 1/16 inch in depth. Maintain broomed surface free from porous spots, irregularities, depressions, and small pockets or rough spots such as may be caused by accidentally disturbing particles of coarse aggregate embedded near the surface.

3.5.3 Edging

At the time the concrete has attained a degree of hardness suitable for edging, carefully finish slab edges, including edges at formed joints, with an edge having a maximum radius of 1/8 inch. When brooming is specified for the final surface finish, edge transverse joints before starting brooming, then operate broom to obliterate as much as possible the mark left by the edging tool without disturbing the rounded corner left by the edger. Clean by removing loose fragments and soupy mortar from corners or edges of slabs which have crumbled and areas which lack sufficient mortar for proper finishing. Refill voids solidly with a mixture of suitable proportions and consistency and refinish. Remove unnecessary tool marks and edges. Smooth remaining edges true to line.

3.6 CURING AND PROTECTION

Protect concrete adequately from injurious action by sun, rain, flowing water, frost, mechanical injury, tire marks and oil stains, and do not allow it to dry out from the time it is placed until the expiration of the minimum curing periods specified herein. Do not use membrane-forming compound on surfaces where its appearance would be objectionable, on surfaces to be painted, where coverings are to be bonded to concrete, or on concrete to which other concrete is to be bonded.

3.6.1 Moist Curing

Maintain concrete to be moist-cured continuously wet for the entire curing period, or until curing compound is applied, commencing immediately after finishing. If forms are removed before the end of the curing period, provide curing on unformed surfaces, using suitable materials. Cure surfaces by ponding, by continuous sprinkling, by continuously saturated burlap or cotton mats, or by continuously saturated plastic coated burlap. Provide burlap and mats that are clean and free from any contamination and completely saturated before being placed on the concrete. Lap sheets to provide full coverage. Provide an approved work system to ensure that moist curing is continuous 24 hours per day and that the entire surface is wet.

3.6.2 White-Burlap-Polyethylene Sheet

Wet entire exposed surface thoroughly with a fine spray of water, saturate burlap but do not have excessive water dripping off the burlap and then cover concrete with White-Burlap-Polyethylene Sheet, burlap side down. Lay sheets directly on concrete surface and overlap 12 inches. Make sheeting not less than 18 inches wider than concrete surface to be cured, and weight

down on the edges and over the transverse laps to form closed joints. Repair or replace sheets when damaged during curing. Check daily to assure burlap has not lost all moisture. If moisture evaporates, resaturate burlap and re-place on pavement (limit re-saturation and re-placing to less than 10 minutes per sheet). Leave sheeting on concrete surface to be cured for at least 7 days.

3.6.3 Liquid Membrane-Forming Compound Curing

Apply compound immediately after surface loses its water sheen and has a dull appearance and before joints are sawed. Agitate curing compound thoroughly by mechanical means during use and apply uniformly in a two-coat continuous operation by suitable power-spraying equipment. Apply a total coverage for the two coats at least [one gallon](#) of undiluted compound per [200 square feet](#) to produce a uniform, continuous, coherent film that will not check, crack, or peel and free from pinholes or other imperfections. The application of curing compound by hand-operated, mechanical powered pressure sprayers is permitted only on odd widths or shapes of slabs and on concrete surfaces exposed by the removal of forms. When the application is made by hand-operated sprayers, apply a second coat in a direction approximately at right angles to the direction of the first coat. Apply an additional coat of compound immediately to areas where film is defective. Respray concrete surfaces that are subject to heavy rainfall within 3 hours after curing compound has been applied in the same manner.

3.6.4 Protection of Treated Surfaces

After the initial saw cut is complete and the slurry has been removed, respray the area with curing compound or restore the white burlap polyethylene sheet to maintain a continuous curing environment in the area of the sawn joints. Keep concrete surfaces to which liquid membrane-forming compounds have been applied free from vehicular traffic and other sources of abrasion for not less than 72 hours. Foot traffic is allowed after 24 hours for inspection purposes. Maintain continuity of coating for entire curing period and repair damage to coating immediately.

3.7 FIELD QUALITY CONTROL

3.7.1 Sampling

Collect samples of fresh concrete in accordance with [ASTM C172/C172M](#) during each working day as required to perform tests specified herein. Make test specimens in accordance with [ASTM C31/C31M](#).

3.7.2 Consistency Tests

Perform [concrete slump tests](#) in accordance with [ASTM C143/C143M](#). Take samples for slump determination from concrete during placement. Perform tests at the beginning of a concrete placement operation and for each batch (minimum) or every [20 cubic yards](#) (maximum) of concrete to ensure that specification requirements are met. In addition, perform tests each time test beams are made.

3.7.3 Flexural Strength Tests

Test for [flexural strength](#) in accordance with [ASTM C78/C78M](#). Fabricate and cure four test specimens in accordance with [ASTM C31/C31M](#) for each set of

tests. Test two specimens at 7 days, and the other two at 28 days. Concrete strength will be considered satisfactory when the minimum of the 28-day test results equals or exceeds the specified 28-day flexural strength, and no individual strength test is less than the tolerance indicated on Table 1. If the ratio of the 7-day strength test to the specified 28-day strength is less than 65 percent, make necessary adjustments for conformance. Fabricate, cure and test a minimum of one set of four beams for each shift of concrete placement. Remove concrete which is determined to be defective, based on the strength acceptance criteria therein, and replace with acceptable concrete.

3.7.4 Air Content Tests

Test air-entrained concrete for [air content](#) at the same frequency as specified for slump tests. Determine percentage of air in accordance with [ASTM C231/C231M](#) on samples taken during placement of concrete in forms.

3.7.5 Surface Testing

Use the profilograph method for all longitudinal testing, except for paving lanes less than [200 feet](#) in length. Use the straightedge method for transverse testing, for longitudinal testing where the length of each pavement lane is less than [200 feet](#), and at the ends of the paving limits for the project. Smoothness requirements do not apply over crowns, drainage structures, or similar penetrations. Maintain detailed notes of the testing results and submit a copy to the Government after each day's testing.

3.7.5.1 Straightedge Testing Method

Test the surface of the pavement with the straightedge to identify all surface irregularities exceeding the tolerances specified in Table 1. Test the entire area of the pavement in both a longitudinal and a transverse direction on parallel lines approximately [15 feet](#) apart. Hold the straightedge in contact with the surface and move ahead one-half the length of the straightedge for each successive measurement. Determine the amount of surface irregularity by placing the straightedge on the pavement surface and allowing it to rest upon the two highest spots covered by its length and measuring the maximum gap between the straightedge and the pavement surface, in the area between these two high points.

3.7.5.2 Profilograph Testing Method

Perform profilograph testing using approved California profilograph and procedures described in [ASTM E1274](#). Utilize electronic recording and automatic computerized reduction of data equipment to indicate "must-grind" bumps and the Profile Index for each [0.1 mile](#) segment of the day's paving. Accommodate grade breaks on parking lots by breaking the profile segment into short sections and repositioning the blanking band on each section. Provide the "blanking band" of [0.2 inch](#) wide and the "bump template" span [1 inch](#) with an offset of [0.4 inch](#). Count the profilograph testing of the last [30 feet](#) of a paving lane in the longitudinal direction from each day's paving operation on the following day's continuation lane. Compute the profile index for each pass of the profilograph (3 per lane) in each [0.1 mile](#) segment. The profile index for each segment is the average of the profile indices for each pass in each segment. Scale and proportion profilographs of unequal lengths to an equivalent [0.1 mile](#) as outlined in the [ASTM E1274](#). Submit a copy of the reduced tapes to the Government at

the end of each day's testing.

3.7.5.3 "Bumps" (Must Grind Areas)

Reduce any bumps ("must grind" areas) shown on the profilograph trace which exceed 0.4 inch in height by diamond grinding in accordance with subparagraph Diamond Grinding until they do not exceed 0.3 inch when retested. Taper such diamond grinding in all directions to provide smooth transitions to areas not requiring diamond grinding.

3.7.5.4 Diamond Grinding

Those performing diamond grinding are required to have a minimum of three years experience in diamond grinding of rigid concrete pavements. In areas not meeting the specified limits for surface smoothness and plan grade, reduce high areas to attain the required smoothness and grade, except as depth is limited below. Reduce high areas by diamond grinding the hardened concrete with an approved equipment after the concrete is at a minimum age of 14 days. Perform diamond grinding by sawing with an industrial diamond abrasive which is impregnated in the saw blades. Assemble the saw blades in a cutting head mounted on a machine designed specifically for diamond grinding that produces the required texture and smoothness level without damage to the concrete pavement or joint faces. Provide diamond grinding equipment with saw blades that are 1/8-inch wide, a minimum of 60 blades per 12 inches of cutting head width, and capable of cutting a path a minimum of 3 ft wide. Diamond grinding equipment that causes ravels, aggregate fractures, spalls or disturbance to the joints is not permitted. The maximum area corrected by diamond grinding the surface of the hardened concrete is 10 percent of the total area of a day's production. The maximum depth of diamond grinding is 1/4 inch. Provide diamond grinding machine equipped to flush and vacuum the pavement surface. Dispose of all debris from diamond grinding operations off Government property. Prior to diamond grinding, submit a Diamond Grinding Plan for review and approval. At a minimum, include the daily reports for the deficient areas, the location and extent of deficiencies, corrective actions, and equipment. Remove and replace all pavement areas requiring plan grade or surface smoothness corrections in excess of the limits specified in Table 1. All areas in which diamond grinding has been performed are subject to the thickness tolerances specified in Table 1.

3.7.6 Plan Grade Testing and Conformance

Within 5 days after each day's paving, test the finished surface of the pavement area by running lines of levels at intervals corresponding with every longitudinal and transverse joint to determine the elevation at each joint intersection. Record the results of this survey and submit a copy to the Government at the completion of the survey.

3.7.7 Edge Slump

Test the pavement surface to determine edge slump immediately after the concrete has hardened sufficiently to permit walking thereon. Perform testing with a minimum 12 foot straightedge to reveal irregularities exceeding the edge slump tolerance specified in Table 1. Determine the vertical edge slump at each free edge of each slipformed paving lane constructed. Place the straightedge transverse to the direction of paving and the end of the straightedge located at the edge of the paving lane. Record measurements at 5 to 10 foot spacings, as directed, commencing at

the header where paving was started. Initially record measurements at 5 foot intervals in each lane. When no deficiencies are present after 5 measurements, the interval may be increased. The maximum interval is 10 feet. When any deficiencies exist, return the interval to 5 feet. In addition to the transverse edge slump determination above, at the same time, record the longitudinal surface smoothness of the joint on a continuous line 1 inch back from the joint line using the minimum 12 foot straightedge advanced one-half its length for each reading. Perform other tests of the exposed joint face to ensure that a uniform, true vertical joint face is attained. Properly reference all recorded measurements in accordance with paving lane identification and stationing, and submit a report within 24 hours after measurement is made. Identify areas requiring replacement within the report.

3.7.8 Reinforcement

Inspect reinforcement prior to installation to verify it is free of loose flaky rust, loose scale, oil, mud, or other objectionable material.

3.7.9 Dowels

Inspect dowel placement prior to placing concrete to verify that dowels are of the size indicated, and are spaced, aligned and painted and oiled as specified. Do not permit dowels to exceed the tolerances shown in paragraph: DOWEL BARS.

-- End of Section --

SECTION 32 16 19

CONCRETE CURBS, GUTTERS AND SIDEWALKS
05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO M 182 (2005; R 2017) Standard Specification for Burlap Cloth Made from Jute or Kenaf and Cotton Mats

ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M (2020) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement

ASTM A1064/A1064M (2017) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete

ASTM C31/C31M (2019a) Standard Practice for Making and Curing Concrete Test Specimens in the Field

ASTM C94/C94M (2020) Standard Specification for Ready-Mixed Concrete

ASTM C143/C143M (2020) Standard Test Method for Slump of Hydraulic-Cement Concrete

ASTM C171 (2020) Standard Specification for Sheet Materials for Curing Concrete

ASTM C172/C172M (2017) Standard Practice for Sampling Freshly Mixed Concrete

ASTM C173/C173M (2016) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method

ASTM C231/C231M (2017a) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method

ASTM C309 (2011) Standard Specification for Liquid

APPENDIX B

Membrane-Forming Compounds for Curing
Concrete

ASTM C920

(2018) Standard Specification for
Elastomeric Joint Sealants

ASTM D1751

(2004; E 2013; R 2013) Standard
Specification for Preformed Expansion
Joint Filler for Concrete Paving and
Structural Construction (Nonextruding and
Resilient Bituminous Types)

ASTM D1752

(2018) Standard Specification for
Preformed Sponge Rubber, Cork and Recycled
PVC Expansion Joint Fillers for Concrete
Paving and Structural Construction

ASTM D5893/D5893M

(2016) Standard Specification for Cold
Applied, Single Component, Chemically
Curing Silicone Joint Sealant for Portland
Cement Concrete Pavements

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1 COMM

(2017) Standard And Commentary Accessible
and Usable Buildings and Facilities

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.

SD-03 Product Data

Concrete

SD-06 Test Reports

Field Quality Control

1.3 ENVIRONMENTAL REQUIREMENTS

1.3.1 Placing During Cold Weather

Do not place concrete when the air temperature reaches 40 degrees F and is falling, or is already below that point. Placement may begin when the air temperature reaches 35 degrees F and is rising, or is already above 40 degrees F. Make provisions to protect the concrete from freezing during the specified curing period. If necessary to place concrete when the temperature of the air, aggregates, or water is below 35 degrees F, placement and protection must be approved in writing. Approval will be contingent upon full conformance with the following provisions. Prepare and protect the underlying material so that it is entirely free of frost when the concrete is deposited. Heat mixing water as necessary to result in the temperature of the in-place concrete being between 50 and 85 degrees F. Methods and equipment for heating must be approved. Use only

APPENDIX B

aggregates that are free of ice, snow, and frozen lumps before entering the mixer. Provide covering or other means as needed to maintain the concrete at a temperature of at least 50 degrees F for not less than 72 hours after placing, and at a temperature above freezing for the remainder of the curing period.

1.3.2 Placing During Warm Weather

The temperature of the concrete as placed must not exceed 85 degrees F except where an approved retarder is used. Cool the mixing water and aggregates as necessary to maintain a satisfactory placing temperature. The placing temperature must not exceed 95 degrees F at any time.

PART 2 PRODUCTS

2.1 CONCRETE

Provide concrete conforming to the applicable requirements of ASTM C94/C94M except as otherwise specified. Concrete must have a minimum compressive strength of 3500 psi at 28 days. Size of aggregate must not exceed 1-1/2 inches. Submit copies of certified delivery tickets for all concrete used in the construction.

2.1.1 Air Content

Use concrete mixtures that have an air content by volume of concrete of 5 to 7 percent, based on measurements made immediately after discharge from the mixer.

2.1.2 Slump

Use concrete with a slump of 3 inches plus or minus 1 inch for hand placed concrete or 1 inch plus or minus 1/2 inch for slipformed concrete as determined in accordance with ASTM C143/C143M.

2.1.3 Reinforcement Steel

Use reinforcement bars conforming to ASTM A615/A615M. Use wire mesh reinforcement conforming to ASTM A1064/A1064M.

2.2 CONCRETE CURING MATERIALS

2.2.1 Impervious Sheet Materials

Use impervious sheet materials conforming to ASTM C171, type optional, except that polyethylene film, if used, must be white opaque.

2.2.2 Burlap

Use burlap conforming to AASHTO M 182.

2.2.3 White Pigmented Membrane-Forming Curing Compound

Use white pigmented membrane-forming curing compound conforming to ASTM C309, Type 2.

2.3 CONCRETE PROTECTION MATERIALS

APPENDIX B

Use concrete protection materials consisting of a linseed oil mixture of equal parts, by volume, of linseed oil and either mineral spirits, naphtha, or turpentine. At the option of the Contractor, commercially prepared linseed oil mixtures, formulated specifically for application to concrete to provide protection against the action of deicing chemicals may be used, except that emulsified mixtures are not acceptable.

2.4 JOINT FILLER STRIPS

2.4.1 Contraction Joint Filler for Curb and Gutter

Use hard-pressed fiberboard contraction joint filler for curb and gutter.

2.4.2 Expansion Joint Filler, Premolded

Unless otherwise indicated, use 1/2 inch thick premolded expansion joint filler conforming to ASTM D1751 or ASTM D1752.

2.5 JOINT SEALANTS

Use cold-applied joint sealant conforming to ASTM C920 or ASTM D5893/D5893M.

2.6 FORM WORK

Design and construct form work to ensure that the finished concrete will conform accurately to the indicated dimensions, lines, and elevations, and within the tolerances specified. Use wood or steel forms that are straight and of sufficient strength to resist springing during depositing and consolidating concrete.

2.6.1 Wood Forms

Use forms that are surfaced plank, 2 inches nominal thickness, straight and free from warp, twist, loose knots, splits or other defects. Use forms with a nominal length of 10 feet. Radius bends may be formed with 3/4 inch boards, laminated to the required thickness.

2.6.2 Steel Forms

Use channel-formed sections with a flat top surface and welded braces at each end and at not less than two intermediate points. Use forms with interlocking and self-aligning ends. Provide flexible forms for radius forming, corner forms, form spreaders, and fillers as needed. Use forms with a nominal length of 10 feet and that have a minimum of 3 welded stake pockets per form. Use stake pins consisting of solid steel rods with chamfered heads and pointed tips designed for use with steel forms.

2.6.3 Sidewalk Forms

Use sidewalk forms that are of a height equal to the full depth of the finished sidewalk.

2.6.4 Curb and Gutter Forms

Use curb and gutter outside forms that have a height equal to the full depth of the curb or gutter. Use rigid forms for curb returns, except that benders or thin plank forms may be used for curb or curb returns with a

radius of 10 feet or more, where grade changes occur in the return, or where the central angle is such that a rigid form with a central angle of 90 degrees cannot be used. Back forms for curb returns may be made of 1-1/2 inch benders, for the full height of the curb, cleated together. In lieu of inside forms for curbs, a curb "mule" may be used for forming and finishing this surface, provided the results are approved.

2.6.5 Biodegradable Form Release Agent

Use form release agent that is colorless and biodegradable and that is composed of at least 87 percent biobased material. Provide product that does not bond with, stain, or adversely affect concrete surfaces and does not impair subsequent treatments of concrete surfaces. Provide form release agent that does not contain diesel fuel, petroleum-based lubricating oils, waxes, or kerosene.

PART 3 EXECUTION

3.1 SUBGRADE PREPARATION

Construct subgrade to the specified grade and cross section prior to concrete placement.

3.1.1 Maintenance of Subgrade

Maintain subgrade in a smooth, compacted condition in conformity with the required section and established grade until the concrete is placed. The subgrade must be in a moist condition when concrete is placed. Prepare and protect subgrade so that it is free from frost when the concrete is deposited.

3.2 FORM SETTING

Set forms to the indicated alignment, grade and dimensions. Hold forms rigidly in place by a minimum of 3 stakes per form placed at intervals not to exceed 4 feet. Use additional stakes and braces at corners, deep sections, and radius bends, as required. Use clamps, spreaders, and braces where required to ensure rigidity in the forms. Remove forms in a manner that will not injure the concrete. Do not use bars or heavy tools against the concrete when removing the forms. Promptly and satisfactorily repair concrete found to be defective after form removal. Clean forms and coat with form oil or biodegradable form release agent each time before concrete is placed. Wood forms may, instead, be thoroughly wetted with water before concrete is placed, except that with probable freezing temperatures, oiling is mandatory.

3.2.1 Sidewalks

Set forms for sidewalks with the upper edge true to line and grade with an allowable tolerance of 1/8 inch in any 10 foot long section. After forms are set, grade and alignment must be checked with a 10 foot straightedge. Sidewalks must have a transverse slope of 1/4 inch per foot. Do not remove side forms less than 12 hours after finishing has been completed.

3.2.2 Curbs and Gutters

Remove forms used along the front of the curb not less than 2 hours nor

more than 6 hours after the concrete has been placed. Do not remove forms used along the back of curb until the face and top of the curb have been finished, as specified for concrete finishing. Do not remove gutter forms while the concrete is sufficiently plastic to slump in any direction.

3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING

3.3.1 Formed Sidewalks

Place concrete in the forms in one layer. When consolidated and finished, the sidewalks must be of the thickness indicated. Use a strike-off guided by side forms after concrete has been placed in the forms to bring the surface to proper section to be compacted. Consolidate concrete by tamping and spading or with an approved vibrator. Finish the surface to grade with a strike off.

3.3.2 Concrete Finishing

After straightedging, when most of the water sheen has disappeared, and just before the concrete hardens, finish the surface with a wood or magnesium float or darby to a smooth and uniformly fine granular or sandy texture free of waves, irregularities, or tool marks. Produce a scored surface by brooming with a fiber-bristle brush in a direction transverse to that of the traffic, followed by edging.

3.3.3 Edge and Joint Finishing

Finish all slab edges, including those at formed joints, with an edger having a radius of $\frac{1}{8}$ inch. Edge transverse joints before brooming. Eliminate the flat surface left by the surface face of the edger with brooming. Clean and solidly fill corners and edges which have crumbled and areas which lack sufficient mortar for proper finishing with a properly proportioned mortar mixture and then finish.

3.3.4 Surface and Thickness Tolerances

Finished surfaces must not vary more than $\frac{5}{16}$ inch from the testing edge of a 10-foot straightedge. Permissible deficiency in section thickness will be up to $\frac{1}{4}$ inch.

3.4 CURB AND GUTTER CONCRETE PLACEMENT AND FINISHING

3.4.1 Formed Curb and Gutter

Place concrete to the required section in a single lift. Consolidate concrete using approved mechanical vibrators. Curve shaped gutters must be finished with a standard curb "mule".

3.4.2 Curb and Gutter Finishing

Approved slipformed curb and gutter machines may be used in lieu of hand placement.

3.4.3 Concrete Finishing

Float and finish exposed surfaces with a smooth wood float until true to grade and section and uniform in texture. Brush floated surfaces with a

fine-hair brush using longitudinal strokes. Round the edges of the gutter and top of the curb with an edging tool to a radius of $1/2$ inch. Immediately after removing the front curb form, rub the face of the curb with a wood or concrete rubbing block and water until blemishes, form marks, and tool marks have been removed. Brush the front curb surface, while still wet, in the same manner as the gutter and curb top. Finish the top surface of gutter to grade with a wood float.

3.4.4 Joint Finishing

Finish curb edges at formed joints as indicated.

3.4.5 Surface and Thickness Tolerances

Finished surfaces must not vary more than $1/4$ inch from the testing edge of a 10-foot straightedge. Permissible deficiency in section thickness will be up to $1/4$ inch.

3.5 SIDEWALK JOINTS

Construct sidewalk joints to divide the surface into rectangular areas. Space transverse contraction joints at a distance equal to the sidewalk width or 5 feet on centers, whichever is less, and continuous across the slab. Construct longitudinal contraction joints along the centerline of all sidewalks 10 feet or more in width. Construct transverse expansion joints at sidewalk returns and opposite expansion joints in adjoining curbs. Where the sidewalk is not in contact with the curb, install transverse expansion joints as indicated. Form expansion joints around structures and features which project through or into the sidewalk pavement, using joint filler of the type, thickness, and width indicated. Expansion joints are not required between sidewalks and curb that abut the sidewalk longitudinally.

3.5.1 Sidewalk Contraction Joints

Form contraction joints in the fresh concrete by cutting a groove in the top portion of the slab to a depth of at least one-fourth of the sidewalk slab thickness. Unless otherwise approved or indicated, either use a jointer to cut the groove or saw a groove in the hardened concrete with a power-driven saw. Construct sawed joints by sawing a groove in the concrete with a $1/8$ inch blade. Provide an ample supply of saw blades on the jobsite before concrete placement is started. Provide at least one standby sawing unit in good working order at the jobsite at all times during the sawing operations.

3.5.2 Sidewalk Expansion Joints

Form expansion joints using $1/2$ inch joint filler strips. Joint filler in expansion joints surrounding structures and features within the sidewalk may consist of preformed filler material conforming to ASTM D1752 or building paper. Hold joint filler in place with steel pins or other devices to prevent warping of the filler during floating and finishing. Immediately after finishing operations are completed, round joint edges using an edging tool having a radius of $1/8$ inch. Remove any concrete over the joint filler. At the end of the curing period, clean the top of expansion joints and fill with cold-applied joint sealant. Use joint sealant that is gray or stone in color. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on

exposed surfaces of the concrete. Apply joint sealing material only when the concrete at the joint is surface dry and atmospheric and concrete temperatures are above 50 degrees F. Immediately remove any excess material on exposed surfaces of the concrete and clean the concrete surfaces.

3.5.3 Reinforcement Steel Placement

Accurately and securely fasten reinforcement steel in place with suitable supports and ties before the concrete is placed.

3.6 CURB AND GUTTER JOINTS

Construct curb and gutter joints at right angles to the line of curb and gutter.

3.6.1 Contraction Joints

Construct contraction joints directly opposite contraction joints in abutting portland cement concrete pavements and spaced so that monolithic sections between curb returns will not be less than 5 feet nor greater than 15 feet in length.

- a. Construct contraction joints (except for slip forming) by means of 1/8 inch thick separators and of a section conforming to the cross section of the curb and gutter. Remove separators as soon as practicable after concrete has set sufficiently to preserve the width and shape of the joint and prior to finishing.
- b. When slip forming is used, cut the contraction joints in the top portion of the gutter/curb hardened concrete in a continuous cut across the curb and gutter, using a power-driven saw. Cut the contraction joint to a depth of at least one-fourth of the gutter/curb depth using a 1/8 inch saw blade.

3.6.2 Expansion Joints

Form expansion joints by means of preformed expansion joint filler material cut and shaped to the cross section of curb and gutter. Construct expansion joints in curb and gutter directly opposite expansion joints of abutting portland cement concrete pavement using the same type and thickness of joints as joints in the pavement. Where curb and gutter do not abut portland cement concrete pavement, provide expansion joints at least 1/2 inch in width at intervals not less than 30 feet nor greater than 120 feet. Seal expansion joints immediately following curing of the concrete or as soon thereafter as weather conditions permit. Seal expansion joints and the top 1 inch depth of curb and gutter contraction-joints with joint sealant. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Concrete at the joint must be surface dry and atmospheric and concrete temperatures must be above 50 degrees F at the time of application of joint sealing material. Immediately remove excess material on exposed surfaces of the concrete and clean concrete surfaces.

3.7 CURING AND PROTECTION

3.7.1 General Requirements

Protect concrete against loss of moisture and rapid temperature changes for at least 7 days from the beginning of the curing operation. Protect unhardened concrete from rain and flowing water. All equipment needed for adequate curing and protection of the concrete must be on hand and ready for use before actual concrete placement begins. Protect concrete as necessary to prevent cracking of the pavement due to temperature changes during the curing period.

3.7.1.1 Mat Method

Cover the entire exposed surface with two or more layers of burlap. Overlap mats at least 6 inches. Thoroughly wet the mat with water prior to placing on concrete surface and keep the mat continuously in a saturated condition and in intimate contact with concrete for not less than 7 days.

3.7.1.2 Impervious Sheeting Method

Wet the entire exposed surface with a fine spray of water and then cover with impervious sheeting material. Lay sheets directly on the concrete surface with the light-colored side up and overlapped 12 inches when a continuous sheet is not used. Use sheeting that is not less than 18-inches wider than the concrete surface to be cured. Secure sheeting using heavy wood planks or a bank of moist earth placed along edges and laps in the sheets. Satisfactorily repair or replace sheets that are torn or otherwise damaged during curing. Sheeting must remain on the concrete surface to be cured for not less than 7 days.

3.7.1.3 Membrane Curing Method

Apply a uniform coating of white-pigmented membrane-curing compound to the entire exposed surface of the concrete as soon after finishing as the free water has disappeared from the finished surface. Coat formed surfaces immediately after the forms are removed and in no case longer than 1 hour after the removal of forms. Do not allow concrete surface to dry before application of the membrane. If drying has occurred, moisten the surface of the concrete with a fine spray of water and apply the curing compound as soon as the free water disappears. Apply curing compound in two coats by hand-operated pressure sprayers at a coverage of approximately 200 square feet/gallon for the total of both coats. Apply the second coat in a direction approximately at right angles to the direction of application of the first coat. The compound must form a uniform, continuous, coherent film that will not check, crack, or peel and must be free from pinholes or other imperfections. If pinholes, abrasion, or other discontinuities exist, apply an additional coat to the affected areas within 30 minutes. Respray concrete surfaces that are subjected to heavy rainfall within 3 hours after the curing compound has been applied by the method and at the coverage specified above. Respray areas where the curing compound is damaged by subsequent construction operations within the curing period. Take precautions necessary to ensure that the concrete is properly cured at sawed joints, and that no curing compound enters the joints. Tightly seal the top of the joint opening and the joint groove at exposed edges before the concrete in the region of the joint is resprayed with curing compound. Use a method used for sealing the joint groove that prevents loss of moisture from the joint during the entire specified curing period. Provide approved standby facilities for curing concrete pavement at a location accessible to the jobsite for use in the event of mechanical failure of the

spraying equipment or other conditions that might prevent correct application of the membrane-curing compound at the proper time. Adequately protect concrete surfaces to which membrane-curing compounds have been applied during the entire curing period from pedestrian and vehicular traffic, except as required for joint-sawing operations and surface tests, and from other possible damage to the continuity of the membrane.

3.7.2 Backfilling

After curing, remove debris and backfill, grade, and compact the area adjoining the concrete to conform to the surrounding area in accordance with lines and grades indicated.

3.7.3 Protection

Protect completed concrete from damage until accepted. Repair damaged concrete and clean concrete discolored during construction. Remove and reconstruct concrete that is damaged for the entire length between regularly scheduled joints. Refinishing the damaged portion will not be acceptable. Dispose of removed material as directed.

3.7.4 Protective Coating

Apply a protective coating of linseed oil mixture to the exposed-to-view concrete surface after the curing period, if concrete will be exposed to de-icing chemicals within 6 weeks after placement. Moist cure concrete to receive a protective coating.

3.7.4.1 Application

Complete curing and backfilling operation prior to applying two coats of protective coating. Concrete must be surface dry and clean before each application. Spray apply at a rate of not more than 50 square yards/gallon for first application and not more than 70 square yards/gallon for second application, except that the number of applications and coverage for each application for commercially prepared mixture must be in accordance with the manufacturer's instructions. Protect coated surfaces from vehicular and pedestrian traffic until dry.

3.7.4.2 Precautions

Do not heat protective coating by direct application of flame or electrical heaters and protect the coating from exposure to open flame, sparks, and fire adjacent to open containers or applicators. Do not apply material at ambient or material temperatures lower than 50 degrees F.

3.8 FIELD QUALITY CONTROL

Submit copies of all test reports within 24 hours of completion of the test.

3.8.1 General Requirements

Perform the inspection and tests described and meet the specified requirements for inspection details and frequency of testing. Based upon the results of these inspections and tests, take the action and submit reports as required below, and additional tests to ensure that the requirements of these specifications are met.

APPENDIX B

3.8.2 Concrete Testing

3.8.2.1 Strength Testing

Take concrete samples in accordance with [ASTM C172/C172M](#) not less than once a day nor less than once for every [250 cubic yards](#) of concrete placed. Mold cylinders in accordance with [ASTM C31/C31M](#) for strength testing by an approved laboratory. Each strength test result must be the average of 2 test cylinders from the same concrete sample tested at 28 days, unless otherwise specified or approved. Concrete specified on the basis of compressive strength will be considered satisfactory if the averages of all sets of three consecutive strength test results equal or exceed the specified strength, and no individual strength test result falls below the specified strength by more than [500 psi](#).

3.8.2.2 Air Content

Determine air content in accordance with [ASTM C173/C173M](#) or [ASTM C231/C231M](#). Use [ASTM C231/C231M](#) with concretes and mortars made with relatively dense natural aggregates. Make two tests for air content on randomly selected batches of each class of concrete placed during each shift. Make additional tests when excessive variation in concrete workability is reported by the placing foreman or the Government inspector. Notify the placing foreman if results are out of tolerance. The placing foreman must take appropriate action to have the air content corrected at the plant. Additional tests for air content will be performed on each truckload of material until such time as the air content is within the tolerance specified.

3.8.2.3 Slump Test

Perform two slump tests on randomly selected batches of each class of concrete for every [250 cubic yards](#), or fraction thereof, of concrete placed during each shift. Perform additional tests when excessive variation in the workability of the concrete is noted or when excessive crumbling or slumping is noted along the edges of slip-formed concrete.

3.8.3 Thickness Evaluation

Determine the anticipated thickness of the concrete prior to placement by passing a template through the formed section or by measuring the depth of opening of the extrusion template of the curb forming machine. If a slip form paver is used for sidewalk placement, construct the subgrade true to grade prior to concrete placement. The thickness will be determined by measuring each edge of the completed slab.

3.8.4 Surface Evaluation

Provide finished surfaces for each category of the completed work that are uniform in color and free of blemishes and form or tool marks.

3.9 SURFACE DEFICIENCIES AND CORRECTIONS

3.9.1 Thickness Deficiency

When measurements indicate that the completed concrete section is deficient in thickness by more than [1/4 inch](#) the deficient section will be removed,

between regularly scheduled joints, and replaced.

3.9.2 High Areas

In areas not meeting surface smoothness and plan grade requirements, reduce high areas either by rubbing the freshly finished concrete with carborundum brick and water when the concrete is less than 36 hours old or by grinding the hardened concrete with an approved surface grinding machine after the concrete is 36 hours old or more. The area corrected by grinding the surface of the hardened concrete must not exceed 5 percent of the area of any integral slab, and the depth of grinding must not exceed 1/4 inch. Remove and replace pavement areas requiring grade or surface smoothness corrections in excess of the limits specified.

3.9.3 Appearance

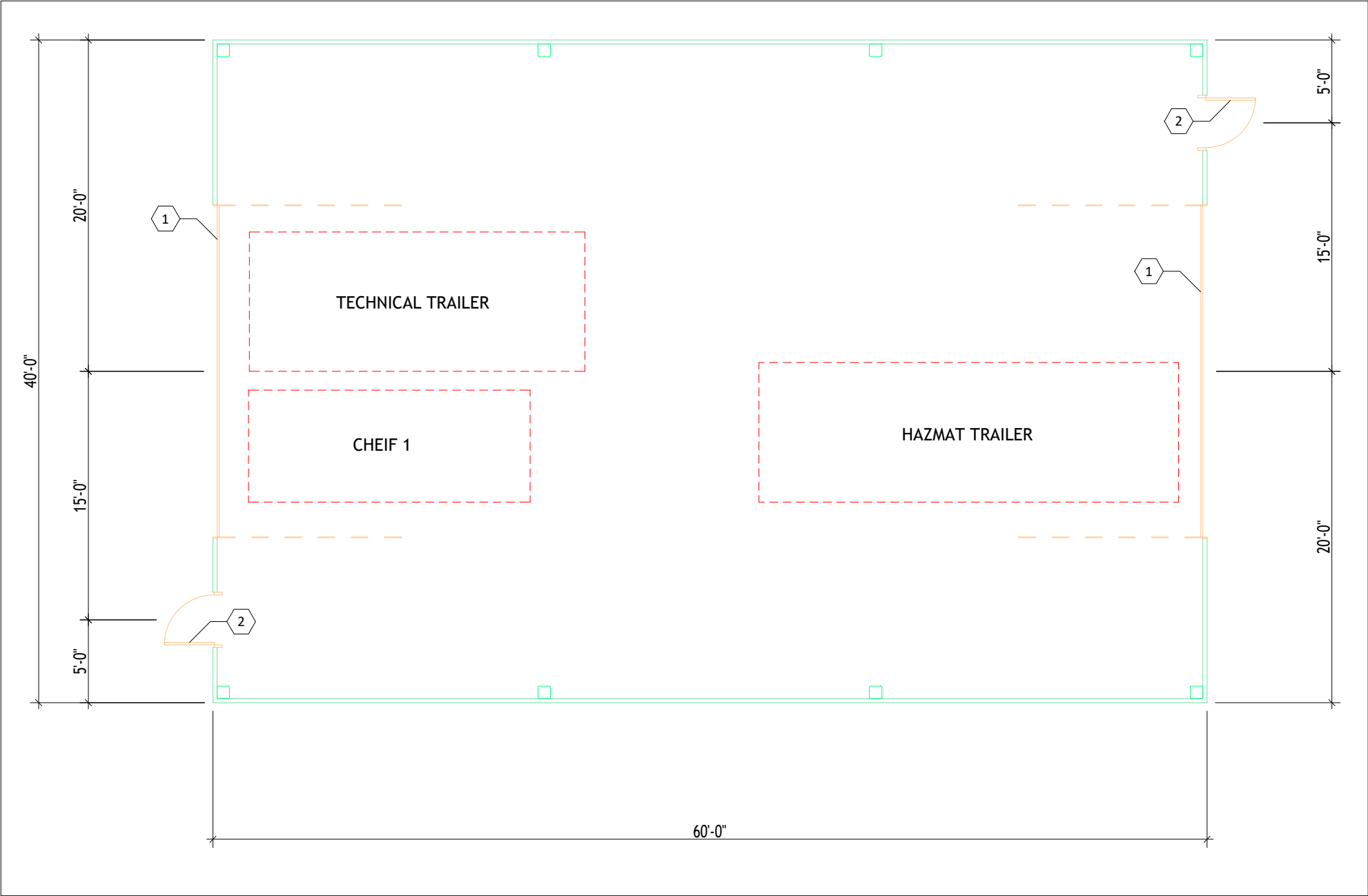
Exposed surfaces of the finished work will be inspected by the Contracting Officer and deficiencies in appearance will be identified. Remove and replace areas which exhibit excessive cracking, discoloration, form marks, or tool marks or which are otherwise inconsistent with the overall appearances of the work.

3.10 DETECTABLE WARNING SYSTEM

Install Detectable Warning Systems required by Contract plans in accordance with ICC A117.1 COMM, Section 705, and by manufacturers' installation instructions.

-- End of Section --

APPENDIX C – DRAWINGS



1 FLOOR PLAN
1/8" = 1'-0"



GENERAL NOTES

- 1. CONTRACTOR IS TO CONFIRM ALL DIMENSIONS.
- 2. ALL WORK, MATERIALS, AND EQUIPMENT SHALL BE IN ACCORDANCE WITH APPLICABLE STATE AND LOCAL CODES.
- 3. REFER TO STATEMENT OF WORK FOR PRE-FABRICATED STRUCTURE DESIGN REQUIREMENTS, DETAILS, AND BASIS OF DESIGN.
- 4. REFER TO STATEMENT OF WORK FOR COLOR SELECTIONS.

KEYNOTES

- 1 20' WIDE X 10' TALL INSULATED MOTORIZED OVERHEAD METAL GARAGE DOOR. SIZE OPENER MOTOR ACCORDING TO DOOR MANUFACTURER RECOMMENDATIONS.
- 2 3' WIDE X 7' TALL INSULATED METAL DOOR.



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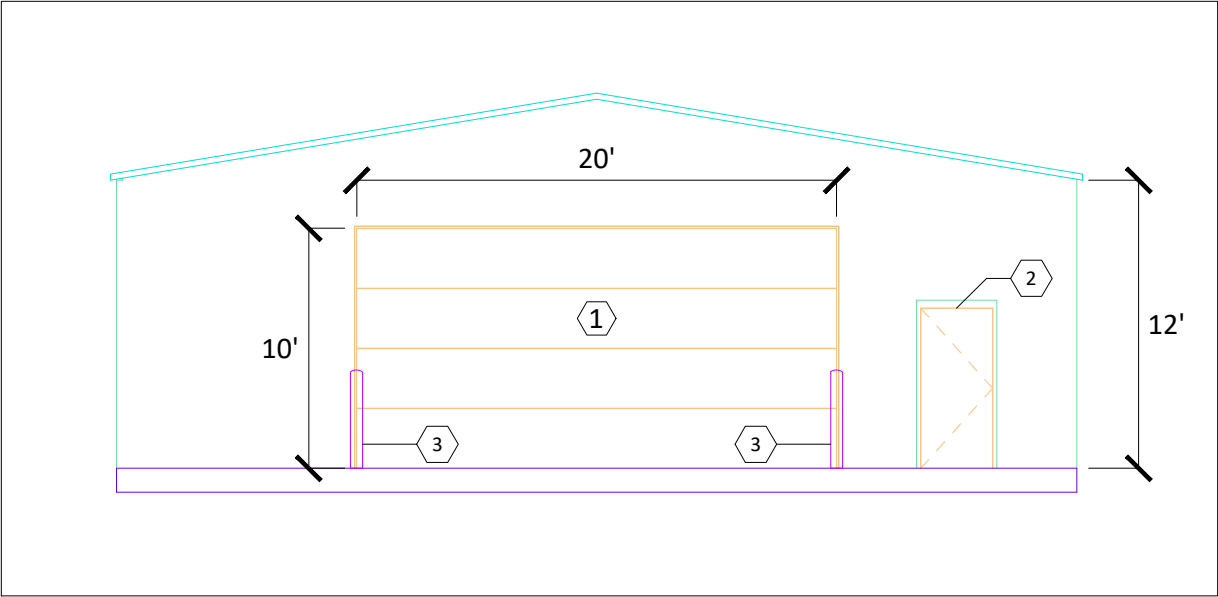


140 CHANNEL STREET
SUITE 138
VANCE AIR FORCE BASE
ENID, OKLAHOMA
580.213.7076

REVISION #	DESCRIPTION	DATE

DRAWN BY: DKM
CHECKED BY: JW
FP NUMBER: FP1024054
DATE: 28APR2026
DESIGN STAGE: 100%

CONSTRUCT METAL
BUILDING, F5067
CONCEPTUAL FLOOR PLAN



1 FRONT AND BACK ELEVATIONS
1/8" = 1'-0"

GENERAL NOTES

- 1. CONTRACTOR IS TO CONFIRM ALL DIMENSIONS.
- 2. CONTRACTOR IS TO INSTALL EQUIPMENT IN STRICT ACCORDANCE WITH MANUFACTURE'S RECOMMENDATIONS.
- 3. ALL WORK, MATERIALS, AND EQUIPMENT SHALL BE IN ACCORDANCE WITH APPLICABLE STATE AND LOCAL CODES.
- 4. REFER TO STATEMENT OF WORK FOR PRE-FABRICATED STRUCTURE DESIGN REQUIREMENTS, DETAILS, AND BASIS OF DESIGN.
- 5. REFER TO STATEMENT OF WORK FOR COLOR SELECTIONS.

KEYNOTES

- 1 20' WIDE X 10' TALL INSULATED MOTORIZED OVERHEAD METAL GARAGE DOOR. SIZE OPENER MOTOR ACCORDING TO OVERHEAD DOOR MANUFACTURER RECOMMENDATIONS.
- 2 3' WIDE X 7' TALL INSULATED METAL DOOR.
- 3 INSTALL BOLLARDS ON EXTERIOR CORNERS OF OVERHEAD ENTRIES. REFER TO STATEMENT OF WORK FOR DETAILS.



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REVISION #	DESCRIPTION	DATE

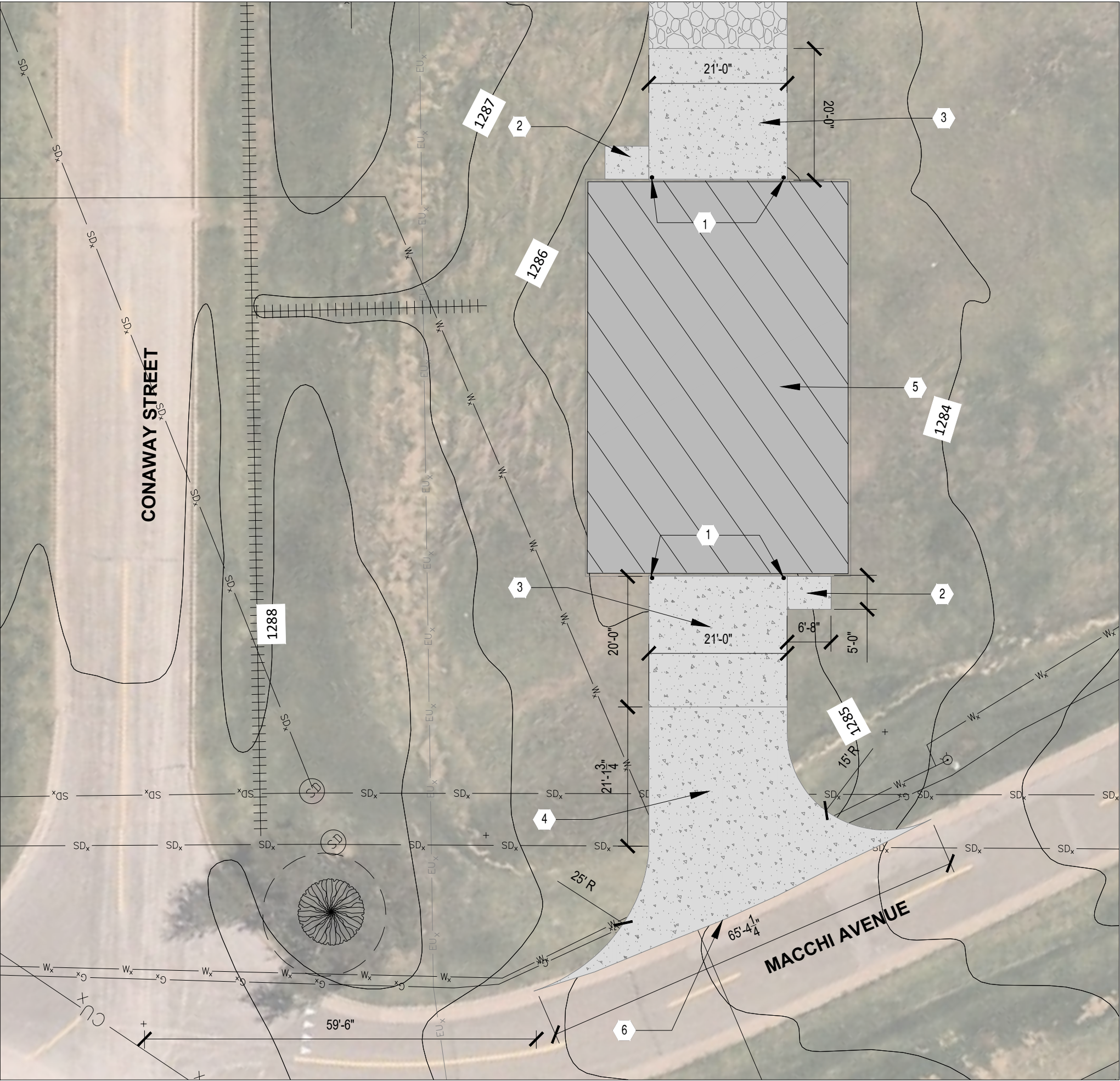
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CHECKED BY: JW
FP NUMBER: FP1024054
DATE: 28APR2026
DESIGN STAGE: 100%

CONSTRUCT METAL
BUILDING, F5067
BUILDING ELEVATIONS



CS101

APPENDIX C



1
1/16" = 1'-0"

GENERAL NOTES

1. REFER TO THE STATEMENT OF WORK TO OBTAIN COMPLETE SCOPE OF WORK TO BE ACCOMPLISHED.
2. THIS DRAWING IS PROVIDED FOR REFERENCE ONLY. THE CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND CONFIRM LOCATION OF WORK.
3. EXISTING UTILITY INFORMATION IS PROVIDED FOR REFERENCE ONLY. THE CONTRACTOR SHALL CONTACT THE CONSTRUCTION MANAGER TO OBTAIN A DIG PERMIT TO CONFIRM LOCATION AND ELEVATION OF UTILITIES PRIOR TO COMMENCING WORK.
4. CONTOUR ELEVATIONS ARE PROVIDED FOR REFERENCE ONLY. THE CONTRACTOR SHALL FIELD VERIFY ALL ELEVATIONS.
5. CONTRACTOR TO PROVIDE POSITIVE DRAINAGE AWAY FROM THE BUILDING SLAB. MINIMUM SLOPE FROM EDGE OF SLAB TO EXISTING GRASS IS 2%.
6. FINAL BUILDING SLAB ELEVATION (TOP) TO BE 1287.5 FT, OR AS NECESSARY TO PROVIDE POSITIVE DRAINAGE AWAY FROM THE BUILDING.

UTILITY SYMBOLS (EXISTING)

- F FIRE SUPPRESSION LINE
- W_x WATER LINE
- SD STORM SEWER
- G NATURAL GAS
- EU_x ELECTRICAL POWER
- ES_x ELECTRICAL POWER SECONDARY
- cu_x COMM LINES
- SS_x SANITARY SEWER
- X BRICK FENCE
- SF SILT FENCE
- 12XX ELEVATION CONTOUR
- ||||| OPEN DRAINAGE

LEGEND

- EXISTING LIGHT POST
- EXISTING FIRE HYDRANT
- NEW CONCRETE SLAB/DRIVE ENTRANCE/SIDEWALK
- NEW STORAGE BUILDING
- GRAVEL DRIVE (BY OTHERS)
- REMOVE EXISTING TREE
- NEW BOLLARDS
- EXISTING TREE
- EXISTING STORM DRAIN MANHOLE
- EXISTING ELECTRICAL SWITCH
- NEW TRANSFORMER (INSTALLED BY OTHERS)

KEYNOTES

1. INSTALL BOLLARDS ON EXTERIOR CORNERS OF OVERHEAD DOOR ENTRIES.
2. CONSTRUCT NEW SIDEWALKS TO ACCESS DOOR ENTRANCE.
3. CONSTRUCT NEW P.C. CONCRETE SLAB.
4. CONSTRUCT NEW P.C. CONCRETE DRIVEWAY APPROACH.
5. CONSTRUCT NEW P.C. CONCRETE SLAB FOR NEW BUILDING.
6. REMOVE EXISTING CURB AND GUTTER.



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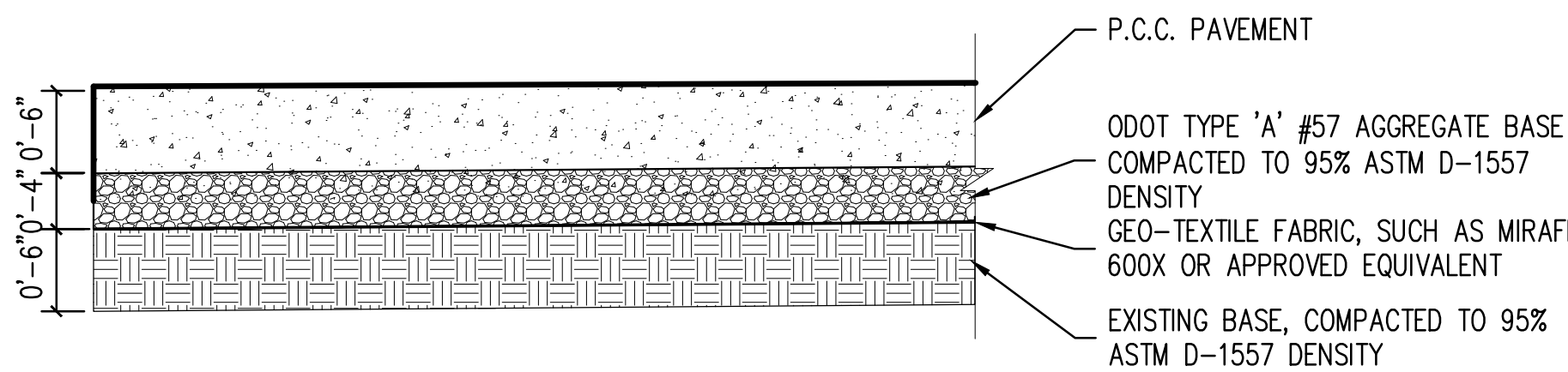
REVISION #	DESCRIPTION	DATE

DRAWN BY: JO
CHECKED BY: CT
FP NUMBER: FP1024054
DATE: 01MAY2026
DESIGN STAGE: 100%

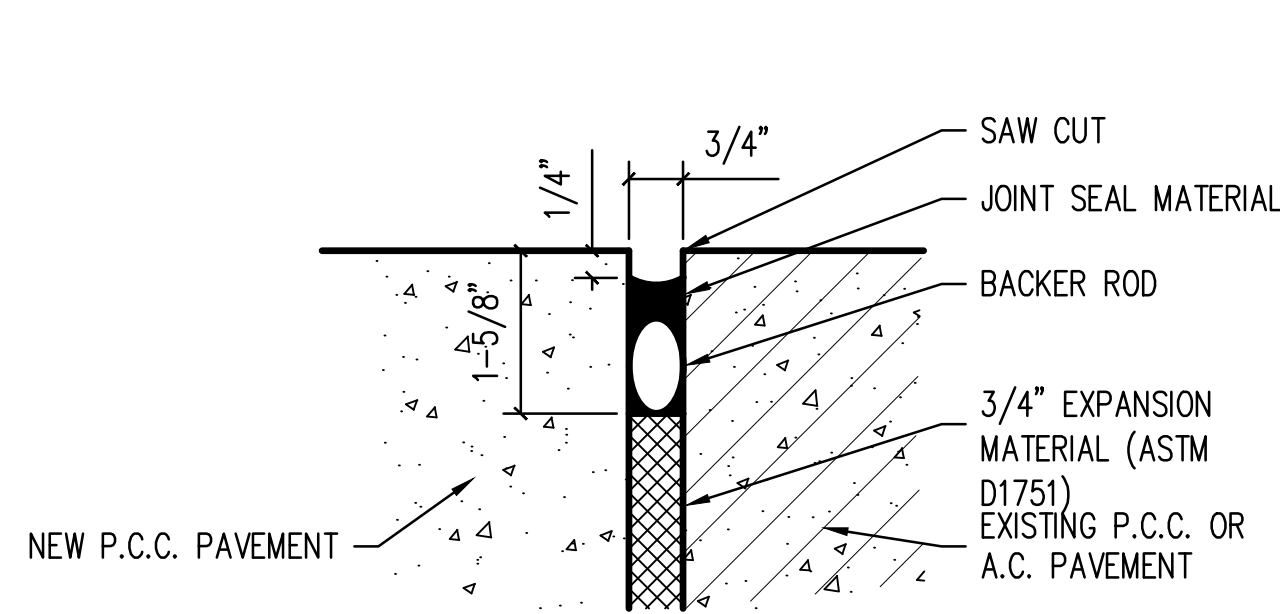
CONSTRUCT METAL
BUILDING, F5067
NEW P.C. CONCRETE LOCATION

CS102
APPENDIX C

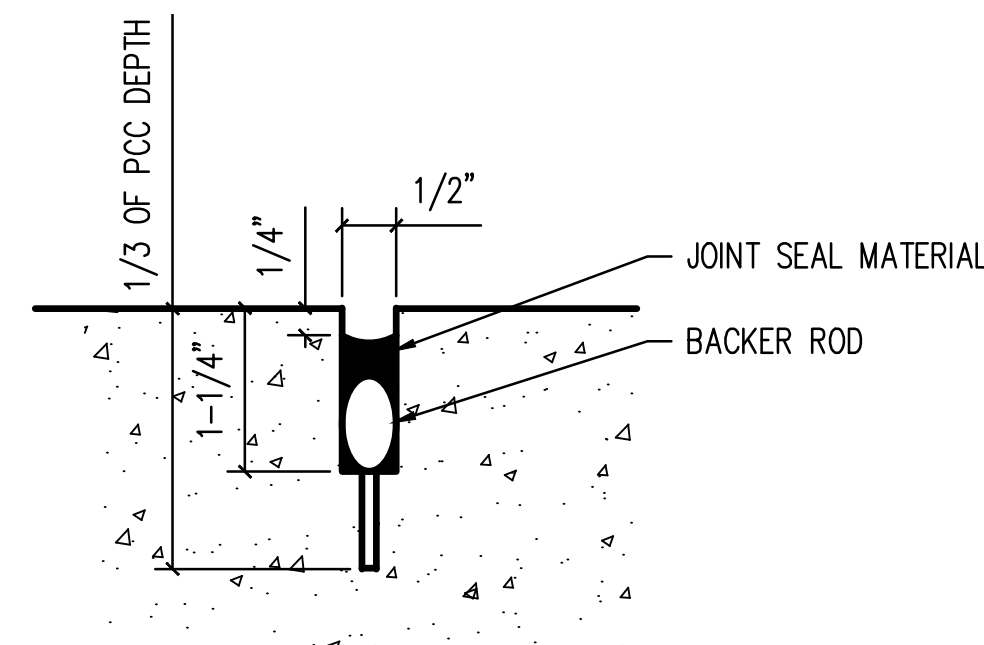
G
F
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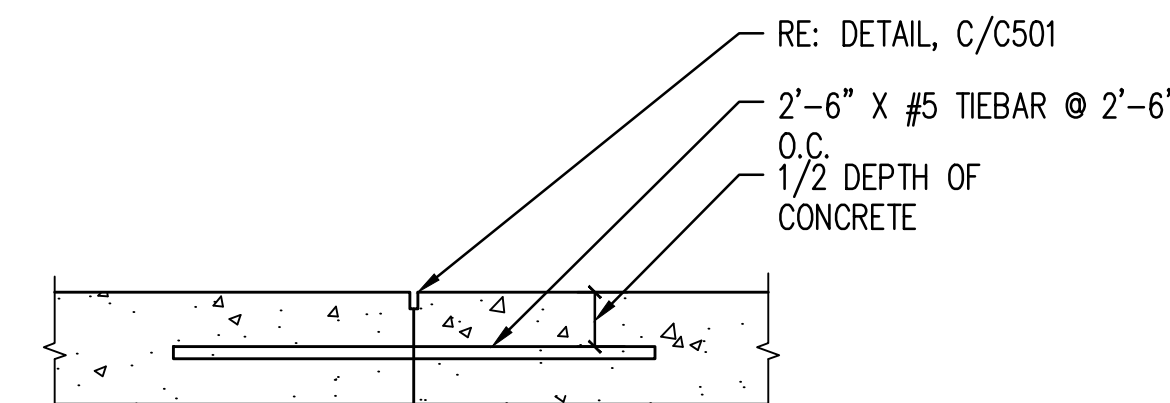
A PORTLAND CEMENT CONCRETE (P.C.C.) ROADWAY SECTION
C501 1" = 1'-0"



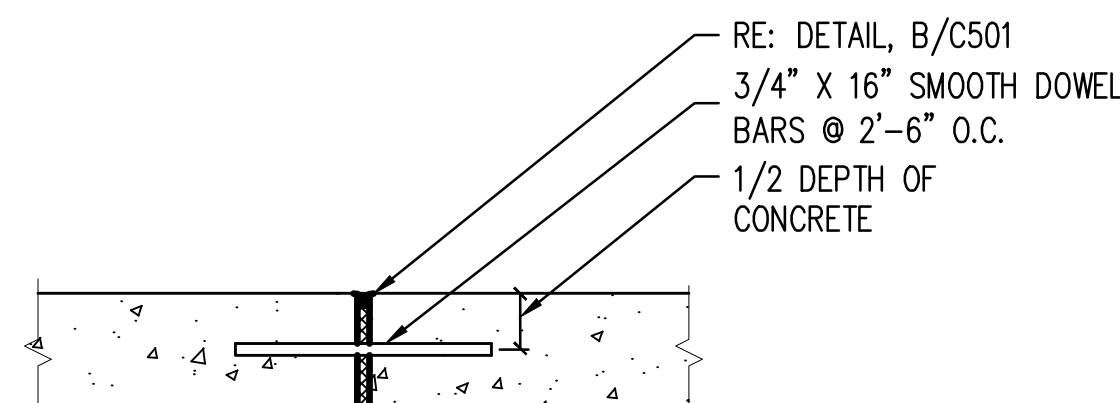
B P.C.C. EXPANSION JOINT - TYP
C501 NTS



C P.C.C. CONTRACTION JOINT - TYP
C501 NTS

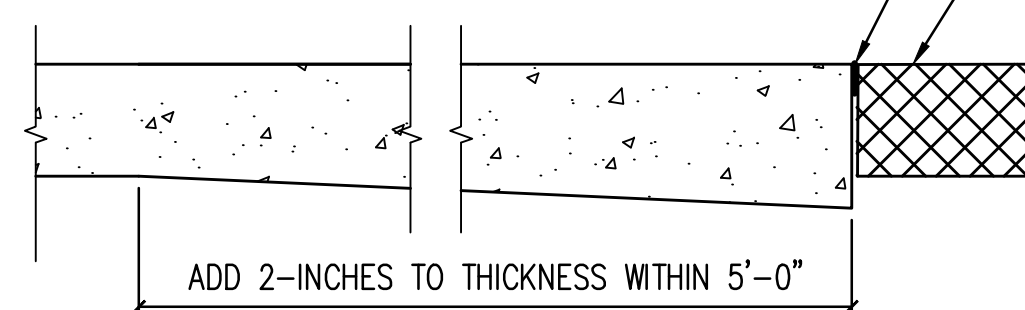


D CONSTRUCTION JOINT - TYP
C501 1" = 1'-0"

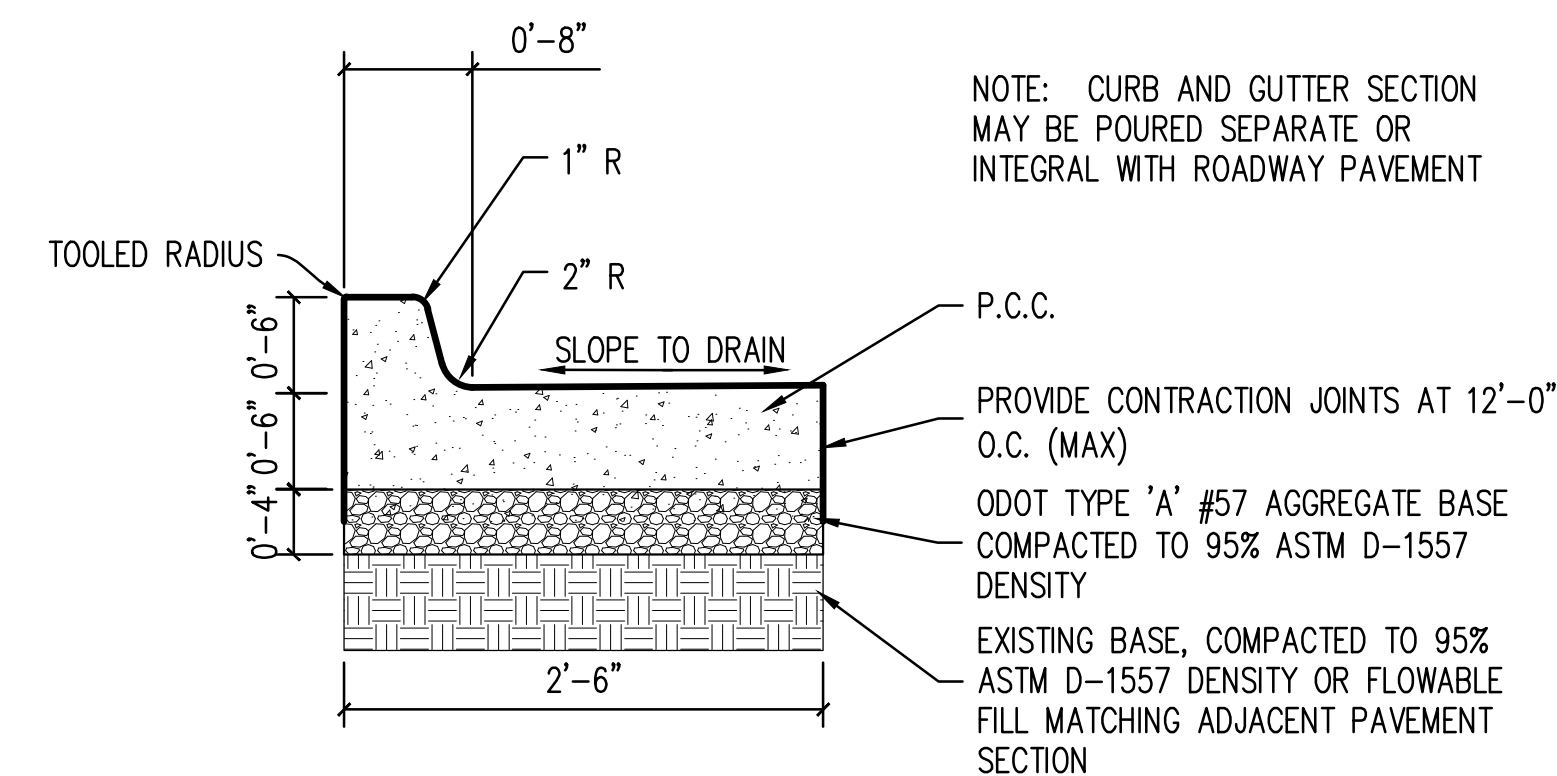


E DOWELED EXPANSION JOINT - TYP
C501 1" = 1'-0"

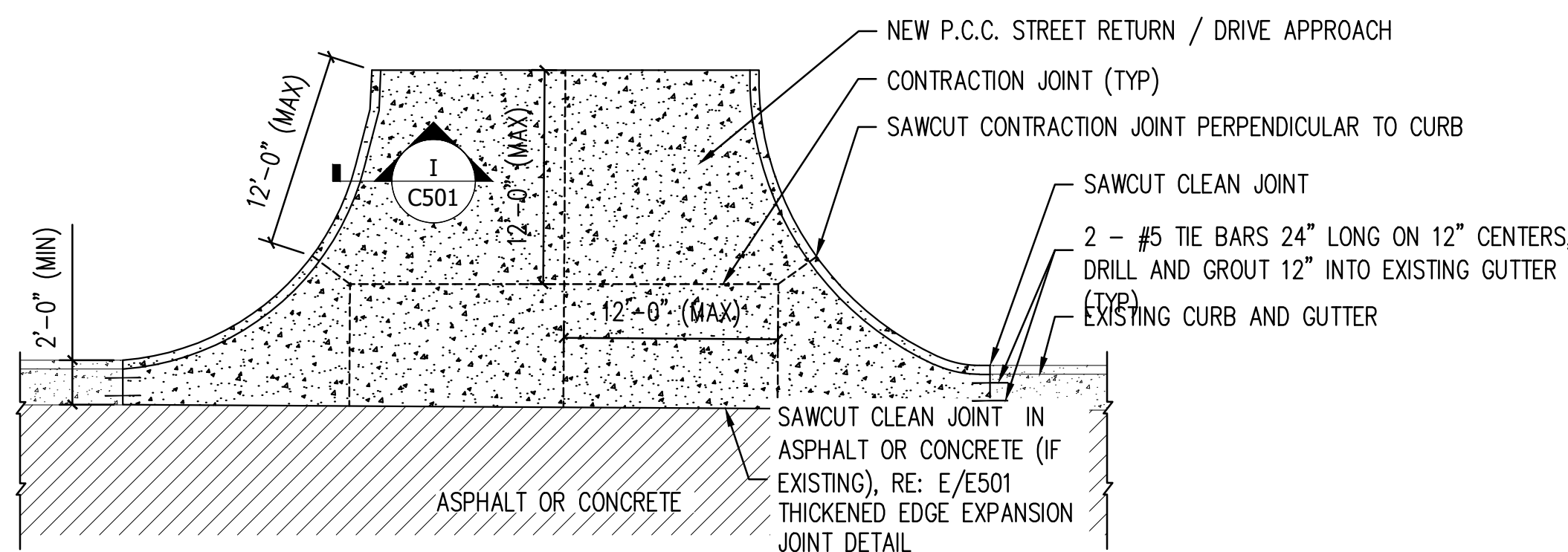
NOTE: WHEN ADJACENT TO NEW P.C.C. ROADWAY PROVIDE THICKENED EDGE ON BOTH THE ROADWAY SIDE AND DRIVE/RETURN SIDE.



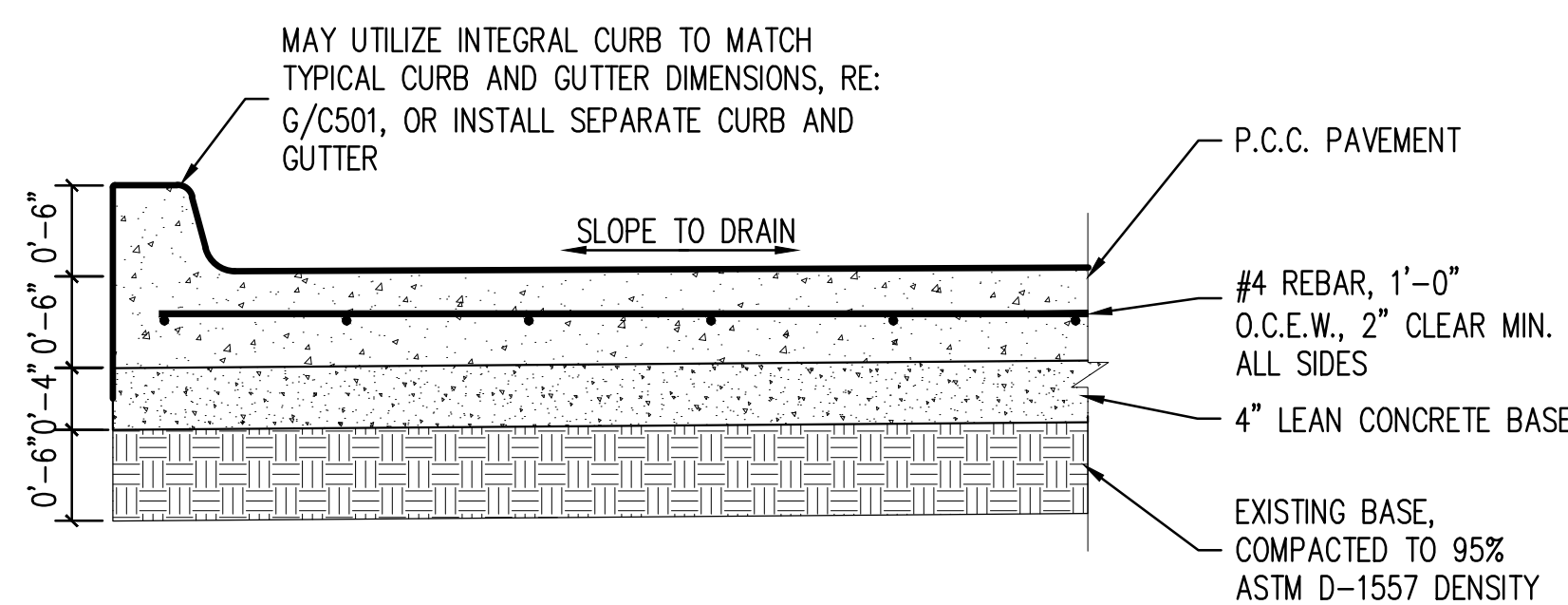
F THICKENED EDGE EXPANSION JOINT - TYP
C501 1" = 1'-0"



G P.C.C. CURB AND GUTTER - TYP
C501 1" = 1'-0"

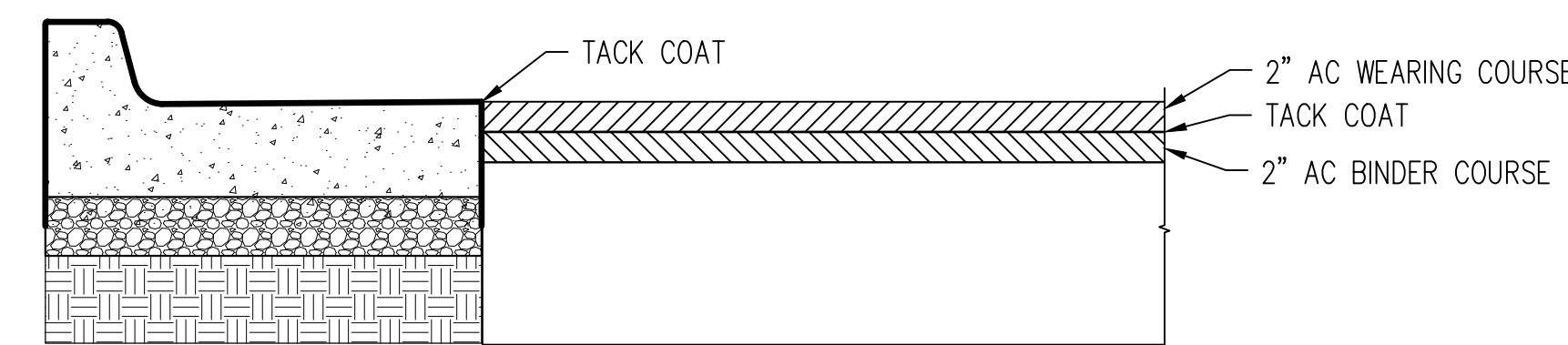


H P.C.C. DRIVE/ROADWAY RETURN (PLAN) - TYP
C501 1/8" = 1'-0"

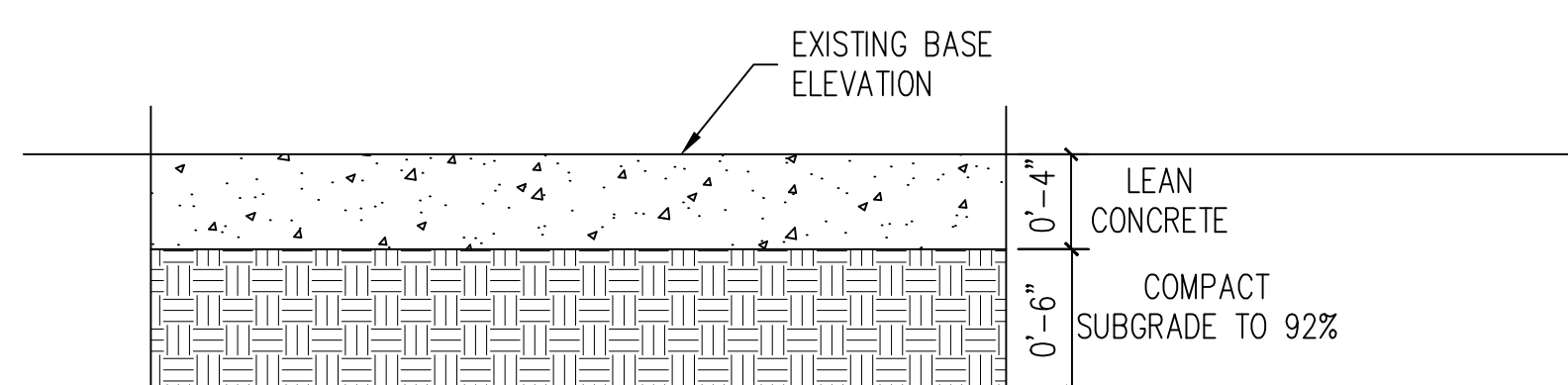


I P.C.C. DRIVE/ROADWAY RETURN (SECTION) - TYP
C501 1" = 1'-0"

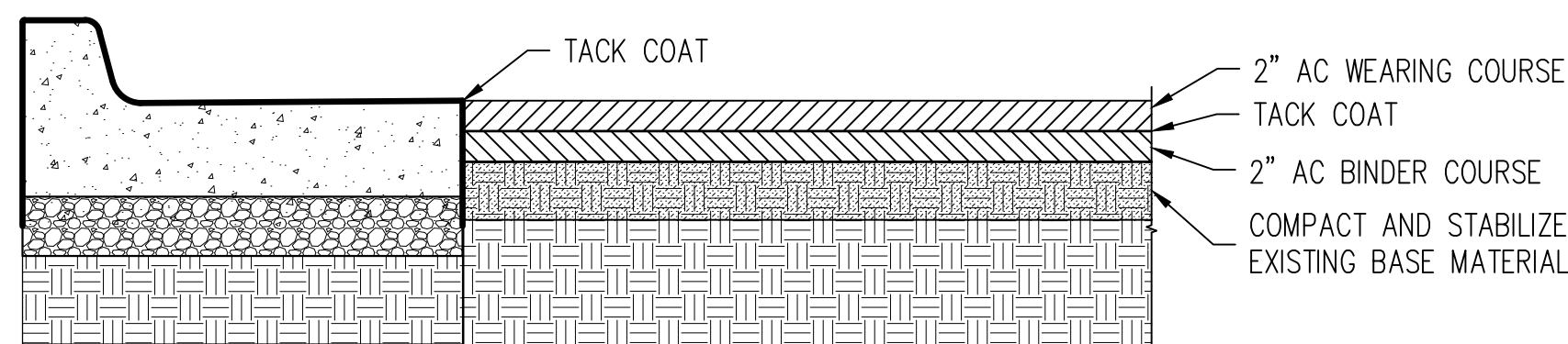
NOTE:
1. MILL 4" EXISTING ASPHALT



J ASPHALTIC CONCRETE (A.C.) MILL AND REPAIR - TYP
C501 1" = 1'-0"



K ROADWAY PAVEMENTS BASE REPAIR - TYP
C501 1-1/2" = 1'-0"



L ASPHALTIC CONCRETE (A.C.) FULL DEPTH REPAIR - TYP
C501 1" = 1'-0"



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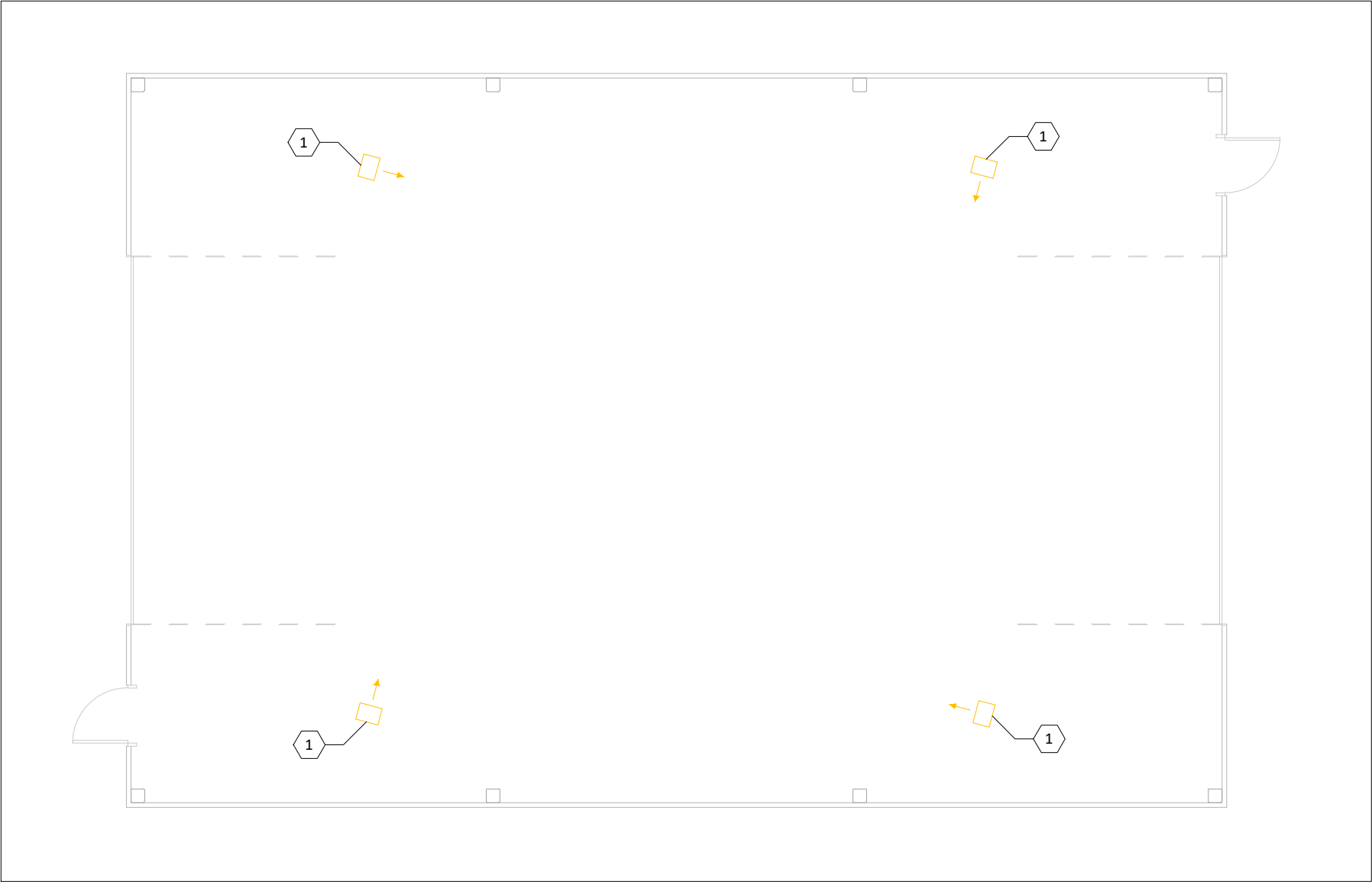
DATE					
DESCRIPTION					
REVISION #					

DRAWN BY: JO
CHECKED BY:
FP NUMBER: NA
DATE: 5/21/24
DESIGN STAGE: 100%

ROADWAY PAVEMENT DETAILS

C501

APPENDIX C



1 FLOOR PLAN
1/8" = 1'-0"



GENERAL NOTES

- 1. CONTRACTOR IS TO CONFIRM ALL DIMENSIONS.
- 2. CONTRACTOR IS TO INSTALL ALL EQUIPMENT IN STRICT ACCORDANCE WITH MANUFACTURE'S RECOMMENDATIONS.
- 3. ALL WORK, MATERIALS, AND EQUIPMENT SHALL BE IN ACCORDANCE WITH APPLICABLE STATE AND LOCAL CODES.
- 4. REFER TO STATEMENT OF WORK FOR DETAILS AND BASIS'S OF DESIGN.
- 5. ENSURE UNIT HEATERS ARE NOT LOCATED WITHIN 3 FEET OF ANY INSULATION, WALLS, OR CEILINGS AND WILL NOT CREATE OBSTRUCTIONS WITH PLANNED VEHICLE PARKING. REFER TO SHEET A101 FOR PLANNED VEHICLE PARKING.
- 6. BUILT-IN THERMOSTATS ARE TO BE SET TO MAINTAIN AN INDOOR TEMPERATURE OF 45°F. IF THERMOSTAT ALLOWS, CONFIGURE THERMOSTAT TO TURN THE HEATERS ON WHEN INDOOR TEMPERATURE FALLS BELOW 40°F AND TURN THE HEATERS OFF WHEN INDOOR TEMPERATURE IS ABOVE 50°F.

KEYNOTES

- 1** 4 KW, 240V SINGLE PHASE, FORCED-AIR ELECTRIC UNIT HEATERS TO PROVIDE FREEZE PROTECTION FOR FACILITY. UNIT HEATERS ARE TO BE SUSPENDED FROM ROOF JOISTS.



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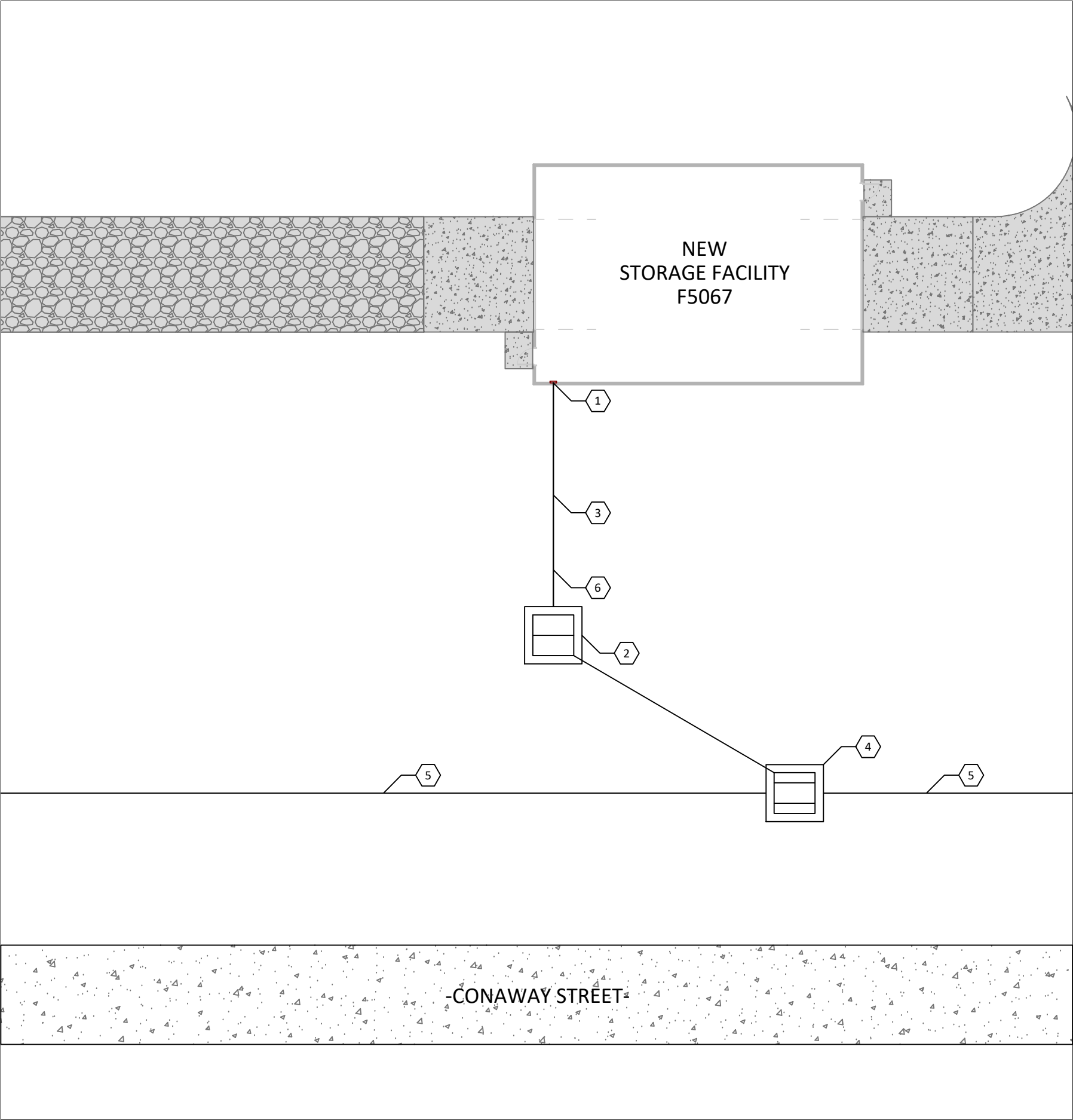


140 CHANNEL STREET
SUITE 138
VANCE AIR FORCE BASE
ENID, OKLAHOMA
580.213.7076

REVISION #	DESCRIPTION	DATE

DRAWN BY: DKM
CHECKED BY: BS
FP NUMBER: FP1024054
DATE: 28APR2026
DESIGN STAGE: 100%

CONSTRUCT METAL
BUILDING, F5067
MECHANICAL PLAN



GENERAL NOTES

- 1. DRAWING IS SCHEMATIC IN NATURE.
- 2. CONTRACTOR IS TO CONFIRM ALL DIMENSIONS.
- 3. CONTRACTOR IS TO INSTALL ALL EQUIPMENT IN STRICT ACCORDANCE WITH MANUFACTURE'S RECOMMENDATIONS.
- 4. ALL WORK, MATERIALS, AND EQUIPMENT SHALL BE IN ACCORDANCE WITH APPLICABLE STATE AND LOCAL CODES.
- 5. REFER TO STATEMENT OF WORK FOR DETAILS AND BASIS OF DESIGN.

KEYNOTES

- 1 NEW "PANEL A" SURFACE MOUNTED SQUARE D (QO) PANEL - 1200A, 240V 1-PHASE - 30 SPACE, with 175A MAIN BREAKER, LOAD CENTER. REFER TO PANEL SCHEDULE IN E701 FOR BREAKER DETAILS.
- 2 TRANSFORMER T-5067, 37.5 kVA, 7200V, 1-PHASE 240/120V TRANSFORMER TO BE INSTALLED BY OWNER PRIOR TO CONSTRUCTION.
- 3 CONTRACTOR TO INSTALL TWO, 2-INCH SCHEDULE 40 PVC CONDUIT, PULL THREE, #2/0 COPPER WIRE, AND PULL ONE, #4 GROUNDING WIRE TO CONNECT AND POWER "PANEL A" VIA 37.5 kVA TRANSFORMER. INSTALL PULL STRING IN SPARE CONDUIT.
- 4 EXISTING SWITCH, SW2-810.
- 5 12.47kV DISTRIBUTION LINE.
- 6 CONTRACTOR TO CONNECT ONTO STUBBED OUT 2-INCH PVC CONDUITS FOR FEEDING POWER TO F5067



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SUITE 138
VANCE AIR FORCE BASE
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580.213.7076

REVISION #	DESCRIPTION	DATE

DRAWN BY: ZM
CHECKED BY: RS
FP NUMBER: FP1024054
DATE: 28APR2026
DESIGN STAGE: 100%

CONSTRUCT METAL
BUILDING, F5067
ELECTRICAL SITE PLAN



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140 CHANNEL STREET
SUITE 138
VANCE AIR FORCE BASE
ENID, OKLAHOMA
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REVISION #	DESCRIPTION	DATE

DRAWN BY: ZM
CHECKED BY: RS
FP NUMBER: FP1024054
DATE: 29APR2026
DESIGN STAGE: 100%

CONSTRUCT METAL
BUILDING, F5067
ELECTRICAL POWER PLAN

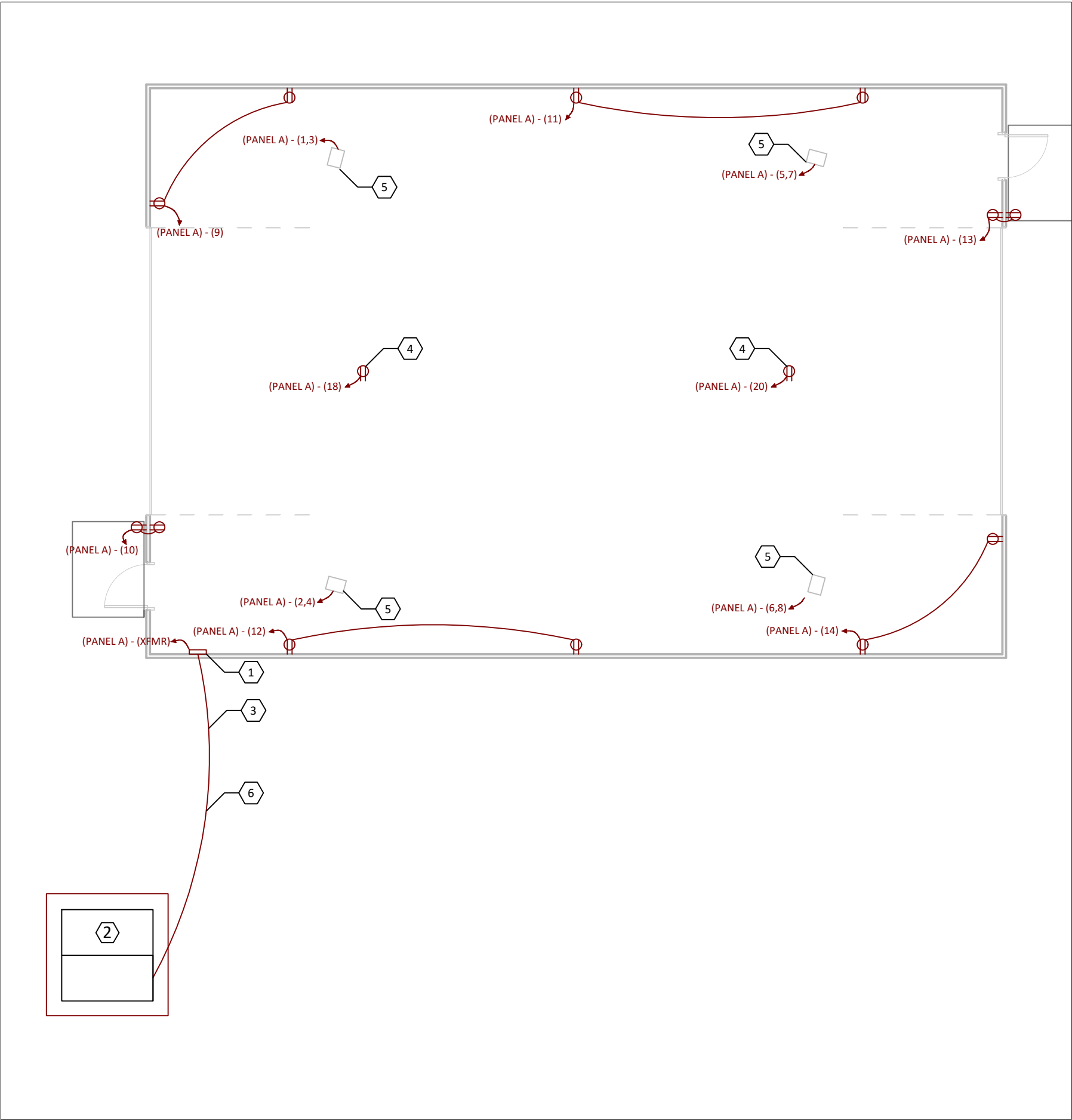
E101
APPENDIX C

GENERAL NOTES

- DRAWING IS SCHEMATIC IN NATURE.
- CONTRACTOR IS TO CONFIRM ALL DIMENSIONS.
- CONTRACTOR IS TO INSTALL ALL EQUIPMENT IN STRICT ACCORDANCE WITH MANUFACTURE'S RECOMMENDATIONS.
- ALL WORK, MATERIALS, AND EQUIPMENT SHALL BE IN ACCORDANCE WITH APPLICABLE STATE AND LOCAL CODES.
- REFER TO STATEMENT OF WORK FOR DETAILS AND BASIS OF DESIGN.
- ALL RECEPTACLES ARE TO BE 20A, DUPLEX RECEPTACLES.

KEYNOTES

- NEW "PANEL A" SURFACE MOUNTED SQUARE D (QO) PANEL - 200A, 240V 1-PHASE - 30 SPACE, WITH 175A MAIN BREAKER, LOAD CENTER. REFER TO PANEL SCHEDULE IN E701 FOR BREAKER DETAILS.
- TRANSFORMER T-5067, 37.5 kVA, 7200V, 1-PHASE 240/120V TRANSFORMER TO BE INSTALLED BY OWNER PRIOR TO CONSTRUCTION.
- CONTRACTOR TO INSTALL TWO, 2-INCH SCHEDULE 40 PVC CONDUIT, PULL THREE, #2/0 COPPER WIRE, AND PULL ONE, #4 GROUNDING WIRE TO CONNECT AND POWER "PANEL A" VIA 37.5 kVA TRANSFORMER. INSTALL PULL STRING IN SPARE CONDUIT.
- CEILING MOUNTED RECEPTACLES TO POWER ROLL-UP DOOR OPENERS.
- CEILING MOUNTED ELECTRIC UNIT HEATERS. REFER TO SHEET M101 FOR HEATER DETAILS.
- CONTRACTOR TO CONNECT ONTO STUBS.





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VANCE AIR FORCE BASE
ENID, OKLAHOMA



140 CHANNEL STREET
SUITE 138
VANCE AIR FORCE BASE
ENID, OKLAHOMA
580.213.7076

REVISION #	DESCRIPTION	DATE

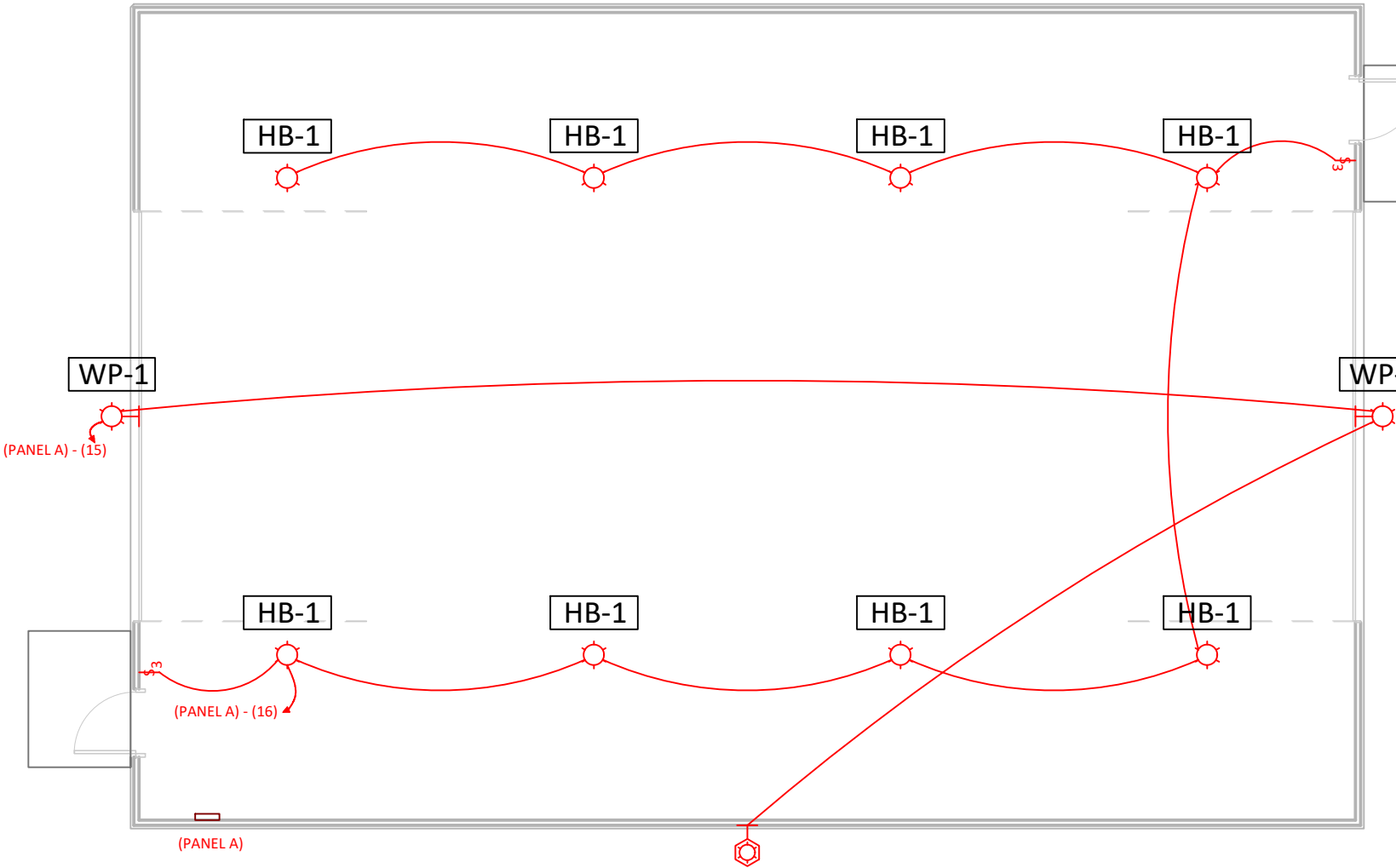
DRAWN BY: ZM
CHECKED BY: RS
FP NUMBER: FP1024054
DATE: 29APR2026
DESIGN STAGE: 100%

CONSTRUCT METAL
BUILDING, F5067
ELECTRICAL LIGHTING PLAN

E102
APPENDIX C

GENERAL NOTES

- DRAWING IS SCHEMATIC IN NATURE.
- CONTRACTOR IS TO CONFIRM ALL DIMENSIONS.
- CONTRACTOR IS TO INSTALL ALL EQUIPMENT IN STRICT ACCORDANCE WITH MANUFACTURE'S RECOMMENDATIONS.
- ALL WORK, MATERIALS, AND EQUIPMENT SHALL BE IN ACCORDANCE WITH APPLICABLE STATE AND LOCAL CODES.
- REFER TO STATEMENT OF WORK FOR DETAILS AND BASIS OF DESIGN.
- LIGHT FIXTURES TO BE CHAIN HUNG 11-FEET FROM FLOOR ELEVATION.
- LIGHT FIXTURES TO HAVE 4-FEET MINIMUM METAL CLAD CABLE WHIPS FROM RAFTER MOUNTED JUNCTION BOXES.
- "HB-1" LIGHT FIXTURES INTERIOR LIGHTING.
- "WP-1" LIGHT FIXTURES EXTERIOR LIGHTING.



LEGEND

	PHOTOCELL
	LIGHT FIXTURE
	3-WAY SWITCH
	DUPLEX RECEPTACLE

LIGHT SCHEDULE

TYPE MARK	FAMILY	MOUNTING TYPE	MOUNTING LOCATION	SHAPE	SIZE	LAMP	LUMINOUS FLUX	INITIAL COLOR TEMPERATURE	FIXTURE VOLTAGE	APPARENT LOAD	COMMENTS	COUNT
HB-1	LED HIGH-BAY	CHAIN/PENDANT	CEILING	RECTANGULAR	14.4" x 11.5"	LED	9,100	4000K	120V	66VA	3-WAY SWITCHED	8
WP-1	LED WALL PACK	SURFACE MOUNTING	WALL MOUNTED	RECTANGULAR	14.2" x 9.3"	LED	13,100	4000K	120V	104VA	PHOTOCELL CONTROLLED	2



Branch Panel: NEW MDP (QO)					QU130M200PQ										
Location: 5067					Volts: 120/240				A.I.C. Rating: 22 kA						
Supply From: 37.5kVA 240/120 TRANSFORMER					Phases: 1				Mains Type: MCB						
Mounting: Surface					Wires: 3				Bus Rating: 200 A						
Enclosure: NEMA 1									MCB Rating: 175 A						
Notes: ALL NEW PANEL & BREAKERS															
ALL receptacles are to be terminated into 20A, GFCI protected circuit breakers.															
CKT	Circuit Description	Trip	Poles	Notes	A		B		Notes	Poles	Trip	Circuit Description	CKT		
1	HEATER #1	25	2		2000	2000				2	25	HEATER #2	2		
3							2000	2000					4		
5					2000	2000							6		
7	HEATER #3	25	2				2000	2000		2	25	HEATER #4	8		
9	NORTHWEST RECEP'S	20	1	1	360	360			1	1	20	SOUTHWEST RECEP'S	10		
11	NORTH RECEP'S	20	1	1			360	360	1	1	20	SOUTH RECEP'S	12		
13	NORTH EAST RECEP'S	20	1	1	360	360			1	1	20	SOUTHEAST RECEP'S	14		
15	EXTERIOR LIGHTING	20	1				207	525		1	20	INTERIOR LIGHTING	16		
17	0	0	0		0	1500			1	1	20	GARAGE D.O. #1	18		
19	0	0	0				0	1500	1	1	20	GARAGE D.O. #2	20		
21	0	0	0		0	0				0	0	0	22		
23	0	0	0				0	0		0	0	0	24		
25	0	0	0		0	0				0	0	0	26		
27	0	0	0				0	0		0	0	0	28		
29	0	0	0		0	0				0	0	0	30		
Total Load:					10940		10952								
Total Amps:					91		91								
Notes Legend:										Panel Totals					
1 20A SINGLE-POLE GFCI CIRCUIT BREAKER										Total Connected Load:		21892		VA	
										Total Connected Current:		91		A	
										Total Estimated Demand:		21892		VA	
										Total Estimated Demand Current:		91		A	

FEEDER SCHEDULE											
AMPERAGE	SETS OF CONDUIT	CONDUIT SIZE	CONDUCTOR QTY/SIZE	SELECT ONE		AMPERAGE	SETS OF CONDUIT	CONDUIT SIZE	CONDUCTOR QTY/SIZE	SELECT ONE	
				EQUIPMENT GROUND QTY/SIZE	SUSE RATED GROUND QTY/SIZE					EQUIPMENT GROUND QTY/SIZE	SUSE RATED GROUND QTY/SIZE
15A	(1)	3/4"	(4) #12	(1) #12	(1) #8	225A	(1)	2 1/2"	(4) #4/0	(1) #4	(1) #2
20A	(1)	3/4"	(4) #12	(1) #12	(1) #8	250A	(1)	3"	(4) #250 KCMIL	(1) #4	(1) #2
25A	(1)	3/4"	(4) #10	(1) #10	(1) #8	300A	(2)	2"	(4) #1/0	(1) #4	(1) #2
30A	(1)	3/4"	(4) #10	(1) #10	(1) #8	400A	(2)	2 1/2"	(4) #3/0	(1) #3	(1) #1/0
40A	(1)	3/4"	(4) #8	(1) #10	(1) #8	450A	(2)	2 1/2"	(4) #4/0	(1) #2	(1) #2/0
45A	(1)	1"	(4) #6	(1) #10	(1) #8	500A	(2)	3"	(4) #250 KCMIL	(1) #2	(1) #2/0
50A	(1)	1"	(4) #6	(1) #10	(1) #8	600A	(2)	3"	(4) #350 KCMIL	(1) #1	(1) #3/0
60A	(1)	1 1/4"	(4) #4	(1) #10	(1) #8	800A	(3)	3"	(4) #300 KCMIL	(1) #1/0	(1) #3/0
70A	(1)	1 1/4"	(4) #4	(1) #8	(1) #8	1000A	(3)	3 1/2"	(4) #400 KCMIL	(1) #2/0	(1) #3/0
80A	(1)	1 1/2"	(4) #3	(1) #8	(1) #8	1200A	(4)	3 1/2"	(4) #350 KCMIL	(1) #3/0	(1) #3/0
90A	(1)	1 1/2"	(4) #2	(1) #8	(1) #8	1600A	(5)	3 1/2"	(4) #400 KCMIL	(1) #4/0	(1) #3/0
100A	(1)	2"	(4) #1	(1) #8	(1) #6	2000A	(6)	3 1/2"	(4) #400 KCMIL	(1) #250 KCMIL	(1) #3/0
125A	(1)	2"	(4) #1	(1) #6	(1) #6	2500A	(7)	3 1/2"	(4) #500 KCMIL	(1) #350 KCMIL	(1) #3/0
150A	(1)	2"	(4) #1/0	(1) #6	(1) #6	3000A	(8)	3 1/2"	(4) #500 KCMIL	(1) #400 KCMIL	(1) #3/0
175A	(1)	2"	(4) #2/0	(1) #6	(1) #4	3500A	(10)	3 1/2"	(4) #500 KCMIL	(1) #500 KCMIL	(1) #3/0
200A	(1)	2 1/2"	(4) #3/0	(1) #6	(1) #4	4000A	(10)	4"	(4) #600 KCMIL	(1) #500 KCMIL	(1) #3/0

FEEDER SIZE GENERAL NOTES:

1. CONDUCTOR QUANTITY BASED ON 3-PHASE, 4-WIRE; FOR EQUIPMENT THAT DOES NOT REQUIRE A NEUTRAL OR IS SINGLE PHASE DEDUCT FROM QUANTITY AS REQUIRED.
2. CONDUCTOR SIZES BASED ON NEC TABLE 310.16 - COPPER
60°C UP TO 100A, 75°C GREATER THAN 100A.
3. GROUND SIZES: EQUIPMENT GROUND BASED ON NEC TABLE 250.122 - COPPER, SUSE RATED GROUND BASED ON NEC TABLE 250.66 - COPPER
4. CONDUIT FILL BASED ON NEC ANNEX C - THW CONDUCTOR INSULATION

GENERAL NOTES

1. CONTRACTOR IS TO CONFIRM ALL DIMENSIONS.
2. CONTRACTOR IS TO INSTALL ALL EQUIPMENT IN STRICT ACCORDANCE WITH MANUFACTURE'S RECOMMENDATIONS.
3. ALL WORK, MATERIALS, AND EQUIPMENT SHALL BE IN ACCORDANCE WITH APPLICABLE STATE AND LOCAL CODES.
4. REFER TO STATEMENT OF WORK FOR DETAILS AND BASIS OF DESIGN.



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TRAINING COMMAND
VANCE AIR FORCE BASE
ENID, OKLAHOMA



140 CHANNEL STREET
SUITE 138
VANCE AIR FORCE BASE
ENID, OKLAHOMA
580.213.7076

DATE				
DESCRIPTION				
REVISION #				

DRAWN BY: ZM

CHECKED BY: RS

FP NUMBER: FP1024054

DATE: 29APR2026

DESIGN STAGE: 100%

CONSTRUCT METAL
BUILDING, F5067

ELECTRICAL SCHEDULES

E701

APPENDIX C

APPENDIX D – SCHEDULE E

APPENDIX D

Schedule E																
FP1024054 - Construct Metal Building											DATE: May 2026					
											To Be Completed by ASRC/CEEC					
	Item or Description of Item	Product Data	Shop Drawings	Samples	O&M	MFG REC	COC	WTY	SDS	MISC	Date Received.	Date Approved	Date Disapproved	Date Resubmitted	Date Approved	Date Disapproved
1	Project Schedule															
2	Schedule of Values															
3	Warranties															
4	O&M Manuals															
5	Hazmat/SDS															
6	Inventory of Installed Property															
7	As-Builts															
8	1.04.C.2.a - Stamped Engineering Drawings for Foundation Plans & Metal Building Plans															
9	1.04.C.2.c - P.C. Concrete															
10	1.04.C.2.c - Base Aggregate															
11	1.04.C.2.d - Steel Pipe Bollards															

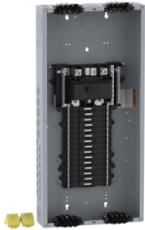
APPENDIX D

Schedule E																
FP1024054 - Construct Metal Building											DATE: May 2026					
	Item or Description of Item	Product Data	Shop Drawings	Samples	O&M	MFG REC	COC	WTY	SDS	MISC	Date Received.	Date Approved	Date Disapproved	Date Resubmitted	Date Approved	Date Disapproved
12	1.04.C.2.e - Metal Panel Materials and Color Samples															
13	1.04.C.2.e - Framing Materials															
14	1.04.C.2.e - Gutters and Downspouts Materials and Color Samples															
15	1.04.C.2.e.1.c.1 - 3'x7' Personnel Door and Door Hardware															
16	1.04.C.2.e.1.c.2 - 10' tall x 20' wide Overhead Door and Door Hardware															
17	1.04.C.2.e.1.c.2 - Overhead Door Electric Opener															
18	1.04.C.2.f- Spray-foam Insulation															
19	1.04.C.2.g.1 - Main Distribution Panel and Breakers															
20	1.04.C.2.g.2 - Underground Wiring and Conduit															
21	1.04.C.2.g - Building Wiring and Conduit															

APPENDIX D

Schedule E																
FP1024054 - Construct Metal Building											DATE: May 2026					
											To Be Completed by ASRC/CEEC					
	Item or Description of Item	Product Data	Shop Drawings	Samples	O&M	MFG REC	COC	WTY	SDS	MISC	Date Received.	Date Approved	Date Disapproved	Date Resubmitted	Date Approved	Date Disapproved
22	1.04.C.2.g.3 - Interior and Exterior Lighting															
23	1.04.C.2.g.3 - Photocell for Exterior Lighting															
24	1.04.C.2.g.3 - Wall and Ceiling Outlets															
25	1.04.C.2.h - Forced-air Electric Unit Heaters															
26	1.04.C.2.i - Fire Extinguishers and Cabinets															

APPENDIX E – BASIS OF DESIGN DETAILS



Load center, QO, 1 phase, 30 spaces, 30 circuits, 200A convertible main breaker, PoN, NEMA1, Qwik-grip

QO130M200PQ

Product availability: Stock - Normally stocked in distribution facility

Main

Marketing Trade Name	QO
Product Type	Load Center
Provided Equipment	Qwik grip assembly kit 1)

Complementary

Tightening Torque	Cover 20 lb.in
PoN Convertible Mains (lugs)	PoN Convertible Mains (breaker)
Rated Current	200 A
Number of Spaces	30
Max Short Circuit Current Rating	22 kA
Maximum Number of Single Pole Circuits	40
Maximum Number of Tandem Breakers	10
Number of Phases	1 phase
Voltage Rating	120/240 V AC
Wire Size	AWG 4/0...250 kcmil aluminium/copper
Ground Bar	Grounding bar (ordered separately)
Busbar Material	Copper: busbar
Enclosure Material	Welded sheet steel
Surface Finish	Baked enamel Gray
Box Number	9Q
Height	29.8 in (758 mm)
Width	14.3 in (362 mm)
Depth	0.006 in (0.15 mm)
Product Weight	12.610 lb(US) (5.720 kg)

Environment

Enclosure Rating	NEMA 1 indoor enclosure
Ambient Air Temperature for Operation	23 °F (-5 °C) 104 °F (40 °C)
Product Certifications	UL Listed

Price is "List Price" and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Ordering and shipping details

Category	US1DE3A10001
Discount Schedule	DE3A
GTIN	785901777915
Returnability	Yes
Country of origin	US

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	31.700 in (80.518 cm)
Package 1 Width	14.700 in (37.338 cm)
Package 1 Length	4.50 in (11.43 cm)
Package weight(Lbs)	18.89 lb(US) (8.57 kg)

Contractual warranty

Warranty (in months)	18
----------------------	----



Environmental Data

Schneider Electric aims to achieve Net Zero status by 2050 through supply chain partnerships, lower impact materials, and circularity via our ongoing “Use Better, Use Longer, Use Again” campaign to extend product lifetimes and recyclability.

[Environmental Data explained >](#)

[How we assess product sustainability >](#)



Environmental footprint

Environmental Disclosure

[Product Environmental Profile](#)

Use Better



Materials and Substances

Packaging made with recycled cardboard

No

Packaging without single use plastic

No

EU RoHS Directive

[Compliant By Exemption](#)

REACH Regulation

[Reference contains Substances of Very High Concern above the threshold](#)

California proposition 65

WARNING: This product can expose you to chemicals including: Lead and lead compounds, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to [www.P65Warnings.ca.gov](#)

Use Longer



Lifetime extension

Repair

No

Use Again



Repack and remanufacture

Circularity Profile

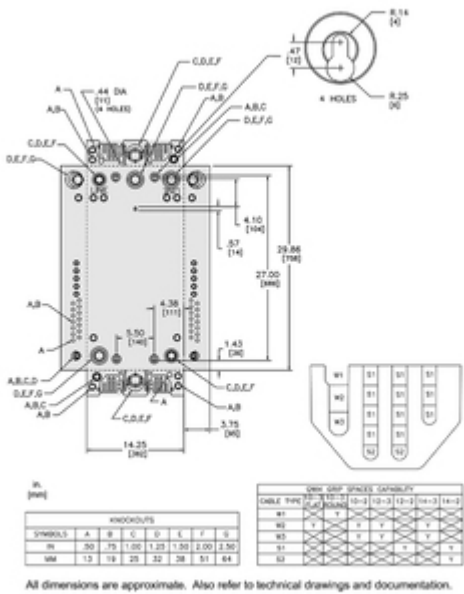
No need of specific recycling operations

Take-back

No

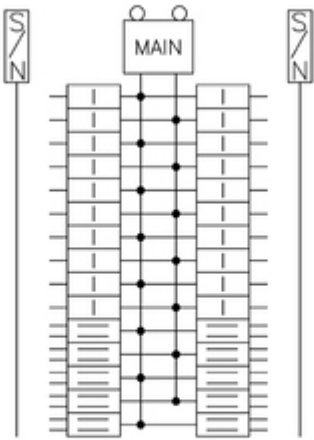
Technical Illustration

Dimensions



Technical Illustration

Wiring diagram



QO130M200PQ



Cover, QO, for 30 space load center, surface, gray

QOC30US

Product availability: Stock - Normally stocked in distribution facility

Main

Marketing Trade Name	QO
Product Type	Load Center Cover

Complementary

Cover Type	Surface cover
Product Compatibility	For indoor load center
Surface Finish	Baked enamel Gray
Number of Spaces	30
Height	29.86 in (758.44 mm)
Width	14.24 in (361.70 mm)
Depth	0.7 in (17.8 mm)
Tightening Torque	Cover 2.3 N.m

Ordering and shipping details

Category	US1DE3A00001
Discount Schedule	DE3A
GTIN	785901746034
Returnability	Yes
Country of origin	US

Packing Units

Unit Type of Package 1	PCE
Nbr. of units in pkg.	1
Package 1 Height	1.300 in (3.302 cm)
Package 1 Width	14.600 in (37.084 cm)
Package 1 Length	30.100 in (76.454 cm)
Package weight(Lbs)	9.050 lb(US) (4.105 kg)
Unit Type of Package 2	PAL
Number of Units in Package 2	81
Package 2 Height	40.000 in (101.600 cm)
Package 2 Width	40.000 in (101.600 cm)
Package 2 Length	48.000 in (121.920 cm)

Price is "List Price" and may be subject to a trade discount – check with your local distributor or retailer for actual price.

Disclaimer: This documentation is not intended as a substitute for and is not to be used for determining suitability or reliability of these products for specific user applications

Package 2 Weight	716.040 lb(US) (324.790 kg)
------------------	-----------------------------

Contractual warranty

Warranty (in months)	18
----------------------	----



Environmental Data

Schneider Electric aims to achieve Net Zero status by 2050 through supply chain partnerships, lower impact materials, and circularity via our ongoing “Use Better, Use Longer, Use Again” campaign to extend product lifetimes and recyclability.

[Environmental Data explained >](#)

[How we assess product sustainability >](#)



Environmental footprint

Total lifecycle Carbon footprint	22 kg CO2 eq.
Carbon footprint of the manufacturing phase [A1 to A3]	21 kg CO2 eq.
Carbon footprint of the distribution phase [A4]	1 kg CO2 eq.
Carbon footprint of the installation phase [A5]	0 kg CO2 eq.
Carbon footprint of the use phase [B2, B3, B4, B6]	0.1 kg CO2 eq.
Carbon footprint of the end-of-life phase [C1 to C4]	0.4 kg CO2 eq.
Environmental Disclosure	Product Environmental Profile

Use Better



Materials and Substances

Packaging made with recycled cardboard	No
Packaging without single use plastic	No
EU RoHS Directive	Compliant
REACH Regulation	Free of Substances of Very High Concern above the threshold
California proposition 65	WARNING: This product can expose you to chemicals including: Lead and lead compounds, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information go to www.P65Warnings.ca.gov

Use Longer



Lifetime extension

Repair	No
--------	----

Use Again



Repack and remanufacture

Circularity Profile	No need of specific recycling operations
Take-back	No
WEEE Label	 The product must be disposed on European Union markets following specific waste collection and never end up in rubbish bins.

Technical Illustration

Dimensions

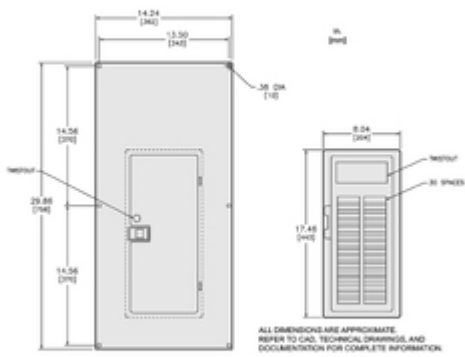


Image of product / Alternate images

Alternative

MFG Model Number
QOC30US



Indoor, surface mount replacement electrical panel cover



Compatible with Square D QO electrical panels*



UL listed and ANSI Certified



* Single-phase

- 100A: 20, 24, 30 spaces
- 200A: 12, 20, 24, 30 spaces

Three-phase

- 100A: 27 spaces (Convertible mains)
- 200A: 18 and 30 spaces (Fixed mains)



MFG Model Number
QOC30US



29.86 in

14.24 in

0.7 in

↑

Dimensions

6

Life Is On | Schneider Electric

Apr 9, 2026

Compatible products

QO 30-Space Indoor Load Center Surface Mount Cover
MFG Model Number: QOC30US





Compatible for use with (150A, 200A) (20, 24, 30 space) QO Indoor Load Center for surface mount installations



Building Details

Building Type: ☒ RF ☐ SS ☐ LT
 Width: 40.000' Peak Offset: 20.000'
 Length: 60.000'
 Sidewall Bay Spacing 3 @ 20.0000'

Eave Ht 12.000' Roof Slope 2.000 in 12 Girt Type Flush
 Front Side: 12.000' Back Side: 12.000' 2.000 in 12 Flush

Frame ID	Frame Type*	Col Type*	Rafter Type*	Frame Line	# Int Col's
<u>1</u>	<u>Rigid Frame</u>	<u>Wide-Flange</u>	<u>Wide-Flange</u>	<u>1</u>	<u>-</u>
<u>2</u>	<u>Rigid Frame</u>	<u>Tapered</u>	<u>Tapered</u>	<u>2 3</u>	<u>-</u>
<u>3</u>	<u>Rigid Frame</u>	<u>Wide-Flange</u>	<u>Wide-Flange</u>	<u>4</u>	<u>-</u>

* May change due to engineering requirements

Building Code

Design Code: IBC'18
 Closed/Open: C
 Exposure: C
 Importance - Wind: 1.00
 Site Class: D
 Importance - Seismic: 1.00
 Seismic Coefficient: 0.32
 Importance - Snow: 1.00

Building Loads:

Dead Load: 2.50 psf
 Live Load: 20.00 psf
 Load Reduction: Yes
 Ground Snow: 15.00 psf
 Collateral: 3.00 psf
 Wind Load: 111.00 mph

Other Loads:

Crane Load? ☐ Yes ☒ No
 Floor Load? ☐ Yes ☒ No
 Parapet / Mansard? ☐ Yes ☒ No

Stepped elevations or structures within 20 feet? ☐ Yes ☒ No

Building Use Classification: **Standard Building**
 Description of building use: **Shop**

Frame Coating**Main Columns & Rafters**

- ☒ Red oxide
- ☐ Hot-Dipped Galv'd

Purlins, Girts, Eave Struts

- ☒ Red oxide
- ☐ Pre-Galvanized
- ☐ Hot-Dipped Galv'd

Door Framing

- ☒ Red oxide
- ☐ Pre-Galvanized
- ☐ Hot-Dipped Galv'd

Base Angle

- ☒ Red oxide
- ☐ Pre-Galvanized
- ☐ Hot-Dipped Galv'd

End Frames

	Left	Right
Expandable	☐	☐
Non-Exp RF	☒	☒
Std. Endwall	☐	☐
CF Endwall	☐	☐

Base Condition

- ☒ Angle
- ☒ Trim
- ☐ Channel
- ☐ Girt
- ☐ GZ Base Fixture

Wall Bracing

Roof:
 Left Endwall:
 Front Sidewall:
 Right Endwall:
 Back Sidewall:

Bracing Details

Diagonal Bracing
Rigid Frame
Diagonal Bracing
Rigid Frame
Diagonal Bracing

Framed Openings

Note: Bolted Clips

Wall	Bay	Open Width	Header Ht	Sill Ht	Offset
<u>Left Endwall</u>	<u>1</u>	<u>20.000'</u>	<u>10.000'</u>	<u>0.000'</u>	<u>10.000'</u>
<u>Right Endwall</u>	<u>1</u>	<u>20.000'</u>	<u>10.000'</u>	<u>0.000'</u>	<u>10.000'</u>

Purlin Extensions

Wall	Surface	Length	Soffit
<u>None</u>			

Eave Extensions and Canopies

Wall	Bay St	Bay End	Height	Width	Slope	Soffit	L Ext	R Ext
<u>None</u>								

Open walls:

Left Endwall: ☐ Yes ☒ No Right Endwall: ☐ Yes ☒ No
 Front Sidewall: ☐ Yes ☒ No Back Sidewall: ☐ Yes ☒ No

Partial Walls and Wainscot:

<u>Wall</u>	<u>Wainscot</u>	<u>Bay Start</u>	<u>Bay End</u>	<u>Open Height</u>	<u>Base</u>	<u>Full Load</u>
None						

Liner Panel

<u>Location</u>	<u>Start</u>	<u>End</u>	<u>Height</u>
None			

Sheets & Trim

<u>Location</u>	<u>Color</u>	<u>Panel</u>	<u>Gauge</u>	<u>Quality</u>	<u>Trim</u>	
Roof	Burnished Slate	PBR	26	30 Yr.	Gable:	Burnished Slate
Walls	Light Stone	PBR	26	30 Yr.	Eave:	Burnished Slate
Soffit	---				Corner:	Burnished Slate
Roof Liner	---				Jamb:	Burnished Slate
Wall Liner	---				Liner:	---
Wainscot	---				Wainscot:	---

Fasteners

<u>Roof</u>		<u>Walls</u>		<u>Soffit</u>	
Panel:	TEK1.25	Panel:	TEK1.25	Panel:	--
Lap:	LAPTEK	Lap:	LAPTEK	Lap:	--
Type:	Painted	Type:	Painted	Type:	--

Ridge Covering

- Peak Sheets
- Ridge Roll

Eave Condition**Front Sidewall**

- Gutter & 4 Downspouts
- Eave Trim

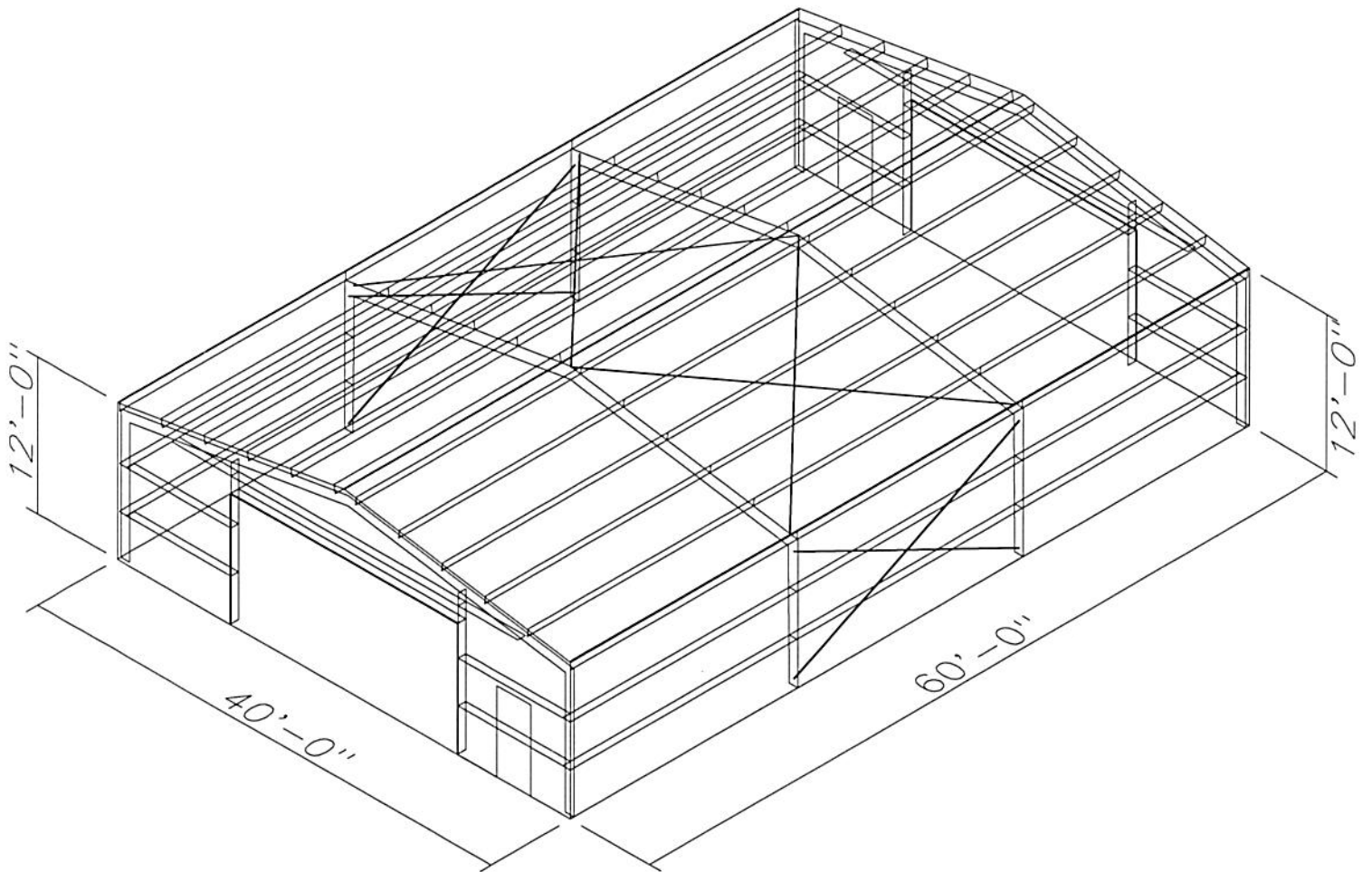
Back Sidewall

- Gutter & 4 Downspouts
- Eave Trim

Accessories

<u>Item Description</u>	<u>Quantity</u>
Alt 1: Wkd 3070 Windstorm 8 1/4" 140m	2
Total Main Bldg	

Special Requirements:



Heavy Duty Unit Heater KB Series

APPENDIX E



Model Code:

KB	48	10	-1	-T	-B1
A	B	C	D	E	F

A: Series

B: 20 - 208V, 24 - 240/208V
27 - 277V, 48 - 480V

C: kiloWatts (4-66)

D: -1 = Single Phase

-3 = Three Phase

E: -T = Built-In Thermostat

F: -B1 = Bracket



- Spiral finned element
- Long life unit bearing motor
- Manual limit
- Aluminum fan blade
- Built-In Thermostat
- Standard color: gray
- High-temp safety limit
- Universal mounting bracket included
- Tested to UL standards for safety
- Adjustable louvers
- 5-year limited warranty
- Hinged access cover



The KB Horizontal Unit Heater

King's KB series unit heaters are designed for use as primary or supplemental heat sources in large or exposed areas, including high-rise buildings, stores, warehouses, garages, water treatment and fire room vaults to prevent freezing. Spiral finned steel elements and high CFM ensure even air distribution producing cooler element operation, prolonging heater life. By combining the highest quality components, with a dynamically balanced fan blade the KB unit heater will provide years of trouble-free service. Includes universal wall/ceiling bracket for simple installation and built-in thermostat for precision heating control.

Ordering Information - single phase

MODEL	UPC	KW	BTUH	PHASE	VOLTS	AMPS **THERMOSTAT CONTROL VOLTAGE	MOTOR HP	CFM	TEMP. RISE	AIR THROW	MAX. MOUNT HEIGHT	SIZE	WT. (lbs.)	
KB2003-1-T-B1	13294	3.3	11.1	1	208	15.8	208	9W	400	36°	15ft.	8ft.	A	30.5
KB2004-1-T-B1	15011	4	13.5	1	208	19.2	208	9W	400	38°	15ft.	8ft.	A	30.5
KB2404-1-T-B1	13356	4	13.5	1	240/208*	16.6/14.4	24	9W	400	38°	15ft.	8ft.	A	30.5
KB2704-1-T-B1	13360	4	13.5	1	277	14.4	277	9W	400	38°	15ft.	8ft.	A	30.5
KB4804-1-T-B1	15017	4	13.5	1	480	8.3	24	9W	400	38°	15ft.	8ft.	A	30.5
KB2005-1-T-B1	13368	5	17	1	208	24	208	9W	400	40°	16ft.	8ft.	A	30.5
KB2405-1-T-B1	13376	5	17	1	240/208*	20.8/18	24	9W	400	40°	16ft.	8ft.	A	30.5
KB2705-1-T-B1	13384	5	17	1	277	18.1	277	9W	400	40°	16ft.	8ft.	A	30.5
KB4805-1-T-B1	13390	5	17	1	480	10.4	24	9W	400	40°	16ft.	8ft.	A	30.5
KB2007-1-T-B2	30780	7.5	25.6	1	208	36.1	208	16W	600	40°	20ft.	9ft.	B	45.0
KB2407-1-T-B2	13436	7.5	25.6	1	240/208*	31.2/27	24	16W	600	40°	20ft.	9ft.	B	45.0
KB2707-1-T-B2	13464	7.5	25.6	1	277	27	277	16W	600	40°	20ft.	9ft.	B	45.0
KB4807-1-T-B2	30781	7.5	25.6	1	480	15.6	24	16W	600	40°	20ft.	9ft.	B	45.0
KB2010-1-T-B2	13540	10	34.1	1	208	48.1	208	25W	725	44°	24ft.	10ft.	B	45.0
KB2410-1-T-B2	13548	10	34.1	1	240/208*	41.6/36.1	24	25W	725	44°	24ft.	10ft.	B	45.0
KB4810-1-T-B2	13562	10	34.1	1	480	20.8	24	25W	725	44°	24ft.	10ft.	B	45.0
KB2012-1-T-B2	13618	12.5	42.7	1	208	60.1	208	35W	825	48°	30ft.	11ft.	C	46.5
KB2412-1-T-B2	13626	12.5	42.7	1	240/208*	52/45.1	24	35W	825	48°	30ft.	11ft.	C	46.5
KB2712-1-T-B2	30782	12.5	42.7	1	277	45.1	277	35W	825	48°	30ft.	11ft.	C	46.5
KB2015-1-T-B2	13674	15	51.2	1	208	72.1	208	50W	925	51°	33ft.	12ft.	C	46.5
KB2415-1-T-B2	13682	15	51.2	1	240/208*	62.5/54.1	24	50W	925	51°	33ft.	12ft.	C	46.5

*Dual rated heaters will draw 13% less amps and 25% less wattage when operated at 208V.

Maximum mounting height is manufacturer's recommendation

**Over 5kw line voltage control is operating the contactor coil only, and will draw under 1 amp.

Ordering Information - three phase

MODEL	UPC	KW	BTUH	PHASE	VOLTS	AMPS	**THERM. CONTROL VOLTAGE	MOTOR HP	CFM	TEMP. RISE	AIR THROW	MAX. MOUNT HEIGHT	*SIZE	WT. (lbs.)
KB4803-3MP-T-B2	13320	3	10.2	1-3	480	6.3	24	5W	400	37°	14ft.	8ft.	C	46.5
KB2005-3MP-T-B2	13372	5	17.1	1-3	208	24 & 13.9	208	9W	400	40°	16ft.	8ft.	C	46.5
KB2405-3MP-T-B2	13380	5	17.1	1-3	240/208*	20.8/18 & 18/15.6	24	9W	400	40°	16ft.	8ft.	C	46.5
KB4805-3MP-T-B2	13394	5	17.1	1-3	480	10.4 & 6	24	9W	400	40°	16ft.	8ft.	C	46.5
KB2007-3MP-T-B2	13452	7.5	25.6	1-3	208	36.1 & 20.8	208	16W	600	40°	20ft.	9ft.	C	46.5
KB2407-3MP-T-B2	13460	7.5	25.6	1-3	240/208*	31.2/27 & 18/15.6	24	16W	600	40°	20ft.	9ft.	C	46.5
KB4807-3MP-T-B2	13474	7.5	25.6	1-3	480	15.6 & 24	24	16W	600	40°	20ft.	9ft.	C	46.5
KB2010-3MP-T-B2	13544	10	34.1	1-3	208	27.7	208	25W	725	44°	24ft.	10ft.	C	46.5
KB2410-3MP-T-B2	13552	10	34.1	1-3	240/208*	41.6/24 & 36.1/20.8	24	25W	725	44°	24ft.	10ft.	C	46.5
KB4810-3MP-T-B2	13566	10	34.1	1-3	480	20.8 & 12	24	25W	725	44°	24ft.	10ft.	C	46.5
KB2012-3-T-B2	30783	12.5	42.7	3	208	34.6	208	35W	825	48°	30ft.	11ft.	C	46.5
KB2412-3-T-B2	30784	12.5	42.7	3	240/208*	30/26	24	35W	825	48°	30ft.	11ft.	C	46.5
KB4812-3MP-T-B2	13646	12.5	42.7	1-3	480	26 & 15	24	35W	825	48°	30ft.	11ft.	C	46.5
KB2015-3-T-B2	30785	15	51.2	3	208	41.6	208	50W	925	51°	33ft.	12ft.	C	46.5
KB2415-3-T-B2	30786	15	51.2	3	240/208*	36/31.2	24	50W	925	51°	33ft.	12ft.	C	46.5
KB4815-3MP-T-B2	13700	15	51.2	1-3	480	18	24	50W	925	51°	33ft.	12ft.	C	46.5
KB2020-3-T-B2	13736	20	68.3	3	208	55.5	208	¼HP	1100	57°	47ft.	13ft.	E	55.0
KB2420-3-T-B2	13740	20	68.3	3	240/208*	48.1/41.6	24	¼HP	1100	57°	47ft.	13ft.	E	55.0
KB4820-3MP-T-B2	13750	20	68.3	1-3	480	24	24	¼HP	1100	57°	47ft.	13ft.	E	55.0

6" minimum clearance to walls and ceilings.

Over 5KW line voltage control is operating the contactor coil only, and will draw under 1 AMP.

*Dual rated heaters will draw 13% less amps and 25% less wattage when operated at 208V.

Field Mount Accessories

MODEL	UPC	DESCRIPTION	WT. (lbs.)
DS32	13480	Disconnect Switch, Field Installed Kit, 32 Amp, 600V	2.0
DS63	13481	Disconnect Switch, Field Installed Kit, 63 Amp, 600V	2.0
DS80	13482	Disconnect Switch, Field Installed Kit, 80 Amp, 600V	2.0
DS100	13483	Disconnect Switch, Field Installed Kit, 100 Amp, 600V	5.0

Replacement Accessories

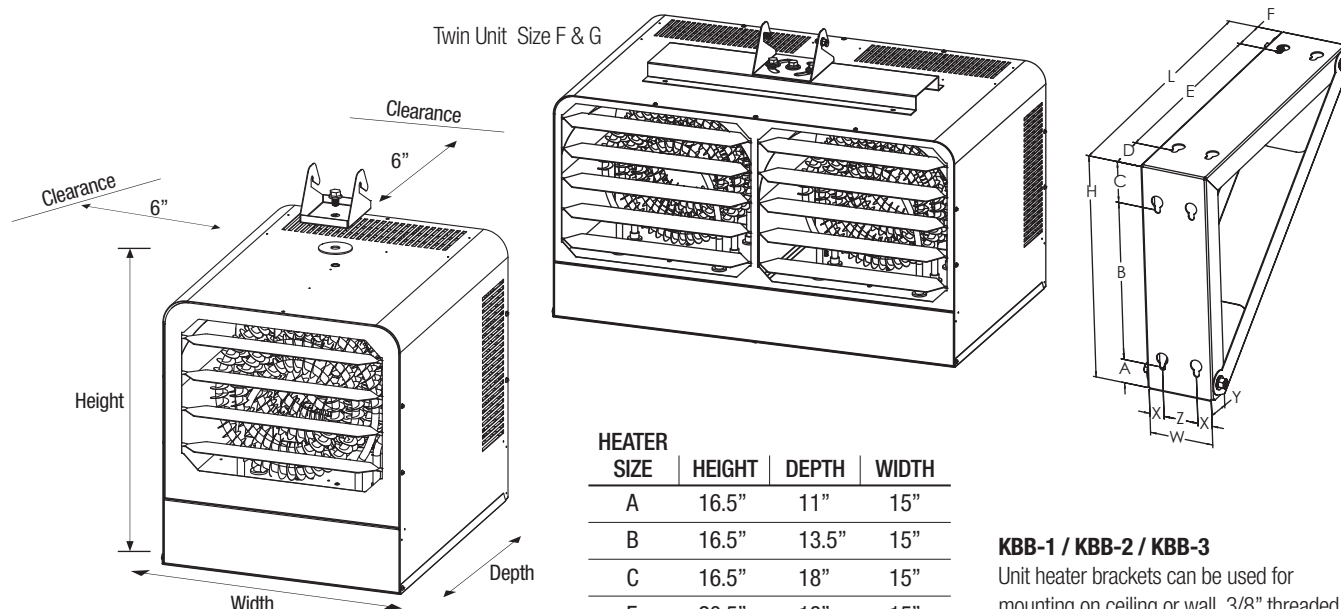
MODEL	UPC	DESCRIPTION	WT. (lbs.)
KBB-1	15192	Universal Wall/Ceiling Small Mounting Bracket (for heaters with size A heaters)	1.9
KBB-2	15195	Universal Wall/Ceiling Large Mounting Bracket (for heaters with size B, C, and E)	2.6
KBHK	13487	KB Replacement Hardware Kit	0.1
KBHS	13486	KB Replacement Hook Swivel	0.3

Bracket Selection Chart

BRACKET #	HEATER KW SIZE USED WITH	BRACKET DIMENSIONS											
		L	W	H	A	B	C	D	E	F	X	Y	Z
KBB-1	3 to 5 kW (1 phase)	12½"	2½/16"	10½/16"	1¾/16"	7"	1¾/8"	1¾/8"	9¼"	1½/8"	¾"	1"	½"
KBB-2	7.5 to 15 kW (1 phase) 3 to 20 kW (3 phase)	16¾/8"	2½/16"	10½/8"	1¼"	7"	1¾/8"	1½/8"	13½/8"	1½/8"	¾"	1"	1½"

Heavy Duty Unit Heater KB Series

Dimensional Data



HEATER SIZE	HEIGHT	DEPTH	WIDTH
A	16.5"	11"	15"
B	16.5"	13.5"	15"
C	16.5"	18"	15"
E	20.5"	18"	15"
F	18.5"	18"	30"
G	16.5"	18"	30"

KBB-1 / KBB-2 / KBB-3

Unit heater brackets can be used for mounting on ceiling or wall. 3/8" threaded hole on top of unit heater can also be used with threaded rod to drop the unit closer to the floor. Refer to bracket selection chart.

Engineering Specifications

Contractor shall supply and install KB Series unit heaters manufactured by King Electrical Mfg. Company. Heaters shall be of the wattage and voltage as indicated on the plans.

Construction: All exterior and interior metal enclosure parts are made from 20 gauge electrogalvanized Steel with a rust inhibiting baked enamel finish. Smooth rounded corners and protective edge trim give an attractive modern appearance.

Adjustable Outlet Louver: Louvers direct air up or down as needed for the heating application.

Rear Intake Screen: Heavy gauge steel screen protects against foreign objects making accidental contact with the rotating fan blade.

Spiral Fin Elements: The metal sheath element is copper brazed with spiral fins then molded into a coil configuration. This combination produces the best heat transfer while eliminating the potential for hot spots by positioning the element in the maximum airflow stream.

Totally Enclosed Fan Motor: Permanently lubricated long life, unit bearing with 20cc of oil. Epoxy-coated motor with enclosed rotor resists moisture and corrosion for long-lasting, trouble-free operation.

Aluminum Fan Blade: Axial flow-type fan mounted directly to the motor shaft for maximum efficiency.

Built-In Thermostat: Single pole factory installed hydraulic capillary tube thermostat for precision heating control. Operating range 40° to 90

Manual Thermal Cutout: Heaters shall be equipped with King's patented thermal overload Smart Limit Protection®, which disconnects elements and motor in the event normal operating temperatures are exceeded. If thermal overload trips due to abnormal operating temperatures, thermal overload shall remain open until manually reset by turning the heater off for 15 minutes. Automatically reset of thermal overloads, which allow the element to continue to cycle under abnormal conditions, will not be accepted. .

Internal Controls: Magnetic contactors are standard on all 480V heaters, all 208V/1-Phase, 240V/1-Phase, 277V/1-Phase heaters above 6kW, and all 3-Phase heaters. The control voltage is equal to the line voltage (208, 240 or 277) except for 480V models where a transformer is provided for 24V control. Over 5kW the line voltage control is operating the contactor coil only and will draw only 1 Amp.

Fusing: Internal circuit fusing is provided when the heater ampacity exceeds 48 Amps to comply with N.E.C. standards. A fan delay is provided standard on models 12.5-kW and above to dissipate residual heat from the heating elements.

Easy Installation: A hinged access panel located on the bottom of the heater allows for quick wiring and easy maintenance for the life of the heater. Includes universal wall/ceiling bracket for simple installation.

Approvals: cULus (E41422)



Catalog Number
Notes
Type

Contractor Select™

COMPACT PRO™

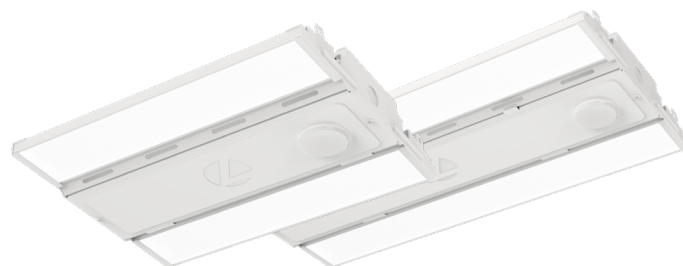
LED High Bay

The Compact Pro High Bay (CPHB) is a budget-oriented high bay designed specifically with the contractor in mind. Its compact design makes it easier and quicker to install. Compact Pro is built with quality to last and performance to meet the needs of the job, making it the best choice for affordable and reliable light-duty industrial applications like warehouses.

FEATURES:

- Compact design saves time and money during installation
- Innovative Glare Control lens meets new DLC 5.1 standards
- Robust 6kV surge protection standard per ANSI standards for Industrial environments
- Operates up to 55°C ambient
- Includes mounting hardware (V-hook & 36" hanger chain)
- 6ft cord included with ALO13 and ALO16 UVOLT fixtures
- Fixture has sensor receptacle with cover to allow for field installable sensor*
- Switchable lumen packages

*ALO13 & ALO16 only



PICTURED WITH SENSOR RECEPTACLE FOR USE WITH FIELD INSTALLABLE TLS SENSOR*



Catalog Number	UPC	Description	Lumens	Input Watts	Color Temperature	Color Rendering	Voltage	Distribution	Pallet Qty.
CPHB ALO22 MVOLT SWW9 80CRI DWH	00197589305388	14" LED High Bay	12000/15000	81/102	4000K/5000K	80CRI	120-277V	Medium	144
CPHB ALO13 MVOLT SWW9 80CRI DWH	00196183428561	14" LED High Bay	12000/15000/18000	83/103/125	4000K/5000K	80CRI	120-277V	Medium	168
CPHB ALO16 MVOLT SWW9 80CRI DWH	00196183428578	22" LED High Bay	24000/27000/30000	159/181/198	4000K/5000K	80CRI	120-277V	Medium	112
CPHB ALO13 UVOLT SWW9 80CRI DWH ‡	00197590826087	14" LED High Bay	12000/15000/18000	83/103/125	4000K/5000K	80CRI	120-347V	Medium	144
CPHB ALO16 UVOLT SWW9 80CRI DWH ‡	00197590826100	22" LED High Bay	24000/27000/30000	159/181/198	4000K/5000K	80CRI	120-347V	Medium	96
CPHB ALO34 UVOLT SWW9 80CRI DWH	198935978010	42" LED High Bay	36000/48000/60000	216/309/411	4000K/5000K	80CRI	120-347V	Medium	52
CPHB ALO13 MVOLT SWW9 80CRI TLS DWH ‡	00199871045125	14" LED High Bay	12000/15000/18000	83/103/125	4000K/5000K	80CRI	120-277V	Medium	168
CPHB ALO16 MVOLT SWW9 80CRI TLS DWH ‡	00199871045217	22" LED High Bay	24000/27000/30000	159/181/198	4000K/5000K	80CRI	120-277V	Medium	112
CPHB 12LM MVOLT 40K	194994608325	14" LED High Bay	12,261	88	4000 K	80 CRI	120-277V	Medium	204
CPHB 12LM MVOLT 50K	194994429548	14" LED High Bay	12,342	88	5000 K	80 CRI	120-277V	Medium	204
CPHB 15LM MVOLT 40K	194994429562	14" LED High Bay	14,857	104	4000 K	80 CRI	120-277V	Medium	204
CPHB 15LM MVOLT 50K	194994429593	14" LED High Bay	14,955	104	5000 K	80 CRI	120-277V	Medium	204
CPHB 18LM MVOLT 40K	194994608295	14" LED High Bay	18,364	134	4000 K	80 CRI	120-277V	Medium	204
CPHB 18LM MVOLT 50K	19499442963	14" LED High Bay	18,485	134	5000 K	80 CRI	120-277V	Medium	204
CPHB 24LM MVOLT 40K	194994429685	22" LED High Bay	24,890	174	4000 K	80 CRI	120-277V	Medium	102
CPHB 24LM MVOLT 50K	194994429753	22" LED High Bay	25,054	174	5000 K	80 CRI	120-277V	Medium	102
CPHB 30LM MVOLT 40K	194994429746	22" LED High Bay	30,298	214	4000 K	80 CRI	120-277V	Medium	102
CPHB 30LM MVOLT 50K	194994429784	22" LED High Bay	30,498	214	5000 K	80 CRI	120-277V	Medium	102



Accessories: Order as separate catalog number.			
Mounting:		Wire guards:	
IBAC120 M100	Aircraft cable 10' with hook (one pair)	WGCPHBSM	Wire guard for CPHB (9LM - 18LM)
IBAC240 M75	Aircraft cable 20' with hook (one pair)	WGCPHBMD	Wire guard for CPHB (24LM - 30LM)
IBHMP	Hook monopoint	Field Installable Sensors:	
CPHBPMPSM	Pendant Monopoint splice box with 3/4" hub (for 9LM - 18LM) ‡	TLS WH	Field Installable PIR twist to lock sensor, white ‡
CPHBMPMD	Pendant Monopoint splice box with 3/4" hub (for 24LM - 30LM) ‡	Field installable battery pack:	
ZACVH	Aircraft 10' V hanger (one pair)	ILBLP CP20 HE SD HV S M5	10x 20W emergency battery pack, Certified in CA Title 20 MAEDBS ‡
THUN J2	Surface mount bracket ‡		



ZACVH
(mounts in holes on fixture channel)

‡ Option Value Ordering Restrictions	
Option value	Restriction
CPHB ALO13 TLS, CPHB ALO16 TLS	TLS WH sensor included in the box, but not installed to the fixture.
CPHBPMPSM/MD	Pendant monopoint splice boxes will require wiring from access plate to splice box KO if power is being dropped through pendant conduit. Fixture does not have a KO in center to pull power out of driver channel through splice box.
ILBLP CP20 HE SD HV S M5	MVOLT only. Not for use with THUN. For use with ALO fixtures.
THUN J2	Order quantity required in multiples of 2. 12LM - 18LM requires one per fixture, 24LM - 30LM require two per fixture.
TLS WH	Available for use with ALO13, ALO16, and ALO34.

Field Installable Sensors			
Stock part number	Stock Part Number Ci Code	UPC	Description
TLS WH	*289JT6	197590692934	Twist-to-lock sensor, white ‡

OUT-OF-BOX SETTINGS:	
Occupancy	10 min
High Trim	100%
Daylight Dimming	Disabled
*IWP Remote used to make changes from these settings	

Require to program TLS sensor.





Specifications

INTENDED USE:

Ideal one-for-one replacement of conventional lighting systems such as HID and fluorescent. For use in light Industrial applications such as warehousing and other large indoor spaces with mounting heights ranging from 10' - permitted. **Certain airborne contaminants can diminish integrity of acrylic and/or polycarbonate.**

[Click here for Acrylic-Polycarbonate Compatibility table for suitable uses.](#)

Certain airborne contaminants, including condensation, may adversely affect the functioning of LEDs and other electronic components, depending on various factors such as concentrations of the contaminants, ventilation, and temperature at the end-user location. [Click here for a list of substances that may not be suitable for interaction with LEDs and other electronic components.](#)

CONSTRUCTION:

Extruded aluminum channels enable superior thermal performance.

Glare Control Lens diffuses light source and reduces glare while protecting LEDs.

Lens meets DLC 5.1 standards for UGR (Unified Glare Rating)

Available in three sizes with optional switchable lumens (12000/15000/18000 or 24000/27000/30000 or 36000/48000/60000) and color temperatures (4000K/5000K

ELECTRICAL:

70% lumen maintenance at > 100,000 hours.

Thermally protected driver standard with 0-10V dimming.

Luminaire surge protection level: designed to withstand up to 6kV/3kA per ANSI C82.77-5-2015.

Multi-volt driver, 120-277V standard. UVOLT driver, 120-347V with switchable versions.

6ft cord included with AL013 and AL016 UVOLT fixtures.

INSTALLATION:

Fixture package includes V-hanger hardware kit with 2- V-hanger brackets and 2- 36" chain lengths.

Fixture is suitable for mounting by chain, cable, surface-mount bracket, or hook monopoint.

Surface mounting available using optional THUN surface mount bracket (order separately).

Designed for use in ambient temperatures ranging from -40°C to 55°C when suspended 18" off ceiling. Max operating temperature of 45°C when surface mounted.

Note: Temperature cycling, or variations can potentially produce condensation; please consult factory.

LISTINGS:

CSA listed. Damp location listed.

DesignLights Consortium® (DLC) Premium qualified product. Not all versions of this product may be DLC Premium qualified. Please check the DLC Qualified Products List at www.designlights.org/QPL to confirm which versions are qualified.

UVOLT Fixtures not NOM Certified.

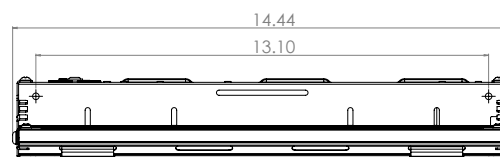
WARRANTY:

5-year limited warranty. This is the only warranty provided and no other statements in this specification sheet create any warranty of any kind. All other express and implied warranties are disclaimed. Complete warranty terms located at: www.acuitybrands.com/support/warranty/terms-and-conditions

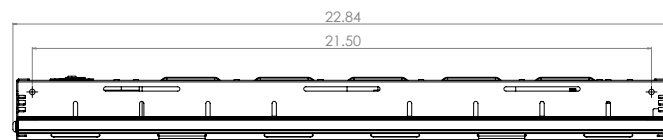
Note: Actual performance may differ as a result of end-user environment and application. All values are design or typical values, measured under laboratory conditions at 25 °C. Specifications subject to change without notice.

Dimensions

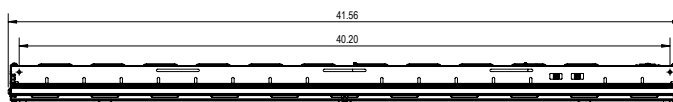
All dimensions are inches (centimeters) unless otherwise indicated.



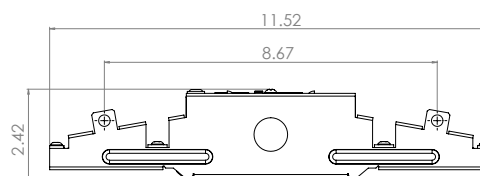
Side View: CPHB 12LM, 15LM, 18LM, AL013, AL022



Side View: CPHB 24LM, 30LM, AL016



Side View: AL034



End View: CPHB 12LM, 15LM, 18LM, 24LM, 30LM, AL013, AL016, AL022, AL034

Lumen package	Length	Width	Depth	Weight
	Dimensions shown in inches (centimeters)			Shown in pounds (kg)
12000LM	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	5 (2.2)
15000LM	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	5 (2.2)
18000LM	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	6.5 (2.9)
24000LM	22.8 (57.9)	11.5 (29.2)	2.3 (5.8)	8 (3.6)
30000LM	22.8 (57.9)	11.5 (29.2)	2.3 (5.8)	8 (3.6)
AL022	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	6.6 (3.0)
AL013	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	6.6 (3.0)
AL016	22.8 (57.9)	11.5 (29.2)	2.3 (5.8)	8.1 (3.7)
AL034	41.5 (107)	11.5 (29.2)	2.3 (5.8)	11.65 (5.3)



FEATURES & SPECIFICATIONS

INTENDED USE — Ideal one-for-one replacement of conventional lighting systems such as HID and fluorescent. For use in tight Industrial applications such as, warehousing and other large indoor spaces with mounting heights ranging from 10' – permitted. **Certain airborne contaminants can diminish integrity of acrylic and LED's.** [Click here for Acrylic Environmental Compatibility table for suitable uses.](#)

Certain airborne contaminants, including condensation, may adversely affect the functioning of LEDs and other electronic components, depending on various factors such as concentrations of the contaminants, ventilation, and temperature at the end-user location. [Click here for a list of substances that may not be suitable for interaction with LEDs and other electronic components.](#)

CONSTRUCTION — Extruded aluminum channels enable superior thermal performance. Glare Control acrylic lens diffuses light source and reduces glare while protecting LEDs. Lens meets DLC 5.1 standards for UGR (Unified Glare Rating).

ELECTRICAL — 70% lumen maintenance at > 100,000 hours. Thermally protected driver standard with 0-10V dimming. Luminaire surge protection level: designed to withstand up to 6kV/3kA per ANSI C82.77-5-2015 Multi-volt driver, 120-277V standard.

INSTALLATION — Fixture is suitable for mounting by chain, cable, surface-mount bracket, pendant monopoint, or hook monopoint. Utilizing appropriate mounting accessory. Designed for use in ambient temperatures ranging from -40°C to 55°C when suspended 18" off ceiling. Surface mount to 45°C ambient operation.

Note: Temperature cycling, or variations can potentially produce condensation; please consult factory.

LISTINGS — CSA listed. Damp location listed.

DesignLights Consortium® (DLC) Premium qualified product. Not all versions of this product may be DLC Premium qualified. Please check the DLC Qualified Products List at www.designlights.org/QPL to confirm which versions are qualified.

GOVERNMENT PROCUREMENT —

BAA — Buy American Act: This product is available in a version that qualifies as a domestic end product under the Buy American Act as implemented in the FAR and DFARS.

BABA — Build America Buy America: This product is available in a version that qualifies as produced in the United States under the definitions of the Build America, Buy America Act, as implemented by 2 C.F.R. Part 184 and other agency implementing regulations, including Department of Transportation regulations. Please refer to www.acuitybrands.com/resources/buy-american for additional information.

WARRANTY — 5-year limited warranty. This is the only warranty provided and no other statements in this specification sheet create any warranty of any kind. All other express and implied warranties are disclaimed. Complete warranty terms located at: www.acuitybrands.com/support/warranty/terms-and-conditions

Note: Actual performance may differ as a result of end-user environment and application.

All values are design or typical values, measured under laboratory conditions at 25 °C.

The product images shown are for illustration purposes only and may not be an exact representation of the product.

Specifications subject to change without notice.

Standard Part Number	Stock Part Number	Stock Part Number CI Code	DLC Product ID	DLC Premium?
CPHB 12000LM SEF GCL MD MVOLT GZ10 40K 80CRI DWH	CPHB 12LM MVOLT 40K	*268SAH	PAMXQBT8	Yes
CPHB 12000LM SEF GCL MD MVOLT GZ10 50K 80CRI DWH	CPHB 12LM MVOLT 50K	*2681J0	P0Q23DMQ	Yes
CPHB 15000LM SEF GCL MD MVOLT GZ10 40K 80CRI DWH	CPHB 15LM MVOLT 40K	*2681J2	PLPPHUG	Yes
CPHB 15000LM SEF GCL MD MVOLT GZ10 50K 80CRI DWH	CPHB 15LM MVOLT 50K	*2681J5	PJ97Z0F9	Yes
CPHB 18000LM SEF GCL MD MVOLT GZ10 40K 80CRI DWH	CPHB 18LM MVOLT 40K	*268SAJ	PWHG1E01	Yes
CPHB 18000LM SEF GCL MD MVOLT GZ10 50K 80CRI DWH	CPHB 18LM MVOLT 50K	*2681J9	PSRNYSS1	Yes
CPHB 24000LM SEF GCL MD MVOLT GZ10 40K 80CRI DWH	CPHB 24LM MVOLT 40K	*2681JE	PC2IBASD	Yes
CPHB 24000LM SEF GCL MD MVOLT GZ10 50K 80CRI DWH	CPHB 24LM MVOLT 50K	*2681JL	P7DUT8S1	Yes
CPHB 30000LM SEF GCL MD MVOLT GZ10 40K 80CRI DWH	CPHB 30LM MVOLT 40K	*2681JM	PRXQVN2Q	Yes
CPHB 30000LM SEF GCL MD MVOLT GZ10 50K 80CRI DWH	CPHB 30LM MVOLT 50K	*2681JP	PDAUA08P	Yes

Catalog Number
Notes
Type

LED High Bay

COMPACT PRO™

9,000 through 60,000 Lumens



Items marked by a shaded background qualify for the Design Select program and ship in 15 days or less. To learn more about Design Select, visit www.acuitybrands.com/designselect.
*See ordering tree for details



Looking for Contractor Select readily available configurations? Click here to visit Contractor Select™ spec sheet or go to www.contractorselect.com

ORDERING INFORMATION

Lead times will vary depending on options selected. Consult with your sales representative.

Example: CPHB 24000LM SEF GCL MD 120 GZ10 40K 80CRI CPSBW SSE PIRHKO HL DWH

CPHB			GCL			GZ10		
Series	Lumens	Performance Package	Lens	Distribution	Voltage	Driver	Color Temperature	Color Rendering Index
CPHB Compact Pro High Bay	9000LM 9,000 lumens 12000LM 12,000 lumens 15000LM 15,000 lumens 18000LM 18,000 lumens ‡ 24000LM 24,000 lumens 30000LM 30,000 lumens 36000LM 36,000 lumens 48000LM 48,000 lumens 60000LM 60,000 lumens	SEF Standard Efficiency HEF High Efficiency	GCL Glare Control Lens	MD Medium Distribution ND Narrow Distribution WD Wide Distribution	MVOLT 120-277V 120 120V 208 208V 240 240V 277 277V 347 347V ‡ 480 480V ‡ HVOLT 347-480V ‡	GZ10 0-10V dimming driver	35K 3500 K 40K 4000 K 50K 5000 K	70CRI 70 CRI 80CRI 80 CRI 90CRI 90 CRI

				DWH
Options				Finish
BAA	Buy America(n) Act and/or Build America Buy America Qualified			DWH Gloss White
Emergency Options: ‡				
E10W	IOTA 10W emergency battery pack, Constant Power, Certified in CA Title 20 MAEDBS, Self-diagnostics (LINK) ‡			Individual Controls: ‡ ‡
E10WCP	10W emergency battery pack, Certified in CA Title 20 MAEDBS ‡			
E15WMCP	15W emergency battery pack, Certified in CA Title 20 MAEDBS ‡			SSE PIRHKO 360° integral high mount motion sensor with standard on/off operation SSE PIRHKO HL 360° integral high mount motion sensor with high/low(off) occupancy operation. For High/Low/(never off) SSE PIRHKO P 360° integral high mount motion sensor with on/off switching photocell SSE APIRHKO 360° integral high mount motion sensor with dimming & switching photocell SSE APIRHKO ANL 360° integral high mount motion sensor with High/Low occupancy dimming & auto-dimming photocell
IE10WMCP	10W emergency battery pack, Certified in CA Title 20 MAEDBS ‡			
IE15WMCP	15W emergency battery pack, Certified in CA Title 20 MAEDBS ‡			*For 360° integral LOW MOUNT sensors, remove "H". For High Mount AISLEWAY sensors, replace "H" in nomenclature with "A".
IE20WCPHE	Iota 20W emergency battery pack, Certified in CA Title 20 MAEDBS ‡			
ETS	Generator Transfer Device ‡			nLight AIR Wireless Controls:
WGX	Wireguard installed			
IMP	Integrated modular plug ‡			NLTAIR2 Embedded wireless controls by nLight NLTAIR2 KO Embedded wireless controls by nLight, Control mounted external to the luminaire via a KO ‡ NLTAIREM2 KO Embedded wireless controls by nLight with UL924 listed emergency operation Control mounted external to the luminaire via a KO ‡ NLTAIRER2 KO Embedded wireless controls by nLight with UL924 listed emergency operation via power sense leads, mounted external to fixture via KO ‡
SPD	Additional 10kV surge protection device ‡			
RRL_	RELOC® Ready luminaire. (Not available with Haleon sensor options). See page 14 for ordering information			NLTAIR2 APIRH* Embedded wireless controls by nLight, Passive Infrared Occ sensor with auto-dimming photocell for high mounting height applications NLTAIREM2 APIRH* Embedded wireless controls by nLight, with UL924 listed emergency operation, Passive Infrared Occ sensor with auto-dimming photocell for high mounting height applications
OCS	RELOC® OnePass selectable cable 6' installed ‡ ‡			
OCS4C	RELOC® One Pass® selectable cable 6' installed. ‡ ‡			*For 360° integral LOW MOUNT sensors, remove "H" from the nomenclature. For AISLE MOUNT sensors, remove "H" replace with "A".
2470P	24/7 operation, factory installed battery with 3-conductor cordset or Reloc, with on/off controlled by sensor rather than switch, the normal hot and unswitched hot wired together in the fixture. Consult local codes to determine if this is allowable			
JP	Job Pack ‡			NLTAIR2 APIRHKO* Embedded wireless controls by nLight, Knock out mounted Passive Infrared Occ sensor with auto-dimming photocell for high mounting height applications NLTAIR2 APIRAKO Embedded wireless controls by nLight, Knock out mounted Passive Infrared Occ sensor with auto-dimming photocell for Aisleway applications NLTAIRER2 APIRKO Embedded wireless controls by nLight with UL924 listed emergency operation via power sense leads, Control mounted external to the luminaire via a KO ‡ NLTAIREM2 APIRHKO* Embedded wireless controls by nLight with UL924 listed emergency operation, Knock out mounted Passive Infrared Occ sensor with auto-dimming photocell for high mounting height applications NLTAIREM2 APIRAKO Embedded wireless controls by nLight with UL924 listed emergency operation, Knock out mounted Passive Infrared Occ sensor with auto-dimming photocell for Aisleway applications
Cord Sets: ‡				
CPSBW	6' white damp location cord with straight blade plug (voltage will match fixture, 120, 208, 240, or 277V only)			NLTAIR2 APIRHKO* Embedded wireless controls by nLight, Knock out mounted Passive Infrared Occ sensor with auto-dimming photocell for high mounting height applications
CPTLW	6' white damp location cord with 15 amp twist-lock plug (voltage will match fixture) ‡			
CPTL20AW	6' white damp location cord with 20 amp twist lock plug (voltage will match fixture) ‡			NLTAIR2 APIRAKO Embedded wireless controls by nLight, Knock out mounted Passive Infrared Occ sensor with auto-dimming photocell for Aisleway applications
CNPW	6' white cord, no plug			
CNP4CW	6' white cord, 4 conductors (for use with battery)			NLTAIRER2 APIRKO Embedded wireless controls by nLight with UL924 listed emergency operation via power sense leads, Control mounted external to the luminaire via a KO ‡ NLTAIREM2 APIRHKO* Embedded wireless controls by nLight with UL924 listed emergency operation, Knock out mounted Passive Infrared Occ sensor with auto-dimming photocell for high mounting height applications NLTAIREM2 APIRAKO Embedded wireless controls by nLight with UL924 listed emergency operation, Knock out mounted Passive Infrared Occ sensor with auto-dimming photocell for Aisleway applications
CNP5CDW	6' white cord, 5 conductors (includes dimming leads)			

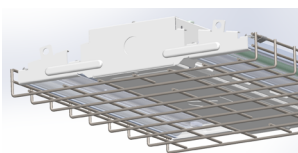
CPHB LED High Bay

Accessories: Order as separate catalog number.Mounting:

IBAC120 M100	Aircraft cable 10' with hook (one pair)
IBAC240 M75	Aircraft cable 20' with hook (one pair)
IBHMP	Hook monopoint
CPHBMPMSM	Pendant Monopoint splice box with 3/4" hub (for 9000LM – 18000LM) ‡
CPHBMPMD	Pendant Monopoint splice box with 3/4" hub (for 24000LM – 30000LM) ‡
CPHBMPPLG	Pendant Monopoint splice box with 3/4" hub (for 36000LM – 60000LM) ‡
ZACVH	Aircraft 10' V hanger (one pair)
HC36	Hanger chain, 36" (one pair)
THUN J2	Tong hanger surface mount bracket ‡

Wire guards:

WGCPHBSM	Wire guard for CPHB (9000LM – 18000LM)
WGCPHBMD	Wire guard for CPHB (24000LM – 30000LM)
WGCPHBLG	Wire guard for CPHB (36000LM – 60000LM)

Cord sets for IMP option: ‡

CS1WIMP	Straight plug, 120V
CS3WIMP	Twist-lock, 120V
CS7WIMP	Straight plug, 277V
CS11WIMP	Twist-lock, 277V
CS93WIMP	600V SO white cord, no plug (no voltage required)

‡ Option Value Ordering Restrictions

Option Value	Ordering Restrictions
18000LM	Not available with HVOLT. Not available with ETS or ER sensor option. When 18000LM requires 347 or 480 volt; the fixture utilizes a step-down transformer.
2470P	Requires cord or reloc, battery and sensor options. Not available with 4 conductor cord sets. CPHB will be wired from the factory for 24/7 operation (via cord or Reloc), with on/off controlled by sensor rather than switch. Consult local codes to determine if this is allowable.
347	Fixture includes an additional 10kV surge protector. When ordered with 18000LM, includes step down transformer.
480	Fixture includes an additional 10kV surge protector. When ordered with 18000LM, includes step down transformer.
Cord Sets	Must specify voltage on cord sets with plugs.
Cord Sets for IMP	Fixture must be ordered with IMP option. All cord sets are 18/3, 6' white.
CPHBMPMSM/MD/LG	Pendant monopoint splice boxes will require wiring from access plate to splice box KO if power is being dropped through pendant conduit. Fixture does not have a KO in center to pull power out of driver channel through splice box. Not for use with battery pack.
CPTL20AW	Required on 347 and 480 volt.
Emergency Options	May alter fixture construction. Consult line art for dimensions. Batteries not for use with pendant mount option. If cord set is needed, use CNP4CW which will have normal hot and unswitched hot exiting cord set. Not available with IMP, ETS, or THUN options.
E10W	Available with MVOLT, 120, 208, 240, or 277. (120, 208, 240, or 277 when any other option ordered such as cord and plug requires voltage to be specified). Available with lumen packages 24000LM – 60000LM.
E10WCP	Available with MVOLT, 120, 208, 240, or 277. (120, 208, 240, or 277 when any other option ordered such as cord and plug requires voltage to be specified). Max ambient operating temperature of 35°C. Includes an additional 10kV surge protector on the unswitched, battery hot. Only available with 9000LM – 18000LM.
E15WMCP	Available with MVOLT, 120, 208, 240, or 277. (120, 208, 240, or 277 when any other option ordered such as cord and plug requires voltage to be specified). Factory installation only. Not approved for field installation. Max ambient operating temperature of 35°C. Includes an additional 10kV surge protector on the unswitched, battery hot.
ETS	MVOLT only. Not available with 18000LM. Not available with cord sets or batteries. When sensor is required, use "ER" sensor option. Utilizes ETS 924 DR component. Includes an additional 10kV surge protector.
HVOLT	Fixture includes an additional 10kV surge protector. Not valid with 18000LM.
IE10WMCP	Available with HVOLT, 347 or 480. (347 and 480 when any other option ordered such as cord with plug requires voltage to be specified). Not available with 18000LM.
IE15WMCP	Available with HVOLT, 347 or 480. (347 and 480 when any other option ordered such as cord with plug requires voltage to be specified). Not available with 9000LM, 12000LM, 15000LM, or 18000LM.
IE20WCPHE	Available with MVOLT, 120, 208, 240, or 277. (120, 208, 240, or 277 when any other option ordered such as cord and plug requires voltage to be specified). Not available for 9LM, 12LM, 15LM and 18LM. Not available with NLTAIR2 KO and Cordset together. Not for use with THUN.
IMP	Not available with battery options, NLTAIR2 KO options, PIRHKO or APIRHKO options, ETS, or NLTAIRER2 APIRHKO options. Sensors will be fixture pre-wired to fixture. Not for use with THUN mounting option.
Individual Controls (LSXR)	Comes standard with SPD. SSE PIRAKO not available as a Design Select option when ordered with 70CRI option. This sensor configuration is suitable for minimum ambient temperature of 14°F (-10°C). Not available with other controls.
JP	Consult table on page 7 for details.
NLTAIR2 APIR, APIRH, or APIRA	Not available with NLTAIRER2 when ordered with 9000LM, 12000LM, 15000LM or 18000LM lumen packages (ER option utilizes ETS).
NLTAIR2 KO	Comes standard with SPD.
NLTAIREM2 KO	Comes standard with SPD. Not available with Emergency battery options.
NLTAIRER2 KO	Comes standard with SPD. Not available with emergency battery options.
OCS	Must specify voltage. Dry Location listed only.
OCS4C	Dry location listed only. Use when unswitched hot is required for battery pack options.
SPD	10kV surge protector standard with HVOLT, 347, 480, ETS, any NLTAIR2, and battery options (on the unswitched, battery hot).
THUN J2	Order quantity required in multiples of two. 9000LM – 18000LM requires one per fixture, 24000LM – 60000LM requires two per fixture. Not for use with IMP or battery pack options.
CNP5CDW	When ordered with a battery pack, 2470P is required.

‡ Design Select Ordering Restrictions

Option Value	Ordering Restrictions
CPTLW	Not on program when combined with 208V, 240V, or 347V
Individual Controls	Not on DS program when ordered as SSE PIRAKO with 70CRI.
OCS, OCS4C (all lengths)	Not on program when combined with 120V, 208V, 240V, or 347V

CORD SET ORDERING INFORMATION

Plug Option	Plug type	Amperage**	Gauge	# of conductors	Color	Location	Length*****
CNP Cord Only	(blank) No Plug Option (for Cord Only option)	(blank) 15 amps	(blank) 18 gauge standard	(blank) 3 conductors (blk/wht/grn)	W White	(blank) Damp Location	(blank) 6 feet
CP Cord with Plug	TL Locking Type	20A 20 amps***		4C 4 conductors; Use with Battery option when unswitched hot is needed			3FT 3 feet
	SB Straight blade*			5CD **** 5 conductors; Use with dimming driver when dimming leads are desired (Not for use with dimming sensors)			10FT 10 feet
							12FT 12 feet
							15FT 15 feet
							20FT 20 feet

* Straight blade on available with 120, 208, 240, and 277 volt.

** Amperage is only configurable for cords with plugs.

*** 20A required with 347 and 480 volt.

**** Not available with plugs.

***** Available in 1' increments from 1 to 30 feet.

NEMA PLUG CONFIGURATION

Voltage	Straight blade	Twist-lock 15 amp plug	Twist-lock 20 amp plug
120	5-15P	L5-15P	L5-20P
208/240	6-15P	L6-15P	L6-20P
277	7-15P	L7-15P	L7-20P
347	X	X	L24-20P
480	X	X	L8-20P

EMBEDDED LUMINAIRE TECHNOLOGY GUIDE

Control type	Sensor nomenclature	Sensor used (click part number to link to sensor spec sheet)	Out of the box settings					Notes
			Unoccupied time until dim/off	Dim to off time delay	Occupied dim level	Unoccupied dim level	Daylighting set point	
Sensor Switch embedded controls	SSE PIRHKO	LSXR 6	10M	n/a	10V/100%	n/a	n/a	
	SSE PIRKO	LSXR 10	10M	n/a	10V/100%	n/a	n/a	
	SSE PIRAKO	LSXR 50	10M	n/a	10V/100%	n/a	n/a	
	SSE PIRHKO HL	LSXR 6 HL	10M	2.5M	10V/100%	1V/10%	n/a	
	SSE PIRKO HL	LSXR 10 HL	10M	2.5M	10V/100%	1V/10%	n/a	
	SSE PIRAKO HL	LSXR 50 HL	10M	2.5M	10V/100%	1V/10%	n/a	
	SSE PIRHKO P	LSXR 6 P	10M	n/a	10V/100%	n/a	4FC	
	SSE PIRKO P	LSXR 10 P	10M	n/a	10V/100%	n/a	4FC	
	SSE PIRAKO P	LSXR 50 P	10M	n/a	10V/100%	n/a	4FC	
	SSE APIRHKO	LSXR 6 ADC	10M	n/a	10V/100%	1V/10%	4FC	ADC only dims based on photocell
	SSE APIRKO	LSXR 10 ADC	10M	n/a	10V/100%	1V/10%	4FC	ADC only dims based on photocell
	SSE APIRAKO	LSXR 50 ADC	10M	n/a	10V/100%	1V/10%	4FC	ADC only dims based on photocell
	SSE APIRHKO ANL	LSXR 6 ANL	10M	n/a	10V/100%	1V/10%	4FC	ANL dims by occupancy and photocell but only turns off due to photocell
	SSE APIRKO ANL	LSXR 10 ANL	10M	n/a	10V/100%	1V/10%	4FC	ANL dims by occupancy and photocell but only turns off due to photocell
	SSE APIRAKO ANL	LSXR 50 ANL	10M	n/a	10V/100%	1V/10%	4FC	ANL dims by occupancy and photocell but only turns off due to photocell
nLight AIR2 embedded controls	NLTAIR2 APIRH	RMS0D 45 180D G2	7.5M	5M	10V/100%	3V/30%	5FC	
	NLTAIREM2 APIRH	RMS0D 45 EM 180D G2	7.5M	5M	10V/100%	3V/30%	5FC	
	NLTAIR2 APIRKO	RLSXR 10 WH G2	7.5M	2.5M	10V/100%	1V/10%	50FC	
	NLTAIR2 APIRHKO	RLSXR 6 WH G2	7.5M	2.5M	10V/100%	1V/10%	50FC	
	NLTAIR2 APIRAKO	RLSXR 50 WH G2	7.5M	2.5M	10V/100%	1V/10%	50FC	
	NLTAIREM2 APIRKO	RLSXR 10 EM WH G2	7.5M	2.5M	10V/100%	1V/10%	50FC	
	NLTAIREM2 APIRHKO	RLSXR 6 EM WH G2	7.5M	2.5M	10V/100%	1V/10%	50FC	
	NLTAIREM2 APIRAKO	RLSXR 50 EM WH G2	7.5M	2.5M	10V/100%	1V/10%	50FC	
	NLTAIRER2 APIRKO	RLSXR 10 WH G2	ETS DR	7.5M	2.5M	10V/100%	1V/10%	50FC
	NLTAIRER2 APIRHKO	RLSXR 6 WH G2	ETS DR	7.5M	2.5M	10V/100%	1V/10%	50FC
	NLTAIRER2 APIRAKO	RLSXR 50 WH G2	ETS DR	7.5M	2.5M	10V/100%	1V/10%	50FC
	NLTAIR2	RIO ZT 90 G2	n/a	n/a	80%	n/a	n/a	
	NLTAIR2 KO	RPP20 D EFP G2	n/a	n/a	100%	n/a	n/a	
	NLTAIREM2 KO	RPP20 D EM EFP G2	n/a	n/a	100%	n/a	n/a	
	NLTAIRER2 KO	RPP20 D ER EFP G2	ETS DR	n/a	n/a	100%	n/a	

CPHB LED High Bay

PHOTOMETRICS

See www.lithonia.com.

PERFORMANCE TABLE

CPHB SEF	Glare Control Lens, Medium Distribution			
	Lumen Package	Wattage	Lumen Output	LPW
Delivered Lumens 4000K, 80CRI	9000LM	61	9,123	150
	12000LM	87	12,273	141
	15000LM	102	15,343	150
	18000LM	133	18,199	137
	24000LM	174	24,873	143
	30000LM	210	29,825	142
	36000LM	235	36,289	163
	48000LM	324	48,342	153
	60000LM	421	58,946	140

Delivered Lumens 5000K, 80CRI	9000LM	61	9,183	150
	12000LM	88	12,354	140
	15000LM	105	15,444	147
	18000LM	133	18,319	138
	24000LM	174	25,037	144
	30000LM	210	30,081	143
	36000LM	235	36,529	155
	48000LM	324	48,661	150
	60000LM	421	59,452	141

*Values shown are at 120V.

SCALING FACTOR TABLES

Multipliers	
ND	1.037
WD	1.004
WGX	0.95

CPHB HEF	Glare Control Lens, Medium Distribution			
	Lumen Package	Wattage	Lumen Output	LPW
Delivered Lumens 4000K, 80CRI	9000LM	55	8,861	161
	12000LM	75	12,056	161
	15000LM	96	14,875	155
	18000LM	127	18,936	149
	24000LM	150	23,740	158
	30000LM	196	29,644	152
	36000LM	210	35,409	169
	48000LM	292	47,480	163
	60000LM	378	59,274	157

Delivered Lumens 5000K, 80CRI	9000LM	55	9,002	164
	12000LM	75	12,248	163
	15000LM	96	15,111	157
	18000LM	127	19,236	151
	24000LM	150	24,117	161
	30000LM	196	30,115	154
	36000LM	210	35,971	171
	48000LM	292	48,234	165
	60000LM	378	60,215	159

CPHB CHARACTERISTICS

Lumen package	Wattage								Length	Width	Depth	Weight	Comparable light source
	Standard Efficiency (SEF)				High Efficiency (HEF)								
	120V	277V	347V	480V	120V	277V	347V	480V	Dimensions shown in inches (centimeters)			Shown in pounds (kg)	
9000LM	61	60	60	60	55	54	54	54	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	5 (2.2)	100W MH, 4-lamp T8 NBF
12000LM	88	88	88	87	75	74	74	74	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	5 (2.2)	175W MH, 4-lamp T8 HBF, 2-lamp T5HO
15000LM	105	104	104	103	96	95	95	94	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	5 (2.2)	200W MH, 6-lamp T8 NBF
18000LM	133	132	144	144	127	126	138	138	14.4 (36.6)	11.5 (29.2)	2.3 (5.8)	6.5 (2.9)	250W MH, 6-lamp T8 HBF, 4-lamp T5HO
24000LM	174	174	173	173	150	149	149	149	22.8 (57.9)	11.5 (29.2)	2.3 (5.8)	8 (3.6)	400W MH, 6-lamp T5HO
30000LM	210	209	208	208	196	194	194	194	22.8 (57.9)	11.5 (29.2)	2.3 (5.8)	8 (3.6)	575W MH, 10-lamp T8 HBF
36000LM	235	235	234	234	210	208	208	208	44 (111.8)	11.5 (29.2)	2.3 (5.8)	14 (6.35)	750W MH, 8-lamp T5HO
48000LM	324	322	320	320	289	287	286	285	44 (111.8)	11.5 (29.2)	2.3 (5.8)	14 (6.35)	875W MH, 10-lamp T5HO
60000LM	421	421	419	419	374	372	371	370	44 (111.8)	11.5 (29.2)	2.3 (5.8)	14 (6.35)	1000W MH

*Dimensions & weights shown for standard product configurations without optional components

PROJECTED LUMEN MAINTENANCE

Operating hours	10,000	20,000	30,000	40,000	50,000	60,000	70,000	80,000	90,000	100,000
Lumen maintenance factor	0.98	0.96	0.93	0.91	0.89	0.87	0.85	0.83	0.81	0.79

BATTERY INFORMATION

EMERGENCY BATTERY PACK OPTIONS

Nomenclature	Part Number	Utilizes BPK option	Field installable (remote mount only)
E10W (MVOLT only)	ILD 10 MVOLT LQD 10-55 PLENUMSS	Yes	Yes
E10WCP (MVOLT only)	PS1055CP	Yes	Yes
E15WMCP (MVOLT only)	PS1555MCP	Yes	No
IE10WMCP (HVOLT only)	ILBHI CP10 HE SD LCSTICK	Yes	No
IE15WMCP (HVOLT only)	ILBHI CP15 HE SD LCSTICK	Yes	No
IE20WCPHE (MVOLT only)	ILB CP20HE SD R	No	Yes

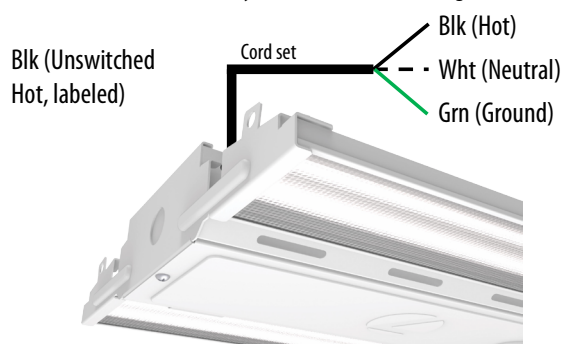
EMERGENCY LUMENS GCL, MD (5000K, 80CRI)			
	Lumen Package	E10W/ E10WCP	E15WMCP
SEF	9000	1784	2738
	12000	1778	2729
	15000	1734	2701
	18000	1796	2744
	24000	1847	2845
	30000	1884	2868
	36000	1890	2915
	48000	1888	2912
	60000	1872	2853

EMERGENCY LUMENS GCL, MD (5000K, 80CRI)			
	Lumen Package	E10W/ E10WCP	E15WMCP
HEF	9000	1916	2911
	12000	1914	2909
	15000	1919	2914
	18000	1916	2898
	24000	1966	2984
	30000	1945	2952
	36000	2058	3125
	48000	2057	3124
	60000	2065	3137

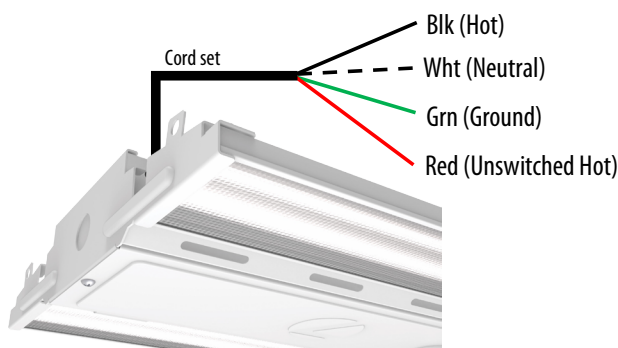
Emergency Batteries & Cord Sets - WIRING

When battery is ordered with a **3-conductor cord set and 24/7 operation**

Ideal for use with battery + SENSOR to control fixture on/off based on occupancy. Order option 247OP to receive wired for this scenario. When 247OP is ordered driver black (Hot) and battery red (Hot) will be wired together inside driver channel.



When battery is ordered with a **4-conductor cord set...**



Emergency Batteries & Cord Set Notes

When a battery is ordered with a 3 conductor cord set, 247OP option is required.

- The 3 conductor cord set will include the hot (BLK), neutral (WHT), and ground (GRN) conductors. 24/7 operation means the design intent is for the fixture to always receive power and never be switched off. This ensures the battery is always being charged. When 247OP is ordered with a 3/c cord set, the red battery hot will be wired to the normal hot inside the driver channel, and the new hot wire lead will exit the fixture with the ground and neutral wires. Consult local codes to determine if this is allowable.

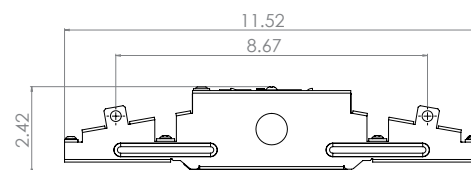
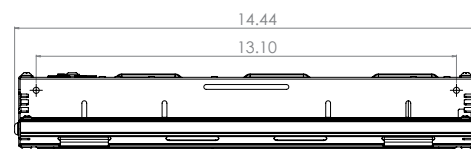
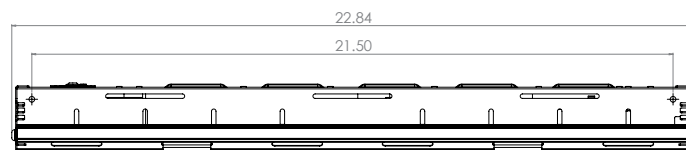
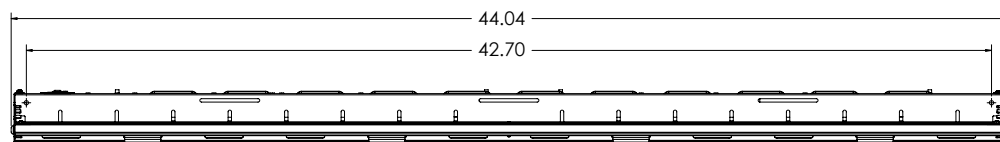
When a battery is ordered with a 4 conductor cord set...

- The unswitched hot will be included in the cord set as a separate conductor (RED).

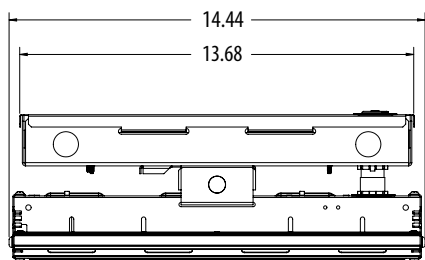
MOUNTING ACCESSORIES

**ZACVH**

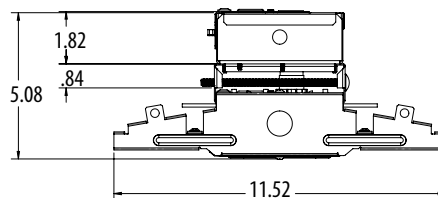
(mounts in holes on fixture channel)

**End View: CPHB 9000LM - 60000LM****Side View: 9000LM, 12000LM, 15000LM, 18000LM****Side View: 24000LM, 30000LM****Side View: 36000LM, 48000LM, 60000LM**

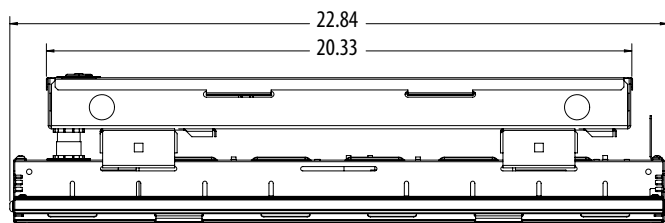
CPHB LED High Bay



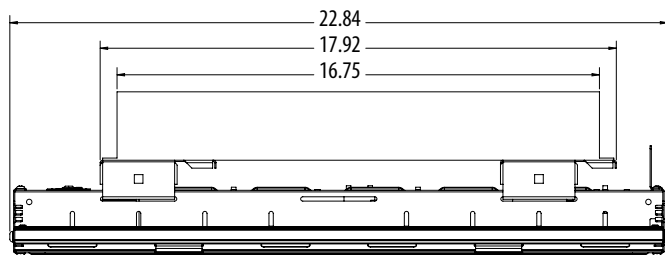
CPHB 9000LM - 18000LM WITH E10WCP/E15WMCP



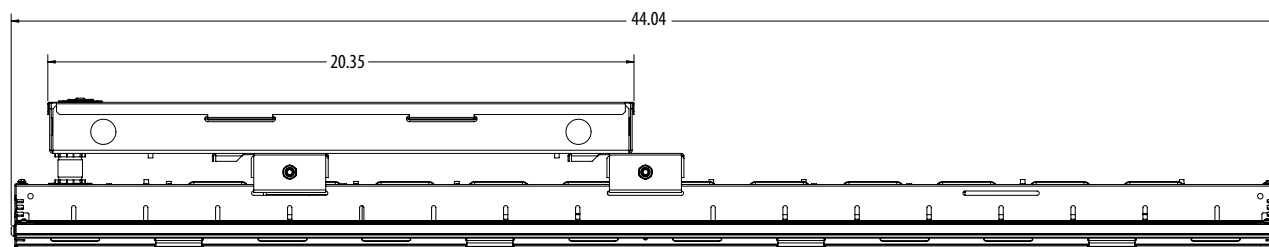
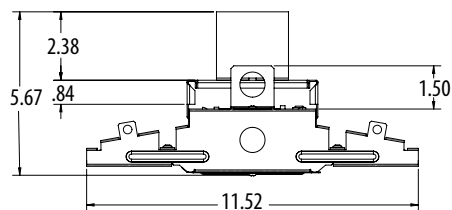
CPHB 9000LM - 60000LM WITH E10WCP/E15WMCP



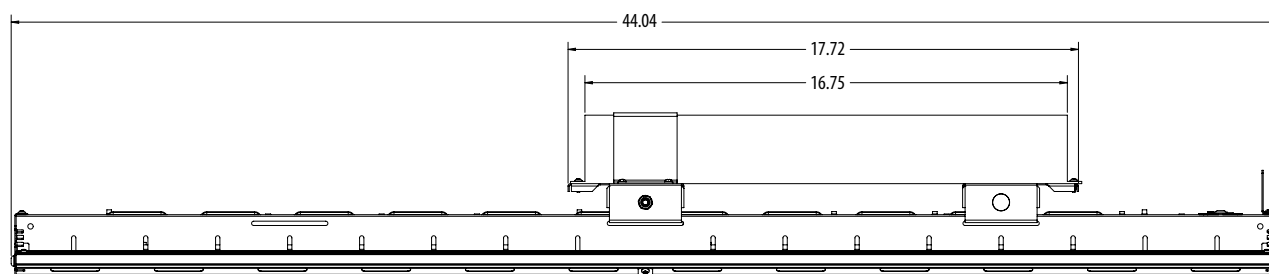
CPHB 24000LM - 30000LM WITH E10WCP/E15WMCP



CPHB 24000LM - 30000LM WITH IE20WCPHE

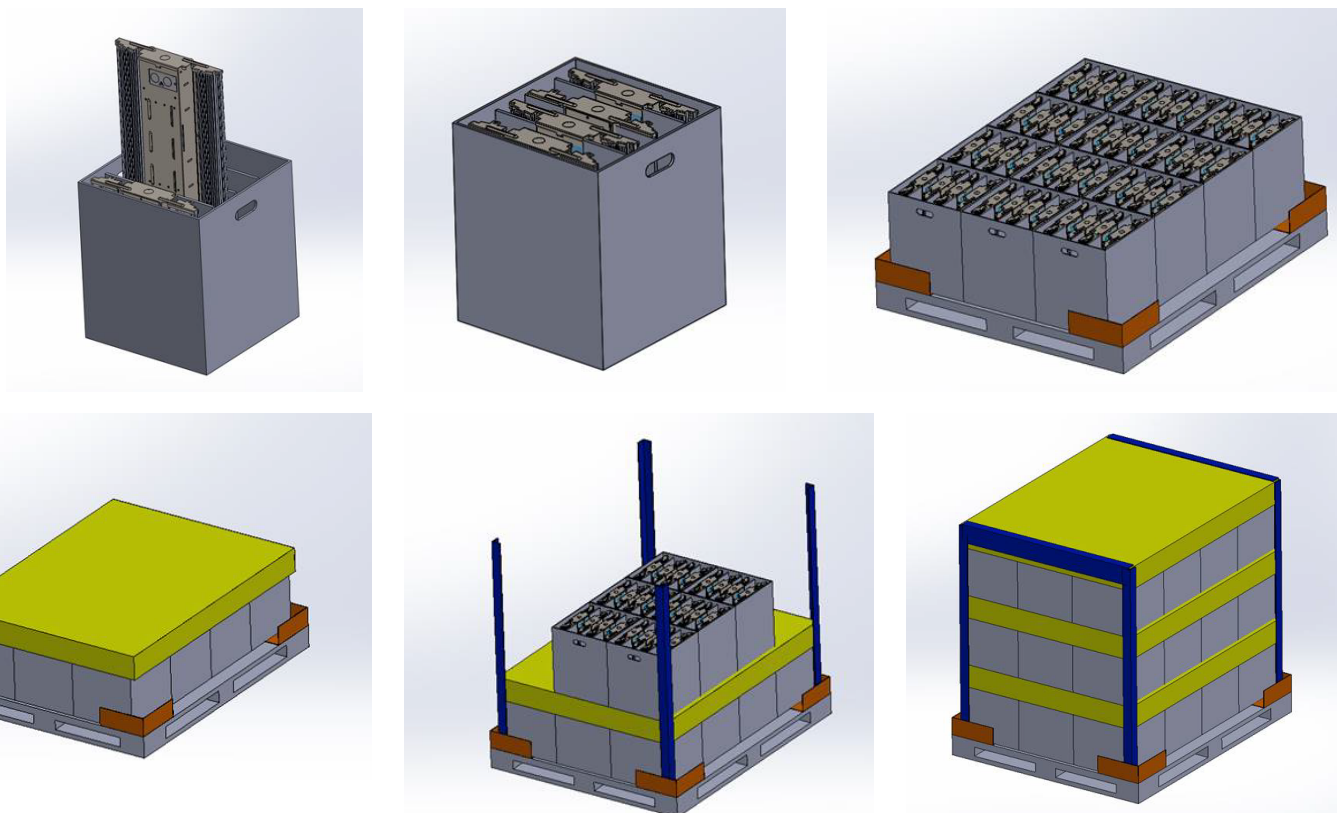


CPHB 36000LM - 60000LM WITH E10WCP/E15WMCP



CPHB 36000LM - 60000LM WITH IE20WCPHE

JOB PACK DETAILS



*Representative of Job Pack design for 14" standard units.

Series	Performance	Lumens	Standard Fixture	Cord Set or RELOC® JP	Wireguard JP	Sensor JP
CPHB	SEF/HEF	9000LM	144	144	144	90*
		12000LM				
		15000LM				
		18000LM				
		24000LM	96	96	96	60*
		30000LM				
		36000LM	45*	30*	30*	30*
		48000LM				
		60000LM				

*Traditional Job Pack

Breakout Example:		
Ordered Line:	Qty: 200	CPHB 24000LM SEF GCL MD MVOLT GZ10 40K 80CRI CNPW DWH JP
		*Above configuration shows 96 units for Job Pack
Breakout Line 1:	Qty: 192	CPHB 24000LM SEF GCL MD MVOLT GZ10 40K 80CRI CNPW DWH JP96
		*Will have 2 pallets of 96 units each in job packs
Breakout Line 2:	Qty: 8	CPHB 24000LM SEF GCL MD MVOLT GZ10 40K 80CRI CNPW DWH
		*Balance will ship in unit cartons

Note: If quantity ordered is less than Job Pack quantity for that configuration, the breakout line will default to unit packs.

STANDARD SENSOR

		Standard Sensor or Control Device (commonly used with Battery Pack Option)	EM Solution (Used when switching single incoming hot to generator power)	ER Solution (Used when switching to generator power via a 2nd hot lead)
Emergency Operation Scenarios	Emergency Lighting Strategy	*Luminaire-integral battery pack and emergency driver *Generator transfer device	*Diesel genset emergency backup supply *Slow transfer inverter (>30ms) emergency backup supply	*Fast Transfer (FT) inverter emergency backup supply *Uninterruptible Power System (UPS) emergency
	Recommended Control Device Option	*Not specifically listed for emergency use. *Wired such that a separately listed emergency device provides emergency lighting power and/or control during loss of normal power scenarios.	*UL 924 listed *EM devices will remain at their high-end trim and ignore wireless lighting control commands, such as in the event of a normal power failure, unless a normal-power-sensed (NPS) broadcast is received at least every 8 seconds. * Using the CLAIRITY+ mobile app, EM devices must be associated with a group that includes a normal power sensing device to receive NPS broadcasts. *Only non-emergency rPP20, rLSXR, rSBOR, rSDGR, and nLight AIR luminaires with version 3.4 or later firmware can provide normal power sensing for EM devices. See specification sheets for control devices and luminaires for more information on options that support normal power sensing.	*UL 924 listed *Utilizes dedicated Normal Power sensing leads to initiate lighting control override during loss of normal power scenarios. *Requires connections to both emergency and normal power circuits.

	Function	Sequence of Operations	Standard Sensor or Control Device	EM Solution (Generator / Inverter powered with 1 hot)	ER Solution (Generator / Inverter on separate emergency circuit – 2 hots)	CPHB Standard Sensor Settings				
						Vacancy Time Out	Dim to Off Time Delay	High Trim	Low Trim	Photocell Set Point
Standalone Sensors (Individual control per fixture)	On/Off	Lights turn on when motion detected; Upon vacancy, Lights turn off after timeout.	PIRHKO	-	PIRHKO ER	10 min	-	-	-	-
	High/Low (Off)	Lights turn on to high trim when presence is detected; Upon vacancy, the lights dim to low trim after timeout and turn off after "Dim To Off" Time Delay. For High/Low (Never Off) function, bypass the relay by bringing power directly into driver rather than wiring hot through LSXR device.	PIRHKO HL	-	PIRHKO HL ER	10 min	2.5 min	100%	10% (Driver Low)	-
	Photocell	Lights turn on unless ambient light level is above set point; If ambient light levels in the space exceed the photocell set point, lights will turn off even during occupancy.	PIRHKO P	-	PIRHKO P ER	-	-	-	-	4 fc
	Dimming + Photocell	Lights turn on when presence is detected unless ambient light level is above set point; Upon vacancy, the lights dim to low trim, then turn off after timeout; During occupancy, automatically raise and lower electric light level to maintain set point and turn off, depending on ambient light.	APIRHKO	-	APIRHKO ER	10 min	2.5	-	-	4 fc
	Dimming + Photocell + High/Low	Lights turn on when presence is detected unless ambient light level is above set point; Upon vacancy, lights dim to low trim after timeout and remain at low trim until presence is detected; Automatically raise and lower electric light level to maintain set point during occupancy and during vacancy keeps lights at low trim if ambient light is not sufficient.	APIRHKO ANL	-	APIRHKO ANL ER	10 min	-	100%	10%	4 fc
Note: For 360° integral Low Mount sensors, replace "6" in nomenclature with "10". Ex. LSXR10 P. For High Aisle Mount sensors, replace "6" with "50".										

nLight AIR Wireless Sensors	Dimming + Photocell + Occupancy	Wirelessly programmable network sensor – On/Off control with dimming, occupancy detection, and daylight harvesting (Sensor embedded in fixture)	NLTAIR2 APIRH	NLTAIREM2 APIRHKO	NLTAIRER2 APIRHKO	7.5 min	-	100%	30%	50 fc
	Dimming	Wirelessly programmable On/Off control with dimming – no sensor (Device embedded in fixture)	NLTAIR2	NLTAIREM2	NLTAIRER2	-	-	100%	10% (driver low)	-
Note: For 360° integral Low Mount sensors, replace "45" in nomenclature with "7". EM sensors/controls are KO-mounted; all others integral. RPP20 D EM de-rates fixture to Damp Location.										

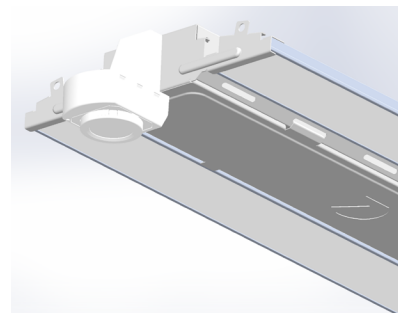
*All ER solutions include standard sensor or control device with a factory-installed Iota ETS#-DR (UL924 bypass device). This device is integral to the fixture and will include a hot and neutral lead for the dedicated emergency circuit. MVOLT, 120-277 only. ER solutions are not available with the IMP option.

Fixture Mount Occupancy Sensor

(see www.AcuityControls.com for additional information)

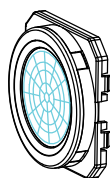
- Three interchangeable lens options to satisfy multiple mounting heights and coverage pattern requirements.
- Integrated mounting bracket drops lens down 3" from chase nipple.
- Single or dual relay versions — designed with robust protection from the harsh switching requirements of T5 and LED loads.
- Photocell and 0-10VDC dimming options.
- No PIR field calibration or sensitivity adjustments required.
- Sensor ambient temperature rating of 14°F (-10°C) to 131°F (55°C).

New sensor nomenclature.	LSXR configuration	Comparable CMRB sensor	Old style sensor nomenclature
For shortest lead times use one of the following LSXR configurations			
	LSXR50 / LC0ZU	CMRB 50	MSI
	LSXR50 HL / LCH0SZU	CMRB 50 D	MSID
	LSXR50 P / LCPZU	CMRB 50 P	MSIPED
	LSXR6 / LA0ZU	CMRB 6	MSI360
	LSXR6 HL / LAH0SZU	CMRB 6 D	MSI360D
	LSXR6 P / LAPZU	CMRB 6 P	MSI360PED



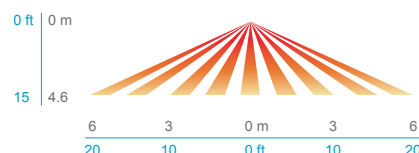
LSXR COVERAGE PATTERNS

HIGH MOUNT 360° LENS (#6)

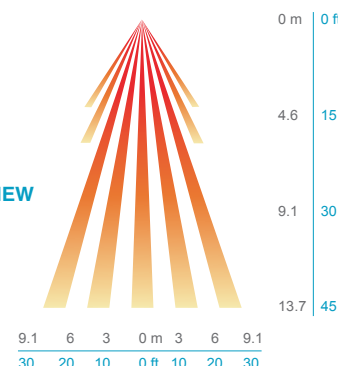


- Best choice for 15 to 45 ft (4.57 to 13.72 m) mounting heights
- 15 to 20 ft (4.57 to 6.10 m) radial coverage overlaps area lit by a typical high bay fixture
- Excellent detection of large motion (e.g. walking) up to a 35 ft (10.76 m) mounting height
- Excellent detection of extra large motion (e.g. forklifts) up to a 45 ft (13.72 m) mounting height

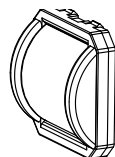
LOW VIEW



HIGH VIEW

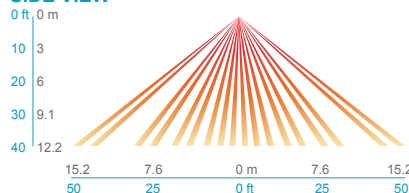


HIGH MOUNT AISLEWAY LENS (#50)

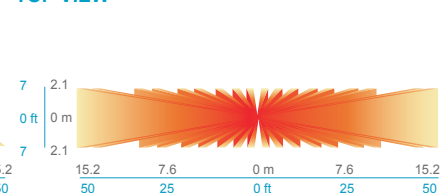


- Provides a bi-directional coverage pattern ideal for warehouse racking
- 1.2x mounting height equals approximate detection range in either direction
- Typical 40 ft (12.19 m) mounting detects 50 ft (15.24 m) in either direction
- Superior aisleway coverage compared to a masked 360° lens

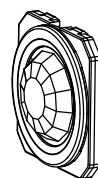
SIDE VIEW



TOP VIEW

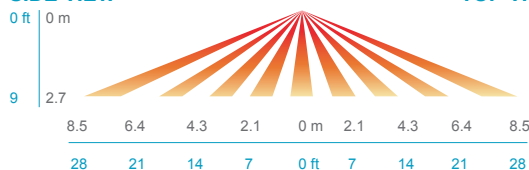


LOW MOUNT 360° LENS (#10)

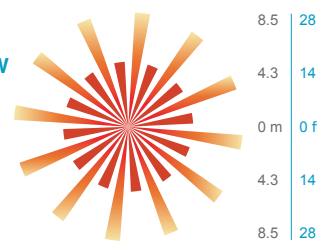


- Best choice for large motion detection (e.g. walking)
- 360° conical shaped pattern
- Provides ~24 ft (7.32 m) radial coverage (~2000 ft2) when mounted at 9 ft (2.74 m)
- 7 to 15 ft (2.13 to 4.57 m) mounting heights provide 16 to 36 ft (4.88 to 10.97 m) radial coverage
- Detection range improves when walking across beams compared to into beams

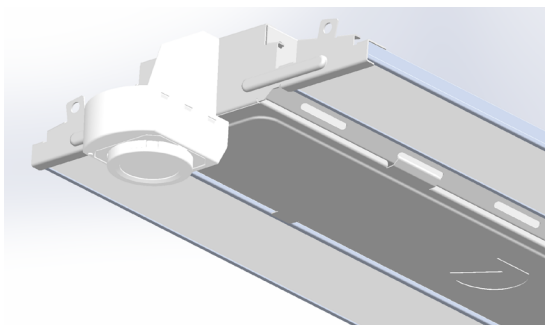
SIDE VIEW



TOP VIEW

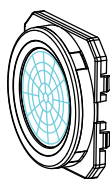


nLIGHT AIR CONTROLS – NLTAIR2 APIRKO, APIRHKO, OR APIRAKO



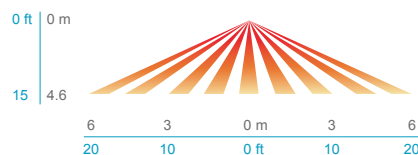
NLTAIR2 APIRKO, APIRHKO, or APIRAKO COVERAGE PATTERNS

HIGH MOUNT 360° LENS (#6)

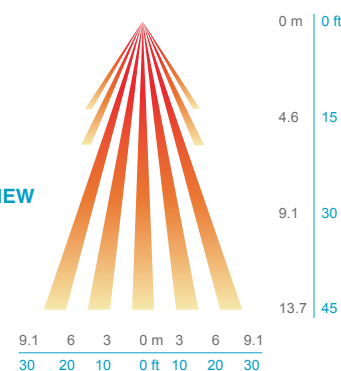


- Best choice for 15 to 45 ft (4.57 to 13.72 m) mounting heights
- 15 to 20 ft (4.57 to 6.10 m) radial coverage overlaps area lit by a typical high bay fixture
- Excellent detection of large motion (e.g. walking) up to a 35 ft (10.76 m) mounting height
- Excellent detection of extra large motion (e.g. forklifts) up to a 45 ft (13.72 m) mounting height

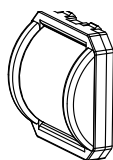
LOW VIEW



HIGH VIEW

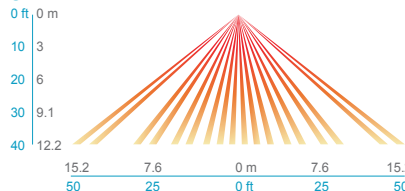


HIGH MOUNT AISLEWAY LENS (#50)

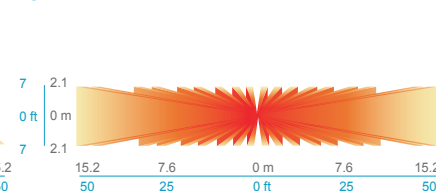


- Provides a bi-directional coverage pattern ideal for warehouse racking
- 1.2x mounting height equals approximate detection range in either direction
- Typical 40 ft (12.19 m) mounting detects 50 ft (15.24 m) in either direction
- Superior aisleway coverage compared to a masked 360° lens

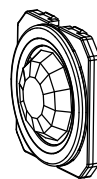
SIDE VIEW



TOP VIEW

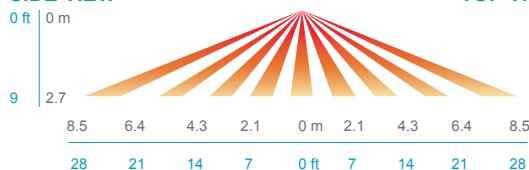


LOW MOUNT 360° LENS (#10)

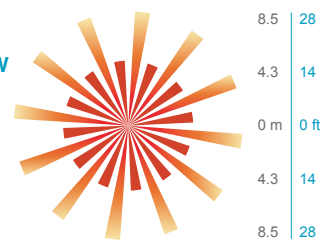


- Best choice for large motion detection (e.g. walking)
- 360° conical shaped pattern
- Provides ~24 ft (7.32 m) radial coverage (~2000 ft²) when mounted at 9 ft (2.74 m)
- 7 to 15 ft (2.13 to 4.57 m) mounting heights provide 16 to 36 ft (4.88 to 10.97 m) radial coverage
- Detection range improves when walking across beams compared to into beams

SIDE VIEW

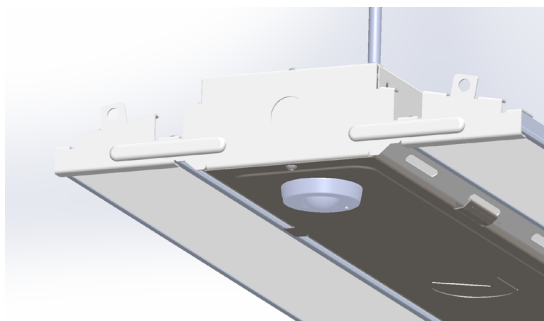


TOP VIEW



NLTAIR2 APIR, APIRH, OR APIRA

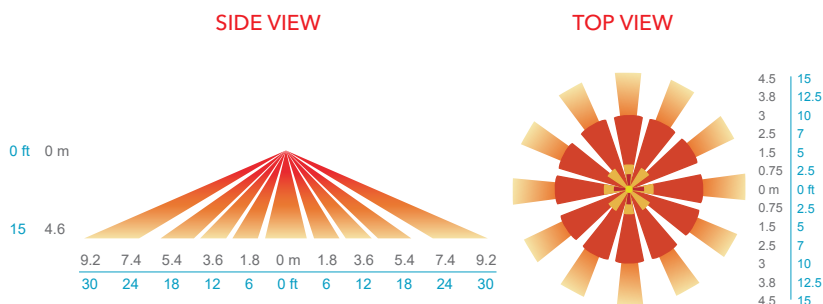
- 100% digital PIR detection
- Combined daylight and occupancy sensor
- Fully dimmable via digital or analog dimming protocols, providing the right amount of light for the application and to optimize energy savings
- Optional UL 924 emergency functionality via EM option, which eliminates wiring for sensing normal power

**COVERAGE PATTERN**

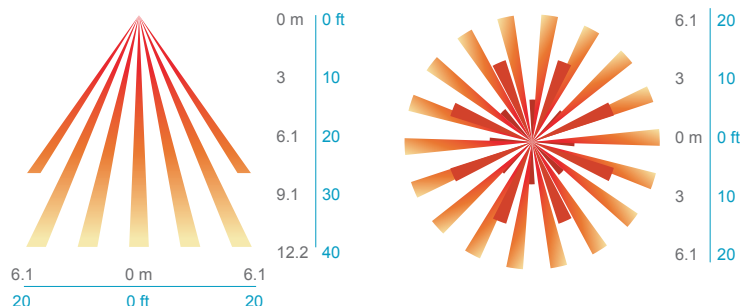
Lens rotates 15 deg to enable adjustment. Initial detection will occur earlier when walking across sensor's field of view than when walking directly at sensor.

7 – MINI LOW-BAY 360°

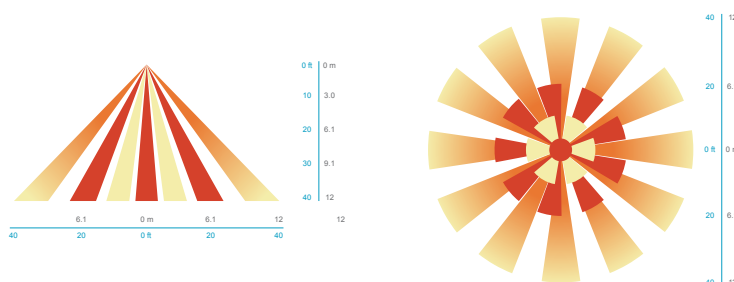
- Recommended for walking motion detection from mounting heights between 8 ft (2.44 m) and 20 ft (6.10 m)
- Initial detection of walking motion along sensor axis at distances of 2x the mounting height up to 15 ft (4.57 m) and 1.75x up to 20ft (6.10 m).
- Provides 12 ft (3.66 m) radial detection of small motion when mounted at 9 ft (2.74 m)

**30 – UNIVERSAL 360°**

- Provides excellent detection of large motion (e.g. walking) when mounted between 15 to 40 ft (4.57 to 12.19 m)
- 15 to 20 ft (4.57 to 6.10 m) radial coverage overlaps area lit by a typical high bay fixture
- Recommended for fixtures that have a 1:1 spacing to mounting height ratio or less (e.g. fixtures 30' on center or less @ a 30' mounting height.)

**45 – HIGH MOUNT 360°**

- Optimized full coverage pattern for 10 – 40 ft. (3.1 – 12 m)
- Reliable detection of large motion (e.g. pedestrian walking traffic) up to 30 ft. (9.1 m) mounting height
- Reliable detection of extra-large motion (e.g. forklift traffic) up to 40 ft. (12 m) mounting height



IMP – Integrated Modular Plug

- The integrated modular plug (IMP) option allows the installer to plug and play a multitude of accessories.
- Cord sets connect quickly to any fixture with IMP option.
- IMP accessories include occupancy sensors, photocells, X-point relays.

IMP compatible cord sets ¹	
CS1WIMP	Straight plug, 120V
CS3WIMP	Twist-lock, 120V
CS7WIMP	Straight plug, 277V
CS11WIMP	Twist-lock, 277V
CS25WIMP	Twist-lock, 347V
CS93WIMP	600V SEOW white cord, no plug
CS97WIMP	Twist-lock, 480V

Ordering Example

Order As: Qty 1 – IBG 12000LM SEF AFL GND 120 GZ10 40K 80CRI IMP CP5BW DWH
Ships As: Qty 1 – IBG 12000LM SEF AFL GND MVOLT GZ10 40K 80CRI DWH
Qty 1 – CS1WIMP

Notes
¹ Cord set required for fixture operation. All cord sets are 18/3, 6' white.

RRL – RELOC®-Ready Luminaire

- RRL connectors to be used with the OnePass system.
- Load side of connector factory installed to luminaire.
- 4-pole mating connector with push-in terminations allows for simple installation.
- Touch-safe design on both halves meets UL/CSA requirement.
- Wiping contact design allows safe disconnect under load.



ORDERING INFORMATION

Lead times will vary depending on options selected. Consult with your sales representative.

Example: RRLA

Series	Wiring instructions
RRL RELOC®-ready luminaire	A Hot conductor wired to position #1 (phase A); non-dimming B Hot conductor wired to position #2 (phase B); non-dimming AE Hot conductor wired to position #1 (phase A), hot conductor #2 wired to position #2 (phase B); non-dimming ¹ C12S Hot conductor in position #1 (phase A), low voltage conductor #1 in position #2, low voltage conductor #2 in position #3; dimming ²

Compatible RELOC® Cables for Industrial Luminaires (ordered and shipped separately)
(click to view RELOC product page for more information)



- Notes
- ¹ AE commercial fixtures should disconnect the TSPL before unplugging the RRL so it does not go into discharge mode. Requires fixture to have battery option.
 - ² C12S option is used with the OnePass for 0-10V/DALI applications. Not for use with dimming sensors.

PLT Wattage and Color Selectable LED Cutoff Wall Pack Spec Sheet


5 YEAR
WARRANTY


► DESCRIPTION

Traditional 35W-120W Semi-Cutoff LED Wall Pack | 120-277V | 4000/5000K Selectable | Glass Lens

► APPLICATION

Building Mount for exterior illumination (perimeters, pathways, loading docks, and other general security lighting requirements)





PLT Wattage and Color Selectable LED Cutoff Wall Pack Spec Sheet

PRODUCT FEATURES

- Up to 82% energy saving compared to traditional fixtures.
- Heavy-duty die cast aluminum housing.
- Architectural bronze finish, anti-corrosion spray.
- Borosilicate glass lens for long life durability, no fading, high light transmittance.
- IP65 rated, waterproofing material is used throughout the fixture.
- DLC 5.1 Premium***
- Dusk to dawn photocell options.
- 4000K/5000K CCT selectable; -40 Deg. to 113 Deg. F operating temperature.
- 100%/80%/60% wattage selectable.
- 5 Years warranty.

*Optional finishes available via Special Order. Call 1-800-624-4488.

**See Selectable Table on Page 5 for full breakdown.

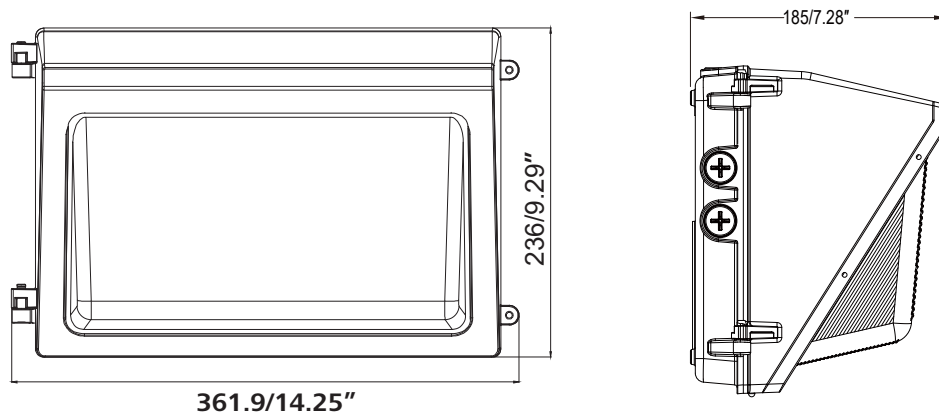
***See Page 5 for full list of DLC part numbers.

ELECTRICAL SPECIFICATIONS

Catalog Number	Photocell	Wattage	Lumens**	Dimming	CCT	Efficacy**	CRI	Housing Color	Input Voltage	LED L70 Life Hours	Wattage Equivalent
PLT-13000	No	21W/ 28W/ 35W	Up to 4800 lm	No	4000K	Up to 149 lm/W	>80	Bronze	120–277V	100,000 hrs	175W MH
					5000K			Black*			
								White*			
PLT-13001	No	72W/ 96W/ 120W	Up to 16,800 lm	No	4000K	Up to 147 lm/W	>80	Bronze	120–277V	100,000 hrs	400W MH
					5000K			Black*			
								White*			

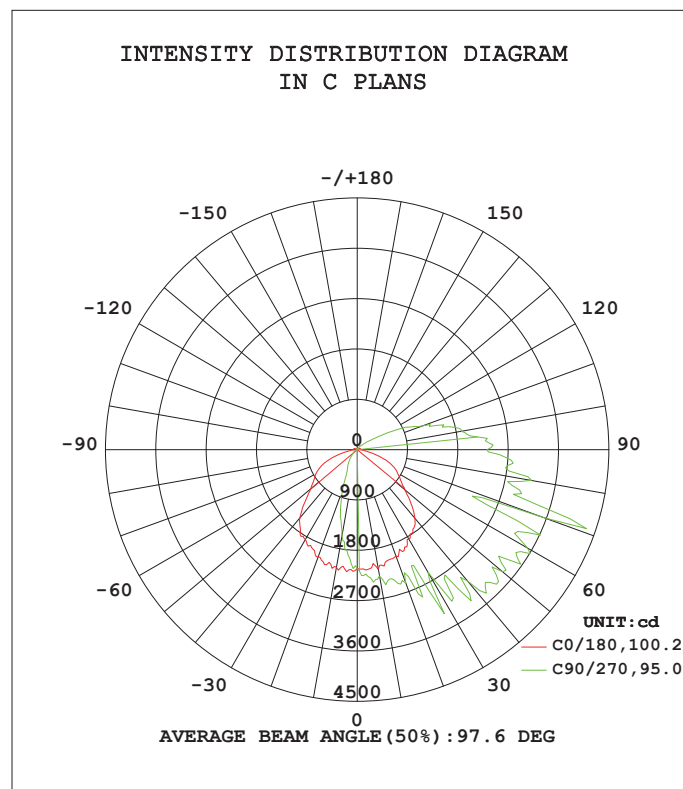
PLT Wattage and Color Selectable LED Cutoff Wall Pack Spec Sheet

DIMENSIONS



35W/120W

PHOTOMETRICS



PLT Wattage and Color Selectable LED Cutoff Wall Pack Spec Sheet

INSTALLATION REFERENCE



ACCESSORIES



Emergency battery back-up

Model NO.	Output Voltage	Emergency Power	Battery Type	For Models
NPS-00189*	81~155Vdc	8W	Li-ion/ Ni-MH	35W-75W
NPS-00188*		15W		100W-120W

*Available via Special Order. Please call 1-800-624-4488.



PLT Wattage and Color Selectable LED Cutoff Wall Pack Spec Sheet

SELECTABLE TABLE

SKU	Wattage	CCT	Lumens	LPW
PLT-13000	35	5000K	4800lm	137lm/w
	35	4000K	4700lm	136lm/w
	28	5000K	3900lm	139lm/w
	28	4000K	3800lm	136lm/w
	21	5000K	3200lm	149lm/w
	21	4000K	3100lm	147lm/w
PLT-13001	120	5000K	16800lm	140lm/w
	120	4000K	16700lm	139lm/w
	96	5000K	13200lm	138lm/w
	96	4000K	13100lm	137lm/w
	72	5000K	10500lm	147lm/w
	72	4000K	10400lm	145lm/w

DLC Part Numbers

SKU	DLC Level	DLC #
PLT-13000	5.1 Premium	PLTBD16
PLT-13001	5.1 Premium	PLTBD56